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Department of Computer Engineering
CEN 302-Software Engineering**

PMS (Police Management System)

Requirements Specification

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1. Executive Summary

1.1 Project Overview

The Police Management System (PMS) is a comprehensive, web-based digital platform designed to modernize and streamline law enforcement operations within police departments. Traditional police record-keeping and case management often rely on paper-based documentation, disconnected databases, and fragmented communication channels between officers, detectives, citizens, experts and administrators. These inefficiencies can lead to delayed investigations, data inconsistencies, loss of critical evidence records, and reduced accountability across departments.

The PMS addresses these challenges by providing a centralized, role-based digital platform that integrates incident reporting, case management, evidence tracking, suspect assignment, officer coordination, and administrative oversight into a single unified system. The platform ensures that each participant in law enforcement operations acts within clearly defined responsibilities, improving accountability, data integrity, and operational efficiency.

The system introduces a multi-role architecture consisting of Police Officers, Detectives, Forensic Experts, Traffic Officers, Citizens, and Administrators. Each role interacts with the system through a tailored interface that reflects the specific permissions and responsibilities associated with their position, enabling efficient collaboration while maintaining strict access control and data security.

Key features of the Police Management System include:

- Secure user authentication and role-based access control
- Incident report creation and management by authorized police officers
- Case management including suspect assignment, evidence tracking, and status updates
- Forensic analysis workflows for DNA, fingerprints, weapons, and multimedia
- Traffic violation registration, fine management, and vehicle control
- Citizen portal for incident reporting, complaint submission, and case status checking
- Structured investigation workflows for detectives
- A real-time administrative dashboard with statistics and system monitoring

By digitizing and automating these workflows, the platform enhances transparency, improves inter-departmental communication, and ensures that all case-related data is stored securely in a centralized and accessible database.

Accessible from any device with internet connectivity, the PMS offers a user-friendly and responsive interface that supports seamless interaction between all user roles. The system ultimately aims to increase investigative efficiency, reduce manual overhead, and provide a reliable, secure, and accountable platform for law enforcement management.

1.2 Purpose and Scope of this Specification

Purpose

This specification document defines the functional and non-functional requirements of the Police Management System. It serves as a comprehensive guide for developers, stakeholders, and evaluators to understand the system's architecture, features, and operational behavior.

The purpose of this project is to design and develop a robust law enforcement management platform that digitalizes and automates police operations, from incident report creation to case closure. The platform facilitates structured collaboration between multiple user roles while ensuring data integrity, security, and transparency throughout all operational processes.

Additionally, this document aims to establish a clear understanding of system functionalities, including user role permissions, workflow processes, and interaction mechanisms. It also supports effective communication within the development team, ensuring alignment with project goals and defined requirements.

Scope

In Scope

The Police Management System includes the following core functionalities:

- **User Authentication and Access Control:** Secure login for all users with role-based permission enforcement throughout the system.
- **Incident Report Management:** Enables Police Officers to create, document, and submit incident reports for recorded crimes.
- **Case Management:** Supports creation, updating, and status tracking of cases (Open, Under Investigation, Closed) by authorized personnel.
- **Suspect and Evidence Management:** Allows Detectives and Officers to link suspects and physical/digital evidence to existing cases.
- **Fine Management:** Allows Traffic Officers add fines and Citizens to be informed about them.
- **User and Role Administration:** Provides Administrators with full control over user accounts, roles, and department management.
- **Statistics Dashboard:** Displays real-time data including number of cases, officers, incidents, and department performance metrics.
- **Arrest Record Management:** Allows Officers to document arrest information and link it to relevant case files.

Out of Scope

The following aspects are not included in the current scope of the Police Management System:

- Integration with external national or international law enforcement databases (e.g., Interpol, national criminal registries).
- Real-time GPS tracking or surveillance integration for patrol officers.
- Automated court case management or legal documentation generation.
- Mobile-native applications; the system is web-based only, though the interface is responsive.
- Biometric authentication systems such as fingerprint or facial recognition.
- Post-deployment maintenance and support services, which will be defined separately.

2. Product / Service Description

2.1 Product Context

The Police Management System is a standalone, self-contained web-based platform that does not depend on or integrate with any external law enforcement systems or third-party databases in its current version. All core workflows — incident report creation, case management, evidence tracking, suspect assignment, officer coordination, and administrative oversight — are handled entirely within the platform.

The system interfaces internally between six distinct user roles: Administrator, Police Officer, Detective, Forensic Expert, Traffic Officer, and Citizen. Each role interacts through a dedicated, permission-restricted interface within the same environment, ensuring that no user can access or perform actions outside their designated responsibilities.

The PMS operates under a client-server architecture, where all users interact with the application through a standard web browser, while data is processed and stored on a central server. This ensures real-time updates, consistent data visibility, and synchronized operations across all roles.

While the platform operates independently in its initial version, it relies on a hosting infrastructure for deployment and availability, and on an email/notification service to deliver real-time alerts to users upon relevant system events such as case assignments, status changes, or report submissions.

2.2 User Characteristics

Police Officer

Represents the frontline law enforcement personnel responsible for documenting criminal incidents and managing field-level case records.

- Registers and logs into the system using department-issued credentials managed by the Administrator.
- Creates new incident reports for reported crimes, entering key details such as location, crime type, description, date, and time.
- Updates case information and status (Open, Under Investigation, Closed) as investigations develop.
- Uploads and attaches physical or digital evidence to existing cases.
- Records and documents arrest information linked to relevant cases.
- Views all cases assigned to them from a personalized dashboard for efficient investigation management.
- Cannot manage user accounts, modify other officers' reports, or access administrative settings.
- Requires basic-to-moderate familiarity with web-based platforms and standard digital record-keeping tools.

Forensic Expert

Represents specialized laboratory personnel responsible for scientific analysis of evidence collected from crime scenes.

- Logs into the system using credentials issued by the Administrator.
- Receives and processes evidence submitted from Police Officers and Detectives.
- Performs DNA analysis on biological samples to support case investigations.
- Analyzes fingerprints and compares them with existing records in the system.
- Conducts forensic examination of weapons, including ballistics and tool mark analysis.
- Reviews and analyzes photos and video materials for potential forensic evidence.
- Maintains proper chain of custody by managing evidence intake, processing, and storage within the laboratory.
- Prepares, writes, and submits detailed forensic reports linked to specific cases.
- Cannot create incident reports, assign suspects, or manage user accounts.
- Requires specialized forensic science knowledge and proficiency with laboratory management systems.

Detective

Represents specialized investigative personnel responsible for managing complex cases, coordinating with officers, and maintaining detailed investigative records.

- Logs into the system using credentials issued by the Administrator.
- Views all cases assigned to them and accesses full case details including incident reports and evidence.
- Assigns and links suspects to existing cases, entering details such as name, age, physical description, and associated evidence.
- Uploads and manages evidence related to ongoing cases in coordination with police officers.
- Reviews and analyzes all information associated with a case before determining investigative direction.
- Cannot create new incident reports, manage user accounts, or perform administrative actions.
- Requires professional investigative knowledge, strong analytical skills, and proficiency with digital case management tools.

Traffic Officer

Represents law enforcement personnel responsible for managing road safety, traffic violations, and vehicle-related incidents.

- Logs into the system using credentials issued by the Administrator.
- Registers traffic accidents by recording details such as location, vehicles involved, injuries, and incident description.
- Issues and records fines for traffic violations within the system.
- Manages and tracks road violations and infractions.
- Performs vehicle control checks including registration, insurance, and inspection validation.
- Maintains and updates driving license records and their validity status.
- Generates traffic-related reports and statistical summaries.
- Views and manages traffic cases assigned to them.
- Cannot access criminal investigation cases, assign suspects, or manage user accounts.
- Requires familiarity with traffic laws, road safety regulations, and reporting systems.

Citizen

Represents the general public who interact with the police system for reporting purposes and status tracking.

- Registers and logs into the system through the citizen portal.
- Reports incidents such as crimes or suspicious activities by submitting relevant details.
- Submits complaints regarding police conduct or services.
- Reports missing persons through the system.
- Checks the status of previously submitted reports and cases.
- Views traffic fines issued to them by Traffic Officers.
- Receives and views system notifications and updates.
- Can only access their own reports, complaints, and fines; no access to other case data.
- Cannot modify official case records or access internal police operations.
- Does not require technical training; the system is designed to be user-friendly for the general public.

Administrator

Oversees the overall integrity of the platform, managing user accounts, roles, and department structure, while also coordinating officer assignments and monitoring investigative activity across the system. The Administrator serves as the highest-authority role and has visibility over all system operations.

- Pre-configured in the system at deployment time; no self-registration or signup flow
- Creates new user accounts for all roles (Police Officer, Detective, Forensic Expert, Traffic Officer, Citizens)
- Updates, deactivates, or reactivates existing user accounts
- Assigns and modifies user roles to ensure correct access levels
- Adds, edits, and manages police department records
- Assigns Police Officers to specific cases
- Monitors all active and ongoing cases within the system
- Views real-time analytics including number of cases, users per role, active incidents, and department metrics
- Cannot create incident reports, upload evidence, or assign suspects to cases
- Requires strong administrative and organizational judgment, familiarity with law enforcement workflows

2.3 Assumptions

- All users are assumed to have basic-to-moderate familiarity with web-based platforms and standard digital tools.
- The system operates under the assumption of a stable and reliable internet connection for all users.
- Police Officers are responsible for providing accurate and complete information when creating incident reports.
- Detectives are assumed to act in good faith when linking suspects and evidence to cases within the boundaries of their role.
- Forensic Experts provide scientifically accurate analysis results

- Traffic Officers are responsible for accurate recording of violations and accidents
- Citizen accounts can be self-registered through the public portal; staff accounts are created by Admin only
- Administrators will complete all pending user verification and account management tasks within a reasonable timeframe.
- All uploaded evidence and documentation is assumed to be authentic and accurately represents the events described.
- All users are responsible for maintaining the confidentiality of their login credentials and not sharing access.
- Notifications and alerts sent by the system are assumed to reach users in a timely manner through the platform's notification mechanism.
- The system assumes that user roles are assigned accurately by the Administrator and reflect actual departmental positions.
- It is assumed that the system will perform regular data backups to prevent data loss in the event of system failure or unexpected issues.
- Users are assumed to follow departmental procedures and legal standards when entering case-related data into the system.

2.4 Constraints

- The platform is web-based and requires a stable internet connection for all operations; offline functionality is not supported.
- The system must comply with applicable data protection and law enforcement data regulations to ensure the privacy, security, and legal integrity of all stored information.
- Role-based access control is strictly enforced; users cannot perform actions outside the permissions defined for their assigned role.
- User accounts for all the roles can only be created by an Administrator; no self-registration flow is available for these roles.
- Incident reports, once formally submitted and saved, should follow a controlled editing process to maintain data integrity.
- The system does not support integration with external national law enforcement databases or third-party systems in its current scope.
- The platform does not support legal contract generation, court case management, or automated digital signing of legal documents.
- The user interface and functional behavior may be constrained by the limitations of the web technologies and frameworks used during development.
- Budget and timeline constraints during development may limit the implementation of certain advanced features or performance optimizations.
- The system is not a mobile-native application; while the interface is designed to be responsive, it is delivered and optimized as a web-based platform.

2.5 Dependencies

- The activation and availability of all Officer, Detective, Forensic Expert, Traffic Officer and Citizens accounts is dependent on the Administrator creating and configuring them correctly.
- The creation of a case is dependent on an Incident Report being submitted first by an authorized Police Officer.

- The assignment of a suspect to a case is dependent on the existence of an active case within the system.
- The upload of evidence is dependent on a case already being created and accessible within the system.
- The generation of dashboard statistics is dependent on accurate and complete data being entered by all users across all roles.
- Forensic analysis workflows are dependent on evidence being properly uploaded and linked to a case.
- Traffic fine records for citizens depend on Traffic Officers accurately entering fine data.
- Notifications and system alerts are dependent on the platform's notification service operating correctly and user email addresses being properly configured.
- The dispute or access issue resolution process is dependent on the Administrator having full visibility into all user accounts, case records, and system logs.
- The platform's overall performance and availability are dependent on the reliability and capacity of the chosen hosting infrastructure.
- The successful development of the project is dependent on the use of specific technologies for backend, frontend, and database management as defined in the technical architecture.

3. Requirements

3.1 Functional Requirements

The following table defines the functional requirements of the Police Management System (PMS), derived from the user scenarios, use cases, and user stories defined in this specification. Each requirement is assigned a unique identifier, a priority level, and describes what the system shall do.

Priority Scale: 1 = Must Have (Highest) | 2 = Should Have | 3 = Nice to Have (Lowest)

Req #	Requirement	Comments	Priority	Date Reviewed	Reviewed/ Approved
FR_01	The system shall provide role-specific dashboards for each user type: Administrator, Police Officer, Detective, Forensic Expert, Traffic Officer, and Citizen.	Every role has its own dashboard displaying only relevant features and data.	1	26/03/2026	Adorela Dashi / Egzon Xheleshi
FR_02	User passwords must be stored securely using a cryptographic hashing algorithm.	Passwords are saved as encrypted hashes rather than plain text.	1	26/03/2026	Ester Alisufi / Kaltrina Lala
FR_03	The system shall authenticate staff users using Admin-assigned credentials; Citizens may self-register.	Upon successful login, the system redirects the user to their role-specific dashboard.	1	26/03/2026	Drini Ziri / Ester Alisufi

Req #	Requirement	Comments	Priority	Date Reviewed	Reviewed/ Approved
FR_04	The system shall lock or flag an account after a defined number of consecutive failed login attempts.	Failed login attempts must be logged; repeated failures trigger protective account lockout.	1	26/03/2026	Amarildo Hazizi / Adorela Dashi
FR_05	The system shall restrict access to all features and data based on the user's assigned role.	Role-based access control enforced at system level; users cannot perform actions outside their permitted scope.	1	26/03/2026	Kaltrina Lala / Egzon Xheleshi
FR_06	Administrator shall be able to create new user accounts for Police Officers, Detectives, Forensic Experts, and Traffic Officers.	Admin provides the new user's full name, username, password, role, and department during account creation.	1	26/03/2026	Ester Alisufi / Petro Boçi
FR_07	Administrator shall be able to update, deactivate, or reactivate existing user accounts.	Status changes take effect immediately; affected users are notified with a documented reason.	1	26/03/2026	Egzon Xheleshi / Drini Ziri
FR_08	Administrator shall be able to assign and modify user roles within the system.	Role changes take effect immediately and update the user's access permissions accordingly.	1	26/03/2026	Adorela Dashi / Kaltrina Lala
FR_09	Administrator shall be able to add and manage police departments within the system.	Departments can be created, edited, and deleted; users are organized by department.	1	26/03/2026	Petro Boçi / Amarildo Hazizi
FR_10	Administrator shall be able to assign Police Officers to specific cases.	Admin selects a case and an available Officer; the Officer is notified of the assignment immediately.	1	26/03/2026	Drini Ziri / Adorela Dashi
FR_11	Administrator shall have access to a real-time analytics dashboard displaying platform-wide metrics.	Dashboard shows total cases per status, registered users per role, active departments, and recent incidents.	3	26/03/2026	Ester Alisufi / Egzon Xheleshi

Req #	Requirement	Comments	Priority	Date Reviewed	Reviewed/ Approved
FR_12	Administrator shall be able to manage access control and permissions across all roles.	Admin can view and adjust what each role is permitted to see and do within the system.	1	26/03/2026	Kaltrina Lala / Amarildo Hazizi
FR_13	Police Officers shall be able to create new incident reports for reported crimes.	The report form requires: location, crime type, detailed description, date, and time of the incident.	1	26/03/2026	Egzon Xheleshi / Petro Boçi
FR_14	Police Officers shall be able to register and document arrest information linked to an existing case.	Arrest records include arrestee details, date/time of arrest, charges, and the arresting Officer's identity.	1	26/03/2026	Amarildo Hazizi / Drini Ziri
FR_15	Police Officers shall be able to manage patrol scheduling and assignments.	Officers can view and update their patrol schedule, including routes and time slots.	2	26/03/2026	Adorela Dashi / Ester Alisufi
FR_16	Police Officers shall be able to register and manage complaints submitted by citizens or in person.	Complaints are linked to a citizen or case record and assigned for follow-up.	1	26/03/2026	Petro Boçi / Drini Ziri
FR_17	Police Officers shall be able to manage records of persons of interest or suspects.	Officer can add, view, and update suspect profiles linked to active cases.	1	26/03/2026	Kaltrina Lala / Adorela Dashi
FR_18	Police Officers shall be able to write and submit investigation reports.	Reports are linked to a specific case and include narrative, findings, and actions taken.	1	26/03/2026	Drini Ziri / Egzon Xheleshi
FR_19	Detectives shall be able to open and manage criminal cases.	Detectives can initiate case records, assign investigators, and track case progress.	1	26/03/2026	Ester Alisufi / Amarildo Hazizi
FR_20	Detectives shall be able to upload and link evidence files to an existing case.	Evidence is stored securely and associated with the correct case.	1	26/03/2026	Egzon Xheleshi / Kaltrina Lala
FR_21	Detectives shall be able to manage witness records and testimonies.	Witness entries include identity details, testimony	1	26/03/2026	Amarildo Hazizi / Ester Alisufi

Req #	Requirement	Comments	Priority	Date Reviewed	Reviewed/ Approved
		content, and confidentiality status.			
FR_22	Detectives shall be able to assign suspects to an existing case by entering the suspect's details.	Required fields: full name, age, physical description, and any related evidence references.	1	26/03/2026	Adorela Dashi / Petro Boçi
FR_23	Detectives shall be able to track and update investigation status.	Status transitions (Open, Under Investigation, Closed) are logged with timestamp and actor identity.	1	26/03/2026	Drini Ziri / Kaltrina Lala
FR_24	Detectives shall be able to write and submit investigation reports.	Reports include case summary, evidence analysis, suspect assessment, and investigative conclusions.	1	26/03/2026	Petro Boçi / Egzon Xheleshi
FR_25	Forensic Experts shall be able to perform and document DNA analysis on submitted samples.	Analysis results are linked to the relevant case and include methodology and findings.	1	26/03/2026	Kaltrina Lala / Ester Alisufi
FR_26	Forensic Experts shall be able to perform and document fingerprint analysis.	Fingerprints are analyzed and compared; results are stored linked to the case.	1	26/03/2026	Egzon Xheleshi / Adorela Dashi
FR_27	Forensic Experts shall be able to perform and document weapon analysis.	Includes ballistics, tool marks, and trace evidence linked to submitted weapons.	1	26/03/2026	Amarildo Hazizi / Petro Boçi
FR_28	Forensic Experts shall be able to analyze and annotate photos and videos submitted as evidence.	Analysis includes metadata extraction, image enhancement notes, and findings.	1	26/03/2026	Ester Alisufi / Drini Ziri
FR_29	Forensic Experts shall be able to manage laboratory evidence (intake, processing, chain of custody).	System tracks evidence from submission through analysis and final storage.	1	26/03/2026	Adorela Dashi / Amarildo Hazizi

Req #	Requirement	Comments	Priority	Date Reviewed	Reviewed/ Approved
FR_30	Forensic Experts shall be able to write and submit forensic reports.	Reports are formally linked to a case and include all analysis findings and conclusions.	1	26/03/2026	Petro Boçi / Ester Alisufi
FR_31	Traffic Officers shall be able to register traffic accidents.	Accident records include location, vehicles involved, injuries, witness information, and description.	1	26/03/2026	Drini Ziri / Amarildo Hazizi
FR_32	Traffic Officers shall be able to issue and register fines for traffic violations.	Fine records include violation type, amount, date, citizen/vehicle identification, and issuing officer.	1	26/03/2026	Kaltrina Lala / Petro Boçi
FR_33	Traffic Officers shall be able to manage road violations and infractions.	Officers can view, update, and close violation records.	1	26/03/2026	Ester Alisufi / Kaltrina Lala
FR_34	Traffic Officers shall be able to perform and document vehicle control checks.	Checks include registration, insurance, and inspection status linked to a vehicle record.	2	26/03/2026	Egzon Xheleshi / Amarildo Hazizi
FR_35	Traffic Officers shall be able to generate traffic reports and statistics.	Reports cover violation trends, accident frequency, and fine collection summaries.	2	26/03/2026	Adorela Dashi / Drini Ziri
FR_36	Traffic Officers shall be able to manage driving license records and validations.	Officers can search, view, and flag license records; suspension notes are recorded.	2	26/03/2026	Amarildo Hazizi / Kaltrina Lala
FR_37	Citizens shall be able to report incidents through the public citizen portal.	Incident report form includes description, location, date/time, and optional photo attachment.	1	26/03/2026	Petro Boçi / Adorela Dashi
FR_38	Citizens shall be able to submit complaints regarding police conduct or services.	Complaint form includes category, description, and the relevant officer or department.	1	26/03/2026	Drini Ziri / Petro Boçi

Req #	Requirement	Comments	Priority	Date Reviewed	Reviewed/ Approved
FR_39	Citizens shall be able to report missing persons.	Report includes personal details, last known location, date missing, and contact information.	1	26/03/2026	Kaltrina Lala / Drini Ziri
FR_40	Citizens shall be able to check the status of their reported cases.	Citizens can view the current status of any case they have submitted.	1	26/03/2026	Ester Alisufi / Adorela Dashi
FR_41	Citizens shall be able to view any traffic fines issued to them by Traffic Officers.	Citizens see fine details including violation type, amount, date, and payment status.	1	26/03/2026	Egzon Xheleshi / Ester Alisufi
FR_42	Citizens shall be able to view system notifications and updates relevant to their reports.	Notifications are delivered when their report status changes or a response is added.	2	26/03/2026	Amarildo Hazizi / Egzon Xheleshi
FR_43	The system shall enforce real-time form validation across all submission forms.	Validation executes client-side in real time without requiring a full page reload.	1	26/03/2026	Petro Boçi / Kaltrina Lala
FR_44	All notifications triggered by system events shall be delivered to relevant parties within 1 minute.	Applies to: case assignments, status changes, suspect additions, and account changes.	1	26/03/2026	Adorela Dashi / Egzon Xheleshi
FR_45	The system shall log all key actions with timestamps for audit purposes.	Logs include: login attempts, account changes, case creation, status updates, evidence uploads.	1	26/03/2026	Drini Ziri / Ester Alisufi
FR_46	The web application shall be responsive and accessible from any modern device.	Interface adapts to desktop, tablet, and mobile screen sizes without loss of functionality.	3	26/03/2026	Kaltrina Lala / Amarildo Hazizi

3.2 Non-Functional Requirements

3.2.1 Product Requirements

3.2.1.1 User Interface Requirements

The user interface for the Police Management System shall be compatible with all modern web browsers and accessible from both desktop and mobile devices. The interface is divided into three role-specific views, each tailored to the responsibilities of the assigned user.

Admin Interface

- A main dashboard displaying real-time platform metrics: total cases per status (Open, Under Investigation, Closed), registered users per role, active departments, and recent incidents
- A user management panel for creating, editing, deactivating, and reactivating Police Officer and Detective accounts, with role and department assignment controls
- A department management panel for adding, editing, and removing police departments, and viewing officers organized by unit
- A case overview panel displaying all cases in the system, filterable by status, assigned officer, department, and date range
- A case assignment panel allowing the Admin to assign available Police Officers to specific open or ongoing cases
- An analytics and statistics section displaying crime trends, case resolution rates, and department performance metrics
- A notifications panel displaying all system-generated alerts in reverse chronological order

Police Officer Interface

- A personal dashboard displaying all cases currently assigned to the Officer, with their statuses, dates, and quick-access links
- An incident report creation form with required fields: location of the incident, crime type, detailed description, date, and time
- A case management panel allowing the Officer to view full case details, update case information, update case status, and add notes
- An evidence upload panel allowing the Officer to attach and manage files (documents, images, videos) linked to a specific case
- An arrest record form allowing the Officer to document arrest details, charges, date/time of arrest, and link the record to the relevant case
- A notifications panel displaying all system-generated alerts relevant to the Officer in reverse chronological order

Detective Interface

- A personal dashboard displaying all cases assigned to the Detective, with full case details and investigation history
- A suspect management panel allowing the Detective to add, view, and manage suspect records linked to an existing case
- A suspect entry form with required fields: full name, age, physical description, and related evidence references
- An evidence panel allowing the Detective to upload and manage evidence files linked to assigned cases
- A case review section where the Detective can view all incident reports, evidence, suspect entries, and arrest records associated with their assigned cases

- A notifications panel displaying all system-generated alerts relevant to the Detective in reverse chronological order

Forensic Expert Interface:

- A dashboard displaying pending and completed forensic analysis requests
- Analysis panels for DNA, fingerprints, weapons, and photo/video evidence
- A laboratory evidence management panel tracking chain of custody
- A forensic reports section for writing and submitting formal forensic reports
- A notifications panel for case-related alerts and new evidence submissions

Traffic Officer Interface:

- A dashboard displaying recent accidents, violations, and pending actions
- An accident registration form (location, vehicles, injuries, description)
- A fine management panel for issuing and tracking traffic fines
- A violations management section for recording and managing road infractions
- A vehicle control section for documenting control checks
- A license management section for validating and flagging driving licenses
- A reports section for generating traffic statistics and summaries

Citizen Interface:

- A public-facing portal accessible after self-registration
- An incident reporting form (description, location, date/time, optional photo)
- A complaints submission form (category, description, relevant officer/departement)
- A missing person report form (personal details, last known location, date missing)
- A case status tracker for monitoring submitted reports
- A fines viewer for checking traffic fines issued to the citizen
- A notifications section for updates on submitted reports

3.2.1.2 Learnability

The Police Management System shall be designed so that users across all three roles can operate the system effectively with minimal prior training. Key learnability features include:

- **Intuitive Role-Based Interfaces:** Each role's dashboard surfaces only the features and data relevant to that role, reducing cognitive load and preventing confusion from irrelevant options
- **Consistent Navigation Patterns:** Common actions such as submitting reports, updating case statuses, and managing suspects follow standardized workflows and button placement across all role interfaces

- **Contextual Guidance:** Key actions such as creating an incident report, adding a suspect, or uploading evidence shall include inline instructions or tooltips explaining required fields and constraints
- **Clear Status Indicators:** All cases shall display their current status (Open, Under Investigation, Closed) prominently and consistently, using standardized labels and color coding
- **Minimal Training Dependency:** The interface shall rely on familiar web design patterns — form submissions, confirmation dialogs, notification badges — to reduce the need for formal onboarding documentation

3.2.1.3 Accessibility

The Police Management System shall adhere to accessibility standards to ensure usability for all users:

- **Standards Compliance:** The web interface shall follow Web Content Accessibility Guidelines (WCAG) 2.1, including sufficient color contrast ratios, full keyboard navigability, and compatibility with screen reader technologies
- **Responsive Design:** The platform shall be fully functional across desktops, tablets, and mobile devices, adapting layouts appropriately to varying screen sizes without loss of functionality
- **Text Alternatives:** All images, icons, and non-text UI elements shall include descriptive alt text for screen reader compatibility
- **Adjustable Text:** The interface shall remain functional and readable when users adjust browser zoom levels or font size settings
- **Accessible Forms:** All report creation, suspect entry, and action forms shall include properly labeled fields, clear error messages, and logical tab order to support users relying on assistive technologies

3.2.1.4 Efficiency

The platform shall be optimized to support fast task completion across all user roles:

- **Page Load Performance:** All pages and dashboards shall load within 3 seconds under normal network conditions; login and messaging interfaces shall load within 2 seconds
- **Case Search Performance:** Case search and filter results shall update within 2 seconds of the user applying or modifying a filter, without requiring a full page reload
- **Notification Delivery:** All system-triggered notifications (case assignments, status changes, account changes) shall be delivered to relevant parties within 1 minute of the triggering event
- **Inline Validation:** All forms shall provide real-time client-side validation, highlighting errors immediately without requiring form submission, reducing correction cycles
- **Optimized Workflows:** Frequently performed actions — such as updating a case status, submitting an incident report, and adding a suspect — shall be accessible within a minimal number of navigation steps from the user's dashboard

3.2.1.5 Dependability and Memorability

The PMS shall ensure that users can return after periods of inactivity and quickly re-establish proficiency:

- **Consistent Layout:** Core interface elements such as navigation menus, dashboards, notification panels, and action buttons shall maintain consistent placement and behavior across all sessions and screen sizes

- **Standardized Visual Language:** Icons, status labels, and color indicators shall be used consistently throughout the platform — for example, case status labels always use the same terminology and color coding
- **Persistent Notification History:** All system-generated notifications shall be retained and accessible from the user's notification panel, allowing users to review past events and case milestones after returning to the platform
- **Case Context Preservation:** Users returning mid-investigation shall find all relevant case history, evidence records, suspect entries, and audit logs intact and accessible from their dashboard without requiring re-navigation
- **Error Recovery Guidance:** When users make errors — such as submitting an incomplete report or attempting an action outside their permissions — the system shall display clear, descriptive messages guiding the user toward a valid next step

3.2.1.6 Errors

The PMS shall minimize user errors and provide clear, structured recovery mechanisms:

- **Preventive Validation:** Input validation shall block logically invalid entries before submission — for example, preventing a case status update to 'Closed' without the required authorization, or a suspect assignment to a non-existent case
- **Descriptive Error Messages:** All error messages shall clearly explain the issue and suggest a corrective action, avoiding technical jargon — for example, 'This case does not exist. Please verify the Case ID before assigning a suspect.'
- **Confirmation Dialogs:** All irreversible or high-impact actions — including account deactivation, case closure, and suspect deletion — shall require explicit user confirmation through a dialog prompt before being processed
- **Permission Boundary Enforcement:** When a user attempts an action outside their role's permitted scope, the system shall block the action at the system level and display a clear, non-technical message informing the user that the action is not available to their role
- **Error Logging:** All system errors and failed user actions shall be logged with timestamps for administrative review and troubleshooting, with no sensitive user data exposed in error outputs
- **Graceful Degradation:** If a non-critical feature such as file upload experiences a temporary failure, the system shall retain all other form data entered by the user, notify them of the specific failure, and prompt a retry without requiring the user to restart the entire form

3.2.1.7 Satisfaction

The Police Management System is designed to provide a clear, professional, and efficient experience for all three user roles — from Police Officers documenting frontline incidents, to Detectives managing complex investigations, to Forensic Experts conducting scientific analysis of evidence, to Traffic Officers handling road safety and violations, to Citizens interacting with the system for reporting and status tracking, and to Administrators overseeing the entire system.

The role-specific dashboards ensure that each user sees only what is relevant to them, reducing friction and making routine tasks — such as filing an incident report, updating a case status, or assigning a suspect — straightforward and immediate. The real-time notification system keeps all parties informed at every stage of an investigation without requiring manual follow-up, reinforcing accountability and operational transparency.

Because the PMS is fully web-based and responsive, users can access their dashboards and take action from any device, ensuring that time-sensitive steps — such as updating a case status immediately after

an arrest or submitting an incident report from the field — are never blocked by device or location constraints.

3.2.1.8 Capacity

The Police Management System shall be developed to support the concurrent operation of all three user roles without performance degradation. All users interact with a shared centralized database, and the system shall process concurrent requests reliably, ensuring that case status changes, new report submissions, and account updates are reflected in real time for all authorized parties.

The platform is web-based and does not integrate with external national law enforcement databases or third-party services in its current scope, which limits external dependency points and reduces the risk of third-party performance bottlenecks. All data — cases, incident reports, evidence, suspects, arrest records, and audit logs — shall be stored in a centralized database with regular automated backups to prevent data loss in the event of system failure.

3.2.2 Organizational Requirements

3.2.2.1 Availability

The Police Management System shall be available at all times to support ongoing criminal investigations, which may require user action outside of standard business hours. Specific availability requirements include:

- The platform shall target a minimum uptime of 99% during active deployment and operational phases
- Scheduled maintenance windows shall be communicated to users in advance where possible and minimized in duration to reduce operational disruption
- Regular automated database backups shall be performed to ensure data integrity and enable full recovery in the event of system failure

3.2.2.2 Latency

Response times for all operations shall meet the following thresholds under normal network conditions:

- Page and dashboard loads: within 3 seconds
- Login interface load: within 2 seconds
- Form submissions (incident report creation, suspect assignment, evidence upload): within 5 seconds
- Case search and filter result updates: within 2 seconds of filter application
- Notification delivery: within 1 minute of the triggering event
- Account status changes and case assignments by Admin: reflected immediately upon confirmation

3.2.2.3 Monitoring

The platform shall support operational monitoring through the following mechanisms:

- The Admin analytics dashboard shall display real-time platform metrics including total cases per status, registered users per role, active departments, and recent incident activity, updated with no more than a 1-minute delay

- All key system events — login attempts, account changes, case creation, status updates, evidence uploads, suspect assignments, arrest records, and case closures — shall be logged with timestamps and accessible for administrative review
- System errors and failed operations shall be logged automatically and retained for troubleshooting and audit purposes

3.2.2.4 Maintenance

- Regular software updates shall be scheduled to address security vulnerabilities, performance improvements, and bug fixes
- System audit logs shall be maintained and retained for the duration of active investigations and for a defined period thereafter to support legal accountability and compliance review
- All case records, incident reports, evidence files, suspect entries, and arrest records shall be retained in the system even after a case is closed, to support future dispute, audit, or legal proceedings

3.2.2.5 Operations

The Police Management System is designed to be operable by users across a range of technical proficiency levels, from Administrators managing the platform daily to Police Officers who may primarily interact with it for report submission and status updates. The interface relies on familiar web interaction patterns and requires no software installation beyond a standard web browser. All core workflows are self-contained within the platform, and no external tools or integrations are required for any user to complete their role-specific tasks.

3.2.2.6 Standards Compliance

The Police Management System shall comply with the following standards and regulations:

- GDPR (General Data Protection Regulation): All personal data collected during account creation and platform use shall be processed and stored in compliance with GDPR requirements, including purpose limitation, data minimization, and user rights regarding their personal data
- WCAG 2.1: The user interface shall conform to Web Content Accessibility Guidelines 2.1 to ensure accessibility for users with disabilities
- Password Security Standards: All user passwords shall be stored as cryptographic hashes using a modern algorithm (e.g., bcrypt); plain-text password storage is strictly prohibited
- Audit Trail Requirements: All key platform actions shall be logged with timestamps and actor identifiers to support accountability and legal compliance in the event of disputes, regulatory review, or court proceedings

3.2.2.7 Portability

- The PMS shall be delivered as a web-based application, ensuring consistent operation across all major operating systems including Windows, macOS, Linux, iOS, and Android
- Users shall require only a device with a modern web browser and a stable internet connection to access the full platform
- The platform shall be fully functional across all commonly used browsers including Chrome, Firefox, Safari, and Edge
- No client-side software installation shall be required for any user role to access or operate the platform

3.2.3 External Requirements

3.2.3.1 Protection

To protect the Police Management System and its users from unauthorized access, data breaches, and misuse, the following security measures shall be implemented:

- All user passwords shall be stored using a cryptographic hashing algorithm; plain-text storage is prohibited under any circumstance
- All data transmitted between the client and server shall be encrypted using HTTPS with TLS to prevent interception
- All user inputs across all forms shall be validated both client-side and server-side to prevent injection attacks, cross-site scripting, and other input-based vulnerabilities
- Role-based access control shall be enforced at the system level, ensuring users can only access data and features within their permitted scope regardless of direct URL access or API manipulation
- All irreversible actions shall require explicit user confirmation to prevent accidental data loss or unauthorized modifications
- Each user shall only be able to access their own assigned cases, reports, and records; no cross-user data visibility exists outside of the defined role permissions
- Session tokens shall expire after a defined period of inactivity, requiring re-authentication to resume access
- The system shall lock or flag an account after a defined number of consecutive failed login attempts to protect against brute-force attacks

3.2.3.2 Authorization and Authentication

- User authentication shall be performed using username and password credentials assigned by the Administrator, validated server-side on each login attempt
- The system shall use JSON Web Tokens (JWT) or an equivalent secure session management mechanism to maintain authenticated sessions
- Authorization shall be enforced based on user role at the system level, ensuring that each user can access only the features, data, and actions defined for their role
- All failed login attempts shall be logged with timestamps and the system shall apply protective lockout measures after a defined threshold of consecutive failures
- Admin accounts shall have the highest level of access; no other role can access user management, department management, or system-wide analytics

3.2.3.3 Legislative Requirements

The Police Management System operates in a law enforcement domain involving sensitive personal data, criminal records, suspect information, and legal evidence. The following legislative requirements apply:

- **Data Protection:** The platform shall comply with applicable personal data protection legislation, including GDPR, ensuring that all user and case data is collected only for defined purposes, stored securely, and not accessible by unauthorized parties
- **Evidence and Record Integrity:** All case records, incident reports, evidence files, and arrest records shall be immutable once formally submitted and locked, as they may carry legal implications in criminal proceedings
- **Audit Accountability:** All key platform actions shall be logged and retained to support accountability requirements in the event of disputes, regulatory review, or legal proceedings

- **Right to Data Access:** Users shall be able to access their own account information and assigned case data through the platform in compliance with data subject access rights defined under applicable legislation

3.3 Domain Requirements

The Police Management System operates within the law enforcement management domain, where the primary goal is to digitize and structure the lifecycle of criminal investigations — from initial incident reporting through case management, evidence tracking, suspect assignment, and final case closure.

The domain imposes specific constraints on the platform's behavior. Criminal investigations are high-stakes processes involving legal obligations, sensitive personal data, and public safety considerations. As a result, the platform prioritizes data integrity, auditability, and strict role separation above convenience features. Arrest records and incident reports are permanently retained once submitted, as they may carry legal implications in criminal proceedings. Case histories and all status transitions are fully logged for the same reason.

The platform does not handle legal contract generation, court case management, or external law enforcement database integration in its current scope, recognizing that these aspects are governed by regulatory and legal frameworks that fall outside the boundaries of a web-based management platform. All formal legal proceedings following case closure are assumed to occur through appropriate external channels.

Access control and data privacy are of particular importance in this domain, as the platform handles sensitive information including suspect details, criminal evidence, and officer activity records. Each user role has strictly defined visibility boundaries — Police Officers can only view cases assigned to them, Detectives cannot access cases outside their assignment, and the Admin has system-wide oversight but cannot perform investigative actions on behalf of Officers or Detectives — reflecting both privacy requirements and the professional accountability structures of law enforcement.

4. User Scenarios/Use Cases

4.1 Requirements Analysis

4.1.1 User Scenarios List

Nr/ ID	Scenario Name	Primary Actor	Description
US-01	Login / Logout	All Users	All Users: All system users log in using their username and password and log out when finished.
US-02	View Dashboard	All Users	All Users: After login, the user views a personalized dashboard with relevant information based on their role.
US-03	Change Own Password	All Users	All Users: The user changes their own password by entering the current password and a new one.

US-04	Create Incident Report	Police Officer	Police Officer: The officer creates a new incident report by filling in details such as location, date, type of incident and description.
US-05	Update Case Status	Police Officer	Police Officer: The officer updates the status of an existing case (e.g. open, under investigation, closed).
US-06	Upload Evidence	Police Officer	Police Officer: The officer uploads physical or digital evidence related to a case, including files, photos or documents.
US-07	Record Arrest Information	Police Officer	Police Officer: The officer records information about an arrested individual, including personal data and charges.
US-08	Manage Patrols	Police Officer	Police Officer: The officer registers and manages patrol schedules, including routes, times and assigned personnel.
US-09	Register Complaint	Police Officer	Police Officer: The officer registers a complaint received from a citizen and links it to the relevant case or department.
US-10	Search and Filter Cases	Police Officer	Police Officer: The officer searches and filters cases by various criteria such as date, status, type or assigned officer.
US-11	Open Criminal Case	Detective	Detective: The detective opens a new criminal case and assigns initial details, evidence and suspects.
US-12	Assign Suspect to Case	Detective	Detective: The detective links a suspect profile to an existing criminal case for further investigation.
US-13	Manage Witnesses	Detective	Detective: The detective adds, edits or removes witness information related to a case.
US-14	Track Investigation Status	Detective	Detective: The detective monitors and updates the progress of an ongoing investigation.
US-15	Write Investigation Report	Detective	Detective: The detective writes a detailed investigation report summarizing findings, evidence and conclusions.
US-16	View and Analyze Case Details	Detective	Detective: The detective reviews all case details including reports, evidence, suspects and witness statements.
US-17	Perform DNA Analysis	Forensic Expert	Forensic Expert: The forensic expert records and processes DNA samples collected from a crime scene and links results to the case.
US-18	Perform Fingerprint Analysis	Forensic Expert	Forensic Expert: The forensic expert analyzes fingerprints collected from evidence and attempts to match them with existing records.
US-19	Perform Weapon Analysis	Forensic Expert	Forensic Expert: The forensic expert examines weapons found at a crime scene and records analysis results.
US-20	Analyze Photos and Videos	Forensic Expert	Forensic Expert: The forensic expert reviews and analyzes photos and videos submitted as evidence.
US-21	Manage Laboratory Evidence	Forensic Expert	Forensic Expert: The forensic expert manages the registration, storage and status of laboratory evidence items.
US-22	Write Forensic Report	Forensic Expert	Forensic Expert: The forensic expert writes an official report documenting the results of all forensic analyses.

US-23	Register Traffic Accident	Traffic Officer	Traffic Officer: The officer registers a traffic accident by entering details such as location, vehicles involved and injuries.
US-24	Issue Traffic Fine	Traffic Officer	Traffic Officer: The officer issues a traffic fine to a driver for a committed road violation.
US-25	Manage Road Violations	Traffic Officer	Traffic Officer: The officer records and manages road traffic violations committed by drivers.
US-26	Perform Vehicle Control	Traffic Officer	Traffic Officer: The officer performs a vehicle inspection and records the results including document checks and vehicle condition.
US-27	Manage Driving Licenses	Traffic Officer	Traffic Officer: The officer manages driving license records including verification, suspension or renewal status.
US-28	Generate Traffic Report	Traffic Officer	Traffic Officer: The officer generates a statistical or summary report about traffic incidents, fines and violations.
US-29	Self-Register as Citizen	Citizen	Citizen: The citizen creates a personal account in the system by providing their personal information.
US-30	Report an Incident	Citizen	Citizen: The citizen submits an incident report through the system, providing details about what occurred.
US-31	Submit a Complaint	Citizen	Citizen: The citizen submits a formal complaint through the portal, which is then reviewed by the relevant officer.
US-32	Report a Missing Person	Citizen	Citizen: The citizen reports a missing person by filling in personal details and last known information.
US-33	Check Case Status	Citizen	Citizen: The citizen checks the current status of a case or complaint they previously submitted.
US-34	View Traffic Fines	Citizen	Citizen: The citizen views their issued traffic fines including details, amounts and payment status.
US-35	Manage Users and Roles	Administrator	Administrator: The administrator creates, edits and deactivates user accounts and assigns roles within the system.
US-36	Manage Departments	Administrator	Administrator: The administrator manages police departments, including adding new departments and updating their information.
US-37	Assign Officer to Case	Administrator	Administrator: The administrator assigns a specific officer to handle or oversee a case.
US-38	Monitor System Statistics	Administrator	Administrator: The administrator views system-wide statistics such as total cases, active users and recent activity.
US-39	Reset User Password	Administrator	Administrator: The administrator resets the password of a user account upon request or for security reasons.
US-40	View System Audit Log	Administrator	Administrator: The administrator reviews the audit log to track all actions performed by users within the system.
US-41	Search Officers by Department	Administrator	Administrator: The administrator searches and filters officers based on their assigned department.

US-42	Deactivate User Account	Administrator	Administrator: The administrator deactivates a user account, preventing the user from accessing the system.
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4.1.2 User Scenarios Extended

US-01: Login / Logout

1. User navigates to login page
2. User enters credentials
3. System validates
4. System redirects to role-based dashboard
5. User performs tasks
6. User logs out
7. Session terminated
8. If invalid credentials: system shows error, user retries
9. If account locked: system prompts to contact Admin

US-02: View Dashboard

1. User logs in successfully
2. System retrieves relevant statistics from the database
3. System displays a role-specific dashboard tailored to the user's role (Admin, Officer, Detective, Forensic Expert, Traffic Officer, Citizen)
4. If no data available: system shows informational message
5. If data loading error: system displays error with refresh option

US-03: Change Own Password

1. User navigates to Account Settings
2. User selects Change Password
3. User enters current and new passwords
4. System validates
5. System updates password
6. Success message displayed
7. If incorrect current password: system shows error, does not update
8. If weak new password: system prompts for stronger password
9. If passwords don't match: system highlights mismatch

US-04: Create Incident Report

1. Officer navigates to Reports
2. Officer clicks Create New Incident Report
3. Officer fills form (location, crime type, description, date/time)
4. Officer submits
5. System validates
6. System saves report
7. System auto-generates linked Case ID
8. Confirmation shown
9. If missing required fields: system highlights empty fields

10. If system error: officer is prompted to retry

US-05: Update Case Status

1. Officer navigates to Cases
2. Officer selects case
3. Officer views details
4. Officer selects new status from dropdown
5. Officer saves
6. System validates and logs change with timestamp
7. Success message shown
8. If unauthorized access: system denies action if case not assigned to officer
9. If same status selected: system notifies user that status is unchanged

US-06: Upload Evidence

1. User navigates to the case
2. User clicks Upload Evidence
3. User selects files and enters description
4. User submits
5. System validates format/size
6. System stores evidence linked to case
7. Success message displayed
8. If invalid file format or size exceeded: error shown, user resubmits with valid file
9. If upload failure: system allows retry

US-07: Record Arrest Information

1. Officer navigates to relevant case
2. Officer selects Add Arrest Record
3. Officer fills form (arrested person name, date, location, charges)
4. Officer submits
5. System validates
6. System saves and links arrest to case
7. Confirmation displayed
8. If incomplete information: system highlights missing required fields
9. If case not found: error displayed, submission prevented

US-08: Manage Patrols

1. Officer navigates to Patrols section
2. Officer views current patrols
3. Officer clicks New Patrol
4. Officer enters route, time slot, area
5. Officer submits
6. System saves patrol record
7. To update: officer selects patrol, modifies details, saves
8. If incomplete patrol data: system highlights missing fields
9. If patrol time conflict: system warns officer of scheduling conflict

US-09: Register Complaint

1. Officer navigates to Complaints
2. Officer clicks Register Complaint
3. Officer enters complainant details, description, category, date
4. Officer submits
5. System validates
6. System saves and links to person/case
7. Success message displayed
8. If missing required fields: system highlights empty fields
9. If case not found when linking: system shows error

US-10: Search and Filter Cases

1. Officer navigates to Cases
2. Officer enters search criteria (case ID, crime type, status, date range)
3. Officer clicks Search/Filter
4. System queries database
5. System displays matching cases
6. Officer selects a case
7. If no cases match: system displays 'No results found'
8. If invalid input: system highlights invalid field and requests correction

US-11: Open Criminal Case

1. Detective navigates to Cases
2. Detective selects existing case or opens new one linked to incident report
3. Detective reviews incident details
4. Detective enters investigation details
5. Detective sets status (Open/Under Investigation)
6. Detective submits
7. System saves
8. If case already open: system notifies detective and prevents duplication
9. If linked incident report not found: system shows error

US-12: Assign Suspect to Case

1. Detective navigates to relevant case
2. Detective selects Add Suspect
3. Detective enters name, age, physical description, associated evidence
4. Detective submits
5. System validates and saves suspect linked to case
6. If case does not exist: system error, submission prevented
7. If missing required fields: system highlights empty fields

US-13: Manage Witnesses

1. Detective navigates to case
2. Detective opens Witnesses section
3. Detective clicks Add Witness
4. Detective enters witness name, contact, testimony, confidentiality status

5. Detective submits
6. System saves and links witness to case
7. If missing required fields: system prevents submission
8. If witness already exists: system alerts detective

US-14: Track Investigation Status

1. Detective navigates to Cases
2. Detective selects case
3. Detective views current status and timeline
4. Detective selects new status
5. Detective adds status note
6. Detective saves
7. System logs change with timestamp
8. System notifies Administrator
9. If attempt to close case without Admin authorization: system blocks action
10. If same status selected: system notifies status unchanged

US-15: Write Investigation Report

1. Detective navigates to case
2. Detective selects Write Investigation Report
3. Detective enters summary, evidence analysis, suspect assessment, conclusions, next steps
4. Detective submits
5. System validates and saves
6. System notifies Administrator
7. If missing required fields: system highlights empty fields
8. If report already exists: system asks detective to confirm new version

US-16: View and Analyze Case Details

1. Detective navigates to Cases
2. Detective selects an assigned case
3. System retrieves all details (crime type, status, officer, suspects, evidence, witnesses, incident reports)
4. Detective reviews case data
5. Detective initiates follow-up actions
6. If case has no data: system displays case with empty sections
7. If unauthorized access: system denies access

US-17: Perform DNA Analysis

1. Expert navigates to Laboratory
2. Expert selects pending DNA analysis request
3. Expert views sample details
4. Expert performs physical analysis
5. Expert returns to system
6. Expert enters methodology, findings, conclusions
7. Expert submits
8. System notifies Detective and Admin

9. If sample not yet received: system shows pending status, prevents analysis entry
10. If missing required fields: system highlights

US-18: Perform Fingerprint Analysis

1. Expert navigates to Laboratory
2. Expert selects fingerprint analysis request
3. Expert views sample details
4. Expert performs physical analysis
5. Expert enters print quality, comparison results, identification status, conclusions
6. Expert submits
7. System saves and notifies
8. If sample too degraded: expert enters 'Non-identifiable' result with notes
9. If missing required fields: system highlights

US-19: Perform Weapon Analysis

1. Expert navigates to Laboratory
2. Expert selects weapon analysis request
3. Expert performs physical analysis (ballistics, tool marks, trace evidence)
4. Expert enters weapon type, condition, results, conclusions
5. Expert submits
6. System saves and notifies
7. If weapon not yet received: system shows pending status
8. If missing required fields: system highlights

US-20: Analyze Photos and Videos

1. Expert opens multimedia evidence section
2. Expert selects photo/video analysis request
3. Expert performs forensic analysis (metadata extraction, image enhancement, authenticity verification)
4. Expert annotates frames
5. Expert enters findings
6. Expert submits
7. If unsupported file format: expert notified, prompted to re-upload
8. If file too large: error with suggestion to split file

US-21: Manage Laboratory Evidence

1. Expert navigates to Lab Evidence Management
2. Expert clicks Register Evidence Intake
3. Expert enters evidence ID, case reference, type, condition, date, analyst
4. Expert submits
5. System creates chain of custody
6. Expert updates status as processing progresses
7. If evidence ID already exists: system warns of potential duplicate
8. If case reference not found: error, submission prevented

US-22: Write Forensic Report

1. Expert navigates to case
2. Expert selects Write Forensic Report
3. System pre-populates form with linked analyses
4. Expert enters summary, methodology, findings, conclusions, case implications
5. Expert submits
6. System notifies Detective and Admin
7. If no completed analyses linked: system warns expert that at least one must be complete
8. If missing required fields: system highlights

US-23: Register Traffic Accident

1. Officer navigates to Traffic Accidents
2. Officer clicks Register Accident
3. Officer enters date, time, location, vehicle details, injuries, witnesses, description
4. Officer submits
5. System validates
6. System saves record
7. System generates accident ID
8. Confirmation displayed
9. If missing required fields: system highlights empty fields
10. If system save error: officer prompted to retry

US-24: Issue Traffic Fine

1. Officer navigates to Fines
2. Officer clicks Issue Fine
3. Officer enters violation type, fine amount, date, time, location, citizen ID or vehicle registration
4. Officer submits
5. System validates
6. System saves fine and links to citizen
7. Fine becomes visible in citizen portal
8. If citizen/vehicle not found: officer manually enters details
9. If missing required fields: system highlights

US-25: Manage Road Violations

1. Officer navigates to Violations
2. Officer registers violation (type, location, date, vehicle details)
3. Officer submits
4. To view: officer filters violations list
5. To update: officer selects violation, modifies status (Open/Resolved), saves
6. If missing required fields: system highlights
7. If violation already closed: system prevents re-opening without authorization

US-26: Perform Vehicle Control

1. Officer navigates to Vehicle Control
2. Officer clicks New Vehicle Control
3. Officer enters registration, make, model, year, owner details, date/time

4. Officer checks registration, insurance, inspection status
5. Officer enters results
6. Officer submits
7. System saves record
8. If vehicle registration not found: officer manually enters details
9. If missing required fields: system highlights

US-27: Manage Driving Licenses

1. Officer navigates to Licenses
2. Officer searches by citizen ID, license number, or name
3. System retrieves license record
4. Officer reviews validity
5. To flag/suspend: officer selects action, enters reason, confirms
6. System updates status and notifies citizen
7. If license not found: system displays 'No license record found'
8. If license already suspended: system notifies officer

US-28: Generate Traffic Report

1. Officer navigates to Traffic Reports
2. Officer selects report type
3. Officer specifies date range and filters
4. Officer clicks Generate Report
5. System processes data
6. System displays report with statistics and visualizations
7. Officer optionally exports as PDF or CSV
8. If no data for selected period: system displays 'No data found'
9. If export failure: error shown with retry option

US-29: Self-Register as Citizen

1. Citizen navigates to public PMS portal
2. Citizen clicks Register
3. Citizen enters full name, national ID, DOB, email, phone, password
4. Citizen submits
5. System validates and checks for duplicates
6. System creates account
7. System sends confirmation email
8. If duplicate account (same national ID or email): system notifies and prevents duplicate
9. If missing fields: system highlights
10. If weak password: system requires stronger

US-30: Report an Incident

1. Citizen navigates to Report Incident
2. Citizen clicks Submit Report
3. Citizen enters description, location, date/time
4. Citizen optionally uploads photo
5. Citizen submits

6. System saves report
7. System assigns to police department
8. Citizen receives reference number
9. If missing required fields: system highlights
10. If file upload failure: system retains form data and prompts retry for file

US-31: Submit a Complaint

1. Citizen navigates to Submit Complaint
2. Citizen clicks New Complaint
3. Citizen enters category, description, date of incident, optionally relevant officer/department
4. Citizen submits
5. System saves and routes to Administrator
6. Citizen receives reference number
7. If missing required fields: system highlights
8. If system save error: system retains form and prompts retry

US-32: Report a Missing Person

1. Citizen navigates to Report Missing Person
2. Citizen clicks File Report
3. Citizen enters missing person details (name, DOB, description, last location, date last seen, contact info)
4. Citizen optionally uploads photo
5. Citizen submits
6. System saves report
7. System assigns to police department
8. Citizen receives reference number
9. If missing required fields: system highlights
10. If photo upload failure: system retains form and prompts retry

US-33: Check Case Status

1. Citizen navigates to My Reports
2. System retrieves all citizen's reports
3. System displays list (type, reference, date, status)
4. Citizen selects specific report
5. System displays full details and current status
6. If no reports found: system displays 'You have not submitted any reports'
7. If status not yet updated: system shows last known status with 'Pending Review' label

US-34: View Traffic Fines

1. Citizen navigates to My Fines
2. System retrieves all fines linked to citizen account
3. System displays list (reference, violation type, amount, date, payment status)
4. Citizen selects fine
5. System displays full details
6. If no fines found: system displays 'No fines have been issued to your account'
7. If data loading error: error shown with refresh option

US-35: Manage Users and Roles

1. Admin navigates to User Management
2. Admin adds user (name, username, password, role, department) OR edits existing user OR deactivates user
3. System validates each action
4. System confirms with success message
5. If duplicate username: system prevents creation with error
6. If missing fields: system highlights
7. If deactivating user with active cases: system warns and requests confirmation

US-36: Manage Departments

1. Admin navigates to Departments
2. Admin adds department (name) OR edits existing department OR deletes (if no staff assigned)
3. System validates
4. System confirms each action
5. If duplicate department name: system shows error
6. If attempt to delete department with assigned staff: system warns and prevents until staff reassigned

US-37: Assign Officer to Case

1. Admin navigates to active case
2. Admin selects Assign Officer
3. System displays available officers
4. Admin selects officer
5. Admin confirms
6. System saves and notifies assigned officer
7. Case details updated
8. If no available officers: system displays message
9. If case already has assigned officer: system asks Admin to confirm reassignment

US-38: Monitor System Statistics

1. Admin navigates to Dashboard
2. System retrieves all platform-wide statistics
3. System displays: departments, users per role, open cases, cases under investigation, incidents, forensic analyses, violations, fines
4. Admin reviews and navigates for detailed info
5. If no data: system displays zero values with informational note
6. If data loading failure: error with refresh option

US-39: Reset User Password

1. Admin navigates to Users
2. Admin selects target user
3. Admin clicks Reset Password
4. System prompts for confirmation
5. Admin confirms

6. System generates temporary password
7. System notifies user with credentials
8. User logs in and changes password
9. If user account not found: system displays error
10. If notification delivery failure: system logs failure and alerts Admin

US-40: View System Audit Log

1. Admin navigates to Audit Log
2. System retrieves all logged actions
3. System displays: action type, user role, username, timestamp, affected record
4. Admin filters by user, date, role, or action type
5. Admin reviews entries
6. If no logs: system displays 'No audit records found'
7. If filter returns no results: system displays 'No results match the selected filters'

US-41: Search Officers by Department

1. Admin navigates to Users
2. Admin selects department filter or role filter
3. System queries database
4. System displays filtered staff list (ID, name, role, department)
5. Admin selects staff member to view or edit profile
6. If no staff in selected department: system displays 'No staff found in this department'
7. If invalid filter: system resets to show all staff

US-42: Deactivate User Account

1. Admin navigates to Users
2. Admin selects target user
3. Admin clicks Deactivate Account
4. System prompts for reason and confirmation
5. Admin enters reason and confirms
6. System sets account to Inactive
7. System terminates active session
8. System logs deactivation
9. If user has ongoing open cases: system warns Admin and requests confirmation before proceeding
10. If Admin cancels: account remains active

4.1.3 User Cases

1. All Users

UC1: Login / Logout	
UC Name	UC1: Login / Logout
Summary	Any authorized user (staff or citizen) logs into the system using their credentials and logs out when finished.

Dependency	None
Actors	Primary: All Users (Administrator, Police Officer, Detective, Forensic Expert, Traffic Officer, Citizen)
Preconditions	- User must have valid credentials (staff: issued by Admin; citizens: self-registered). - System must be accessible via web browser.
Description of Main Sequence	- User navigates to the login page. - System displays the login form (Username, Password). - User enters their username and password. - User clicks the Login button. - System validates the credentials. - System grants access and redirects the user to their role-based dashboard. - User performs their tasks. - User clicks Logout. - System ends the session and returns to the login page.
Description of Alternative Sequence	- Invalid credentials: System displays an error message. User may retry. - Account locked/inactive: System displays an error indicating the account is not active and prompts contacting the Administrator. - Citizen account not yet verified: System notifies the citizen to complete registration.
Non-Functional Requirements	- Login should complete within 3 seconds under normal load. - Passwords must be encrypted (hashed) in the database. - System must enforce session timeout after inactivity.
Postconditions	- User is authenticated and their session is active. - On logout, the session is terminated and no sensitive data remains accessible.

UC2: View Dashboard

UC Name	UC2: View Dashboard
Summary	After logging in, any user can view a role-specific dashboard showing relevant statistics and quick-access actions.
Dependency	UC1 (User logs in)
Actors	Primary: All Users (Administrator, Police Officer, Detective, Forensic Expert, Traffic Officer, Citizen)
Preconditions	- User is authenticated and logged in. - User has a role-specific dashboard configured.

Description of Main Sequence	- User is redirected to the dashboard after successful login. - System retrieves relevant statistics from the database. - System displays the dashboard tailored to the role: Admin: total cases, users per role, departments, incidents; Officer: assigned cases, recent incidents, patrols; Detective: assigned cases, open investigations, suspects; Forensic Expert: pending analyses, submitted reports; Traffic Officer: accidents, fines, violations; Citizen: submitted reports, fine status, notifications.
Description of Alternative Sequence	- No data available: System displays a message that no statistics are currently available. - Data loading error: System shows an error message and provides an option to refresh.
Non-Functional Requirements	- Dashboard must load within 5 seconds. - Statistics must reflect real-time data. - Dashboard layout must be responsive and accessible from any device.
Postconditions	- The dashboard is displayed with up-to-date statistics for the user's role. - User can navigate to specific sections from the dashboard.

UC3: Change Own Password

UC Name	UC3: Change Own Password
Summary	Any logged-in user can change their own account password for security purposes.
Dependency	UC1 (User logs in)
Actors	Primary: All Users (Administrator, Police Officer, Detective, Forensic Expert, Traffic Officer, Citizen)
Preconditions	- User is authenticated and logged in. - User knows their current password.
Description of Main Sequence	- User navigates to Account Settings. - User selects 'Change Password'. - System displays the change password form. - User enters current password, new password, and confirms new password. - User submits the form. - System validates current password and checks new password strength. - System updates the password in the database. - System confirms with a success message.
Description of Alternative Sequence	- Incorrect current password: System displays an error and does not update. - New password too weak: System prompts the user to choose a stronger password.

	New password and confirmation do not match: System highlights the mismatch.
Non-Functional Requirements	- Passwords must meet minimum security requirements (length, complexity). - New password must not be the same as the current one. - Password change must be logged in the audit trail.
Postconditions	- Password is updated securely in the system. - User session remains active or user is prompted to log in again.

2. Police Officer

UC4: Create Incident Report	
UC Name	UC4: Create Incident Report
Summary	A Police Officer creates a new incident report to document a reported crime in the system.
Dependency	UC1 (User logs in)
Actors	Primary: Police Officer
Preconditions	- Police Officer is logged in and has permission to create incident reports. - The system is operational.
Description of Main Sequence	- Officer navigates to the Reports section. - Officer clicks 'Create New Incident Report'. - System displays the incident report form. - Officer enters the required details: location, crime type, description, date and time. - Officer submits the form. - System validates all required fields. - System saves the incident report in the database. - System automatically generates a new case linked to the incident report. - System displays a confirmation message with the Case ID.
Description of Alternative Sequence	- Missing required fields: System highlights empty fields and displays an error message. Officer must complete them before submitting. - System error during save: System displays an error and prompts the officer to try again.
Non-Functional Requirements	- The form must be simple and completable within 5 minutes. - System must validate data formats (e.g., date/time format, text length). - Data must be stored securely with access restricted by role.
Postconditions	- A new incident report is saved in the system. - A new case is automatically created and linked to the report. - The case appears in the Cases list with 'Open' status.

UC5: Update Case Status

UC Name	UC5: Update Case Status
Summary	A Police Officer updates the status of an existing case (Open, Under Investigation, or Closed).
Dependency	UC1 (User logs in), UC4 (Create Incident Report)
Actors	Primary: Police Officer
Preconditions	- Officer is logged in. - The case exists in the system and is assigned to the officer.
Description of Main Sequence	- Officer navigates to the Cases section. - Officer selects the case to update. - System displays the case details. - Officer selects a new status from the dropdown (Open, Under Investigation, Closed). - Officer saves the update. - System validates the status change. - System saves the updated status with a timestamp. - System confirms the update with a success message.
Description of Alternative Sequence	- Unauthorized access: If the officer tries to update a case not assigned to them, the system denies the action. - Same status selected: System notifies the user that the status is unchanged.
Non-Functional Requirements	- Status update should be saved within 2 seconds. - Status change history should be logged for audit purposes. - Access control must prevent unauthorized status changes.
Postconditions	- The case status is updated in the database. - The change is reflected immediately in the case list and dashboard.

UC6: Upload Evidence

UC Name	UC6: Upload Evidence
Summary	A Police Officer or Detective uploads evidence files related to an existing case.
Dependency	UC1 (User logs in), UC4 (Create Incident Report)
Actors	Primary: Police Officer Secondary: Detective
Preconditions	- User is logged in with permission to upload evidence. - A case already exists in the system.

Description of Main Sequence	• User navigates to the specific case. • User clicks 'Upload Evidence'. • System displays the evidence upload form. • User selects the evidence file(s) and enters a description. • User submits the upload. • System validates the file format and size. • System stores the evidence and links it to the case. • System confirms the upload with a success message.
Description of Alternative Sequence	• Invalid file format or size exceeded: System displays an error and requests a valid file. • Upload failure: System shows an error and allows retry.
Non-Functional Requirements	• Supported file formats: PDF, JPG, PNG, MP4, AVI. • Maximum file size per upload should be enforced. • Evidence files must be stored securely with restricted access.
Postconditions	• Evidence is stored securely in the system. • Evidence is linked to the correct case and visible in case details.

UC7: Record Arrest Information

UC Name	UC7: Record Arrest Information
Summary	A Police Officer records the details of an arrest and links them to the relevant case.
Dependency	UC1 (User logs in), UC4 (Create Incident Report)
Actors	Primary: Police Officer
Preconditions	• Officer is logged in. • A case exists in the system. • An arrest has been made related to an existing case.
Description of Main Sequence	• Officer navigates to the relevant case. • Officer selects 'Add Arrest Record'. • System displays the arrest information form. • Officer fills in: arrested person's name, date of arrest, location, charges. • Officer submits the form. • System validates the input. • System saves the arrest record and links it to the case. • System confirms with a success message.
Description of Alternative Sequence	• Incomplete information: System highlights missing required fields. • Case not found: System displays an error and prevents submission.
Non-Functional Requirements	• Arrest records must be immutable once submitted (controlled edit process).

	• Data must comply with law enforcement data regulations. • Records must be audit-logged.
Postconditions	• Arrest record is saved and linked to the case. • Case details are updated to reflect the arrest.

UC8: Manage Patrols

UC Name	UC8: Manage Patrols
Summary	A Police Officer views, registers, and updates patrol assignments and schedules.
Dependency	UC1 (User logs in)
Actors	Primary: Police Officer
Preconditions	• Officer is logged in. • Patrol data exists or the officer has permission to create new patrols.
Description of Main Sequence	• Officer navigates to the Patrols section. • Officer views the list of current patrols and their status. • To register a patrol: Officer clicks 'New Patrol', enters route, time slot, and area. Officer submits the form. System saves the patrol. • To update a patrol: Officer selects the patrol, modifies the details, and saves. • System confirms each action with a success message.
Description of Alternative Sequence	• Incomplete patrol data: System highlights missing fields. • Patrol time conflict: System warns the officer of a scheduling conflict.
Non-Functional Requirements	• Patrol data must be saved within 2 seconds. • Patrol schedules must be visible to all relevant officers and administrators. • All patrol changes must be audit-logged.
Postconditions	• Patrol record is saved and linked to the officer. • Patrol schedule is visible in the officer's dashboard. • Changes are reflected immediately for administrators.

UC9: Register Complaint

UC Name	UC9: Register Complaint
Summary	A Police Officer registers a complaint received from a citizen or filed in person at the department.
Dependency	UC1 (User logs in)

Actors	Primary: Police Officer
Preconditions	- Officer is logged in. - A complaint has been received or reported in person.
Description of Main Sequence	- Officer navigates to the Complaints section. - Officer clicks 'Register Complaint'. - System displays the complaint form. - Officer enters: complainant details, complaint description, category, and date. - Officer submits the form. - System validates the input. - System saves the complaint and links it to the relevant person or case (if applicable). - System confirms with a success message.
Description of Alternative Sequence	- Missing required fields: System highlights empty fields. - Case not found (when linking to a case): System shows an error.
Non-Functional Requirements	- Complaint records must be immutable once submitted. - Complaint data must be stored securely. - Complaints must be audit-logged with timestamp and officer identity.
Postconditions	- Complaint is saved and accessible in the Complaints section. - Complaint is linked to a citizen or case record where applicable.

UC10: Search and Filter Cases

UC Name	UC10: Search and Filter Cases
Summary	A Police Officer searches for cases using filters such as status, crime type, date, or case ID.
Dependency	UC1 (User logs in)
Actors	Primary: Police Officer
Preconditions	- Officer is logged in. - One or more cases exist in the system.
Description of Main Sequence	- Officer navigates to the Cases section. - Officer enters search criteria: case ID, crime type, status, or date range. - Officer clicks 'Search' or 'Filter'. - System queries the database with the given criteria. - System displays the filtered list of matching cases. - Officer selects a case to view its details.

Description of Alternative Sequence	- No cases match the criteria: System displays a 'No results found' message. - Invalid search input: System highlights the invalid field and requests correction.
Non-Functional Requirements	- Search results must be returned within 2 seconds. - Filters must be combinable (e.g., status AND crime type). - Only cases accessible to the officer's role are shown in results.
Postconditions	- A filtered list of cases matching the search criteria is displayed. - Officer can select and open any case from the results.

3. Detective

UC11: Open Criminal Case	
UC Name	UC11: Open Criminal Case
Summary	A Detective opens a new criminal case or takes ownership of a case linked to an incident report for formal investigation.
Dependency	UC1 (User logs in), UC4 (Create Incident Report)
Actors	Primary: Detective
Preconditions	- Detective is logged in. - An incident report has been submitted by a Police Officer. - The case is assigned to the detective.
Description of Main Sequence	- Detective navigates to the Cases section. - Detective selects an existing case or opens a new case linked to an incident report. - System displays the case creation / case details form. - Detective reviews the incident report and enters additional investigation details. - Detective sets the initial status (Open or Under Investigation). - Detective submits the case record. - System saves the case and links it to the incident report. - System confirms with a success message and case ID.
Description of Alternative Sequence	- Case already open: System notifies the detective that the case already exists and prevents duplication. - Linked incident report not found: System shows an error.
Non-Functional Requirements	- Case creation must complete within 3 seconds. - Cases must be linked to an incident report to ensure traceability. - All case creation actions must be audit-logged.
Postconditions	- A criminal case is opened and saved in the system. - The case is visible in the detective's case list with the assigned status.

- The case is linked to the originating incident report.

UC12: Assign Suspect to Case

UC Name	UC12: Assign Suspect to Case
Summary	A Detective links a suspect to an existing case by entering suspect details into the system.
Dependency	UC1 (User logs in), UC4 (Create Incident Report)
Actors	Primary: Detective
Preconditions	• Detective is logged in. • A case exists in the system and is assigned to the detective. • There is a potential suspect related to the case.
Description of Main Sequence	• Detective navigates to the relevant case. • Detective selects 'Add Suspect'. • System displays the suspect information form. • Detective enters: name, age, physical description, and associated evidence. • Detective submits the form. • System validates the input. • System saves the suspect information and links it to the case. • System confirms with a success message.
Description of Alternative Sequence	• Case does not exist: System displays an error and prevents submission. • Missing required fields: System highlights empty fields and requires completion.
Non-Functional Requirements	• Suspect data must be stored securely with restricted access. • Suspect records must be clearly linked to the correct case. • Input validation must prevent invalid data formats.
Postconditions	• Suspect information is saved and linked to the case. • Suspect appears in the case details under the Suspects section.

UC13: Manage Witnesses

UC Name	UC13: Manage Witnesses
Summary	A Detective adds, views, and manages witness records and testimonies linked to an existing case.
Dependency	UC1 (User logs in), UC11 (Open Criminal Case)
Actors	Primary: Detective
Preconditions	• Detective is logged in.

	A case is assigned to the detective.
Description of Main Sequence	- Detective navigates to the relevant case. - Detective opens the Witnesses section. - Detective clicks 'Add Witness'. - System displays the witness form. - Detective enters: witness name, contact details, testimony summary, and confidentiality status. - Detective submits the form. - System saves the witness record and links it to the case. - To view: Detective selects a witness from the list; system displays full witness profile. - System confirms each action with a success message.
Description of Alternative Sequence	- Missing required fields: System highlights missing fields and prevents submission. - Witness already exists for the case: System alerts the detective to avoid duplication.
Non-Functional Requirements	- Witness data must be stored securely with restricted access. - Confidentiality status must determine visibility to other roles. - All witness management actions must be audit-logged.
Postconditions	- Witness record is saved and linked to the case. - Witness appears in the case details under the Witnesses section.

UC14: Track Investigation Status

UC Name	UC14: Track Investigation Status
Summary	A Detective updates and tracks the investigation status of an assigned case.
Dependency	UC1 (User logs in), UC11 (Open Criminal Case)
Actors	Primary: Detective
Preconditions	- Detective is logged in. - A case is assigned to the detective.
Description of Main Sequence	- Detective navigates to the Cases section. - Detective selects the case to update. - System displays current case status and investigation timeline. - Detective selects the new status: Open, Under Investigation, or Closed. - Detective adds a status update note explaining the change. - Detective saves the update. - System saves the new status with timestamp and detective identity. - System sends a notification to the Administrator. - System confirms with a success message.

Description of Alternative Sequence	- Attempt to close a case without Admin authorization: System prevents the action and displays a message. - Same status selected: System notifies the detective that the status has not changed.
Non-Functional Requirements	- Status transitions must be logged with timestamp and detective identity. - Closed cases require Administrator authorization to reopen. - Status updates must be reflected in real time for all authorized parties.
Postconditions	- The case status is updated in the system. - The status change is visible in the case timeline and dashboard. - Administrator is notified of the status change.

UC15: Write Investigation Report

UC Name	UC15: Write Investigation Report
Summary	A Detective writes and submits a formal investigation report summarizing findings, evidence analysis, and investigative conclusions for an assigned case.
Dependency	UC1 (User logs in), UC11 (Open Criminal Case)
Actors	Primary: Detective
Preconditions	- Detective is logged in. - A case is assigned to the detective and has sufficient data for reporting.
Description of Main Sequence	- Detective navigates to the relevant case. - Detective selects 'Write Investigation Report'. - System displays the report form. - Detective enters: case summary, evidence analysis, suspect assessment, investigative conclusions, and recommended next steps. - Detective submits the report. - System validates all required fields. - System saves the report and links it to the case. - System notifies the Administrator of the new report. - System confirms with a success message.
Description of Alternative Sequence	- Missing required fields: System highlights empty fields. - Report already exists for the case: System asks the detective to confirm whether to create a new version.
Non-Functional Requirements	- Reports must be immutable once formally submitted. - Reports must be linked to the correct case. - All report submissions must be audit-logged with timestamp and detective identity.
Postconditions	- Investigation report is saved and linked to the case. - Report is visible to the Administrator and authorized parties.

- Submission is recorded in the audit log.

UC16: View and Analyze Case Details

UC Name	UC16: View and Analyze Case Details
Summary	A Detective reviews all information associated with a case including incident reports, evidence, witnesses, and suspects.
Dependency	UC1 (User logs in)
Actors	Primary: Detective
Preconditions	• Detective is logged in. • One or more cases are assigned to the detective.
Description of Main Sequence	• Detective navigates to the Cases section. • Detective selects a case assigned to them. • System retrieves and displays all case details: crime type, status, assigned officer, suspects, evidence, witnesses, and incident reports. • Detective reviews the information. • Detective may add notes or initiate follow-up actions (e.g., assign suspect, upload evidence).
Description of Alternative Sequence	• Case has no data yet: System displays the case with empty sections. • Unauthorized access to a case not assigned to the detective: System denies access.
Non-Functional Requirements	• Case details must load within 3 seconds. • All sensitive case data must be accessible only to authorized roles. • The view must display all sections clearly: Suspects, Evidence, Witnesses, Incident Reports.
Postconditions	• Detective has full visibility into all case-related data. • Detective can proceed to take further investigative actions from the case view.

4. Forensic Expert

UC17: Perform DNA Analysis

UC Name	UC17: Perform DNA Analysis
Summary	A Forensic Expert performs and documents DNA analysis on biological samples submitted as evidence for a case.
Dependency	UC1 (User logs in), UC6 (Upload Evidence)
Actors	Primary: Forensic Expert

Preconditions	- Forensic Expert is logged in. - A DNA sample has been submitted and linked to an active case.
Description of Main Sequence	- Forensic Expert navigates to the Laboratory section. - Expert selects the pending DNA analysis request linked to a case. - System displays the analysis form with the sample details. - Expert performs the physical DNA analysis in the laboratory. - Expert returns to the system and enters the analysis results: methodology, sample condition, findings, and conclusions. - Expert submits the analysis. - System validates all required fields. - System saves the analysis results and links them to the case. - System notifies the assigned Detective and Administrator. - System confirms with a success message.
Description of Alternative Sequence	- Sample not yet received: System shows a pending status and prevents analysis entry. - Missing required fields: System highlights empty fields.
Non-Functional Requirements	- DNA analysis records must be immutable once submitted. - All analysis data must be stored securely. - Results must be audit-logged with timestamp and expert identity.
Postconditions	- DNA analysis results are saved and linked to the case. - Results are visible to authorized parties (Detective, Administrator). - Submission is recorded in the audit log.

UC18: Perform Fingerprint Analysis

UC Name	UC18: Perform Fingerprint Analysis
Summary	A Forensic Expert analyzes fingerprint samples collected from a crime scene and documents the findings.
Dependency	UC1 (User logs in), UC6 (Upload Evidence)
Actors	Primary: Forensic Expert
Preconditions	- Forensic Expert is logged in. - Fingerprint samples have been submitted and linked to an active case.
Description of Main Sequence	- Forensic Expert navigates to the Laboratory section. - Expert selects the fingerprint analysis request. - System displays the fingerprint analysis form with sample details. - Expert performs the physical fingerprint analysis. - Expert enters findings: print quality, comparison results, identification status, and conclusions. - Expert submits the analysis. - System saves the results and links them to the case. - System notifies the assigned Detective and Administrator.

	• System confirms with a success message.
Description of Alternative Sequence	• Sample too degraded for analysis: Expert enters a 'Non-identifiable' result with notes. • Missing required fields: System highlights empty fields.
Non-Functional Requirements	• Fingerprint analysis records must be immutable once submitted. • Comparison results must indicate whether a match was found. • All submissions must be audit-logged.
Postconditions	• Fingerprint analysis results are saved and linked to the case. • Results are visible to authorized parties.

UC19: Perform Weapon Analysis

UC Name	UC19: Perform Weapon Analysis
Summary	A Forensic Expert performs ballistic, tool mark, and trace evidence analysis on weapons submitted as evidence.
Dependency	UC1 (User logs in), UC6 (Upload Evidence)
Actors	Primary: Forensic Expert
Preconditions	• Forensic Expert is logged in. • A weapon has been submitted and linked to an active case.
Description of Main Sequence	• Forensic Expert navigates to the Laboratory section. • Expert selects the weapon analysis request. • System displays the weapon analysis form. • Expert performs the physical weapon analysis (ballistics, tool marks, trace evidence). • Expert enters findings: weapon type, condition, ballistic results, trace evidence, and conclusions. • Expert submits the analysis. • System saves the results and links them to the case. • System notifies the assigned Detective and Administrator. • System confirms with a success message.
Description of Alternative Sequence	• Weapon not yet received in the lab: System shows pending status. • Missing required fields: System highlights empty fields.
Non-Functional Requirements	• Weapon analysis records must be immutable once submitted. • All analysis data must be linked to the correct case. • Submissions must be audit-logged.
Postconditions	• Weapon analysis results are saved and linked to the case. • Results are visible to authorized parties.

UC20: Analyze Photos and Videos

UC Name	UC20: Analyze Photos and Videos
Summary	A Forensic Expert performs forensic analysis on photos and videos submitted as evidence, including metadata extraction and image enhancement.
Dependency	UC1 (User logs in), UC6 (Upload Evidence)
Actors	Primary: Forensic Expert
Preconditions	- Forensic Expert is logged in. - Photos or videos have been uploaded and linked to an active case.
Description of Main Sequence	- Forensic Expert navigates to the multimedia evidence section of the relevant case. - Expert selects the photo/video analysis request. - System displays the analysis form with the uploaded media. - Expert performs forensic analysis: metadata extraction, image enhancement, authenticity verification. - Expert enters findings and annotates specific frames or regions if applicable. - Expert submits the analysis. - System saves the results and links them to the case. - System notifies the assigned Detective and Administrator. - System confirms with a success message.
Description of Alternative Sequence	- File format not supported for analysis: System notifies the expert and prompts re-upload. - File too large for processing: System shows an error and suggests file splitting.
Non-Functional Requirements	- Analysis results must be stored alongside the original media files. - Metadata extraction must be performed automatically where possible. - Submissions must be audit-logged with timestamp and expert identity.
Postconditions	- Photo/video analysis results are saved and linked to the case. - Results and annotations are visible to authorized parties.

UC21: Manage Laboratory Evidence

UC Name	UC21: Manage Laboratory Evidence
Summary	A Forensic Expert manages the intake, processing, and chain of custody of physical evidence received in the laboratory.
Dependency	UC1 (User logs in)
Actors	Primary: Forensic Expert
Preconditions	Forensic Expert is logged in.

	Physical evidence has been received by the laboratory.
Description of Main Sequence	- Forensic Expert navigates to the Lab Evidence Management section. - Expert clicks 'Register Evidence Intake'. - System displays the intake form. - Expert enters: evidence ID, case reference, type, condition on arrival, receiving date, and responsible analyst. - Expert submits the intake record. - System saves the entry and creates a chain of custody record. - As the evidence is processed, Expert updates its status (Received, In Analysis, Analysis Complete, Stored). - System logs each status change with timestamp and expert identity.
Description of Alternative Sequence	- Evidence ID already exists: System warns the expert of a potential duplicate. - Case reference not found: System shows an error and prevents submission.
Non-Functional Requirements	- Chain of custody records must be immutable and fully traceable. - All status changes must be logged with timestamp and user identity. - Evidence records must comply with law enforcement data regulations.
Postconditions	- Evidence intake is recorded and linked to the case. - Chain of custody is established and traceable from receipt to storage. - All status transitions are logged in the audit trail.

UC22: Write Forensic Report

UC Name	UC22: Write Forensic Report
Summary	A Forensic Expert writes and submits a formal forensic report summarizing all analysis results for a specific case.
Dependency	UC1 (User logs in), UC17/UC18/UC19/UC20 (Any analysis)
Actors	Primary: Forensic Expert
Preconditions	- Forensic Expert is logged in. - At least one analysis has been completed and linked to the case.
Description of Main Sequence	- Forensic Expert navigates to the relevant case. - Expert selects 'Write Forensic Report'. - System displays the report form, pre-populated with linked analyses. - Expert enters: report summary, methodology description, all analysis findings, conclusions, and case implications. - Expert submits the report. - System validates all required fields. - System saves the report and links it to the case. - System notifies the assigned Detective and Administrator. - System confirms with a success message.

Description of Alternative Sequence	- No completed analyses linked to the case: System warns the expert that at least one analysis must be complete. - Missing required fields: System highlights empty fields.
Non-Functional Requirements	- Forensic reports must be immutable once formally submitted. - Reports must be linked to all related analyses. - Submissions must be audit-logged.
Postconditions	- Forensic report is saved and linked to the case. - Report is visible to the Detective and Administrator. - Submission is recorded in the audit log.

5. Traffic Officer

UC23: Register Traffic Accident	
UC Name	UC23: Register Traffic Accident
Summary	A Traffic Officer registers a traffic accident, documenting vehicles involved, injuries, witnesses, and location.
Dependency	UC1 (User logs in)
Actors	Primary: Traffic Officer
Preconditions	- Traffic Officer is logged in. - A traffic accident has occurred and been reported.
Description of Main Sequence	- Officer navigates to the Traffic Accidents section. - Officer clicks 'Register Accident'. - System displays the accident registration form. - Officer enters: date, time, location, vehicle details (registration, make, model), number of people involved, injury description, witness information, and accident description. - Officer submits the form. - System validates all required fields. - System saves the accident record. - System generates an accident report ID. - System confirms with a success message.
Description of Alternative Sequence	- Missing required fields: System highlights empty fields. - System save error: System displays an error and prompts retry.
Non-Functional Requirements	- Accident records must be immutable once submitted (controlled edit process). - Data must comply with traffic law and data protection regulations. - Records must be audit-logged with timestamp and officer identity.
Postconditions	Accident record is saved and assigned a unique ID.

- Accident is visible in the Traffic Accidents list.
- Record is accessible for reporting and statistics.

UC24: Issue Traffic Fine

UC Name	UC24: Issue Traffic Fine
Summary	A Traffic Officer issues and registers a fine for a traffic violation to a citizen or vehicle.
Dependency	UC1 (User logs in)
Actors	Primary: Traffic Officer
Preconditions	• Traffic Officer is logged in. • A traffic violation has been observed or reported.
Description of Main Sequence	• Officer navigates to the Fines section. • Officer clicks 'Issue Fine'. • System displays the fine registration form. • Officer enters: violation type, fine amount, date, time, location, citizen ID or vehicle registration, and description. • Officer submits the form. • System validates all required fields. • System saves the fine record and links it to the citizen or vehicle. • System makes the fine visible to the citizen in their portal. • System confirms with a success message.
Description of Alternative Sequence	• Citizen or vehicle not found in the system: System allows the officer to manually enter the details. • Missing required fields: System highlights empty fields.
Non-Functional Requirements	• Fine records must be immutable once issued. • Fines must be immediately visible to the citizen in their portal. • All fine issuances must be audit-logged.
Postconditions	• Fine is saved and linked to the citizen or vehicle. • Fine is visible to the citizen in their portal. • Fine issuance is recorded in the audit log.

UC25: Manage Road Violations

UC Name	UC25: Manage Road Violations
Summary	A Traffic Officer records, views, and manages road violations and infractions.
Dependency	UC1 (User logs in)
Actors	Primary: Traffic Officer

Preconditions	- Traffic Officer is logged in. - A road violation has been observed or reported.
Description of Main Sequence	- Officer navigates to the Violations section. - To register a violation: Officer clicks 'Register Violation', enters violation type, location, date, vehicle details, and description. Officer submits the form. System saves the violation record. - To view violations: Officer browses or filters the violations list by date, type, or status. - To update a violation: Officer selects a violation, modifies the status (Open, Resolved), and saves. - System confirms each action with a success message.
Description of Alternative Sequence	- Missing required fields: System highlights empty fields. - Violation already closed: System prevents re-opening without authorization.
Non-Functional Requirements	- Violation records must be saved within 2 seconds. - Closed violations must be audit-logged before status change. - Access control prevents unauthorized modifications.
Postconditions	- Violation record is saved and accessible in the Violations list. - Status updates are reflected immediately. - All changes are audit-logged.

UC26: Perform Vehicle Control

UC Name	UC26: Perform Vehicle Control
Summary	A Traffic Officer performs and documents a vehicle control check, verifying registration, insurance, and inspection status.
Dependency	UC1 (User logs in)
Actors	Primary: Traffic Officer
Preconditions	- Traffic Officer is logged in. - A vehicle has been stopped for a control check.
Description of Main Sequence	- Officer navigates to the Vehicle Control section. - Officer clicks 'New Vehicle Control'. - System displays the control form. - Officer enters: vehicle registration number, make, model, year, owner details, date and time of check. - Officer checks registration status, insurance validity, and inspection certificate. - Officer enters the control results and any notes. - Officer submits the form. - System saves the control record and links it to the vehicle. - System confirms with a success message.

Description of Alternative Sequence	- Vehicle registration not found: System allows manual entry of vehicle details. - Missing required fields: System highlights empty fields.
Non-Functional Requirements	- Vehicle control records must be immutable once submitted. - Results must be clearly linked to the vehicle and date of control. - All control records must be audit-logged.
Postconditions	- Vehicle control record is saved and linked to the vehicle. - Record is accessible in the Vehicle Controls list. - Control results are audit-logged.

UC27: Manage Driving Licenses

UC Name	UC27: Manage Driving Licenses
Summary	A Traffic Officer searches, views, and manages driving license records, including flagging suspended or expired licenses.
Dependency	UC1 (User logs in)
Actors	Primary: Traffic Officer
Preconditions	1. Traffic Officer is logged in.
Description of Main Sequence	- Officer navigates to the Licenses section. - Officer searches for a license by citizen ID, license number, or name. - System retrieves and displays the license record: license number, category, issue date, expiry date, status. - Officer reviews the license validity. - To flag a license: Officer selects 'Flag / Suspend', enters the reason, and confirms. System updates the license status. - System confirms each action with a success message.
Description of Alternative Sequence	- License not found: System displays a 'No license record found' message. - License already suspended: System notifies the officer.
Non-Functional Requirements	- License status changes must be audit-logged with timestamp and officer identity. - Citizens must be notified if their license is flagged or suspended. - License data must be stored securely with role-restricted access.
Postconditions	- License status is updated and visible in the system. - Citizen is notified of any flag or suspension. - Status change is audit-logged.

UC28: Generate Traffic Report	
UC Name	UC28: Generate Traffic Report
Summary	A Traffic Officer generates statistical reports covering violations, accidents, and fine collection for a specified period.
Dependency	UC1 (User logs in)
Actors	Primary: Traffic Officer Secondary: Administrator
Preconditions	- Traffic Officer is logged in. - Traffic data (accidents, violations, fines) exists in the system.
Description of Main Sequence	- Officer navigates to the Traffic Reports section. - Officer selects the report type: Violations Summary, Accident Statistics, or Fine Collection Report. - Officer specifies the date range and any additional filters. - Officer clicks 'Generate Report'. - System processes the data. - System displays the report with summary statistics and optional visualizations. - Officer can export the report (PDF or CSV).
Description of Alternative Sequence	- No data available for the selected period: System displays a 'No data found' message. - Export failure: System shows an error and allows retry.
Non-Functional Requirements	- Reports must generate within 10 seconds. - Reports must be accurate and sourced from live database data. - Exported files must be formatted clearly and comply with department standards.
Postconditions	- Traffic report is generated and displayed. - Report can be exported if needed. - Report generation is logged in the audit trail.

6. Citizen

UC29: Self-Register as Citizen	
UC Name	UC29: Self-Register as Citizen
Summary	A member of the public creates a citizen account on the PMS portal to access public-facing services.
Dependency	None
Actors	Primary: Citizen
Preconditions	The citizen portal is publicly accessible.

	The person does not already have a citizen account.
Description of Main Sequence	- Citizen navigates to the public PMS portal. - Citizen clicks 'Register'. - System displays the citizen registration form. - Citizen enters: full name, national ID number, date of birth, email address, phone number, and password. - Citizen submits the form. - System validates all required fields and checks for duplicate accounts. - System creates the citizen account. - System sends a confirmation notification or email. - System confirms with a success message.
Description of Alternative Sequence	- Duplicate account (same national ID or email): System notifies the citizen and prevents duplicate registration. - Missing required fields: System highlights empty fields. - Weak password: System requires a stronger password.
Non-Functional Requirements	- Registration must complete within 5 seconds. - Citizen account data must be stored securely and comply with GDPR. - Citizens must only have access to their own submitted data.
Postconditions	- Citizen account is created and active. - Citizen can log in and access the citizen portal. - Registration is recorded in the system.

UC30: Report an Incident

UC Name	UC30: Report an Incident
Summary	A Citizen submits an incident report through the public portal to notify the police of a crime or suspicious activity.
Dependency	UC1 (Citizen logs in)
Actors	Primary: Citizen
Preconditions	- Citizen is logged in to their portal account. - The incident has occurred.
Description of Main Sequence	- Citizen navigates to the 'Report Incident' section. - Citizen clicks 'Submit Report'. - System displays the incident report form. - Citizen enters: incident description, location, date and time, and optionally uploads a photo. - Citizen submits the form. - System validates all required fields. - System saves the incident report. - System assigns the report to the police department for review.

	System confirms with a success message and provides a reference number.
Description of Alternative Sequence	- Missing required fields: System highlights empty fields. - File upload failure: System retains the form data and prompts retry for the file only.
Non-Functional Requirements	- Incident reports must be saved within 3 seconds. - Citizens must receive a reference number for tracking. - Report data must be stored securely.
Postconditions	- Incident report is saved and submitted to the police department. - Citizen receives a reference number for tracking. - Report is accessible in the citizen's portal for status tracking.

UC31: Submit a Complaint

UC Name	UC31: Submit a Complaint
Summary	A Citizen submits a formal complaint regarding police conduct, services, or a specific officer through the public portal.
Dependency	UC1 (Citizen logs in)
Actors	Primary: Citizen
Preconditions	- Citizen is logged in to their portal account. - The citizen has a complaint to submit.
Description of Main Sequence	- Citizen navigates to the 'Submit Complaint' section. - Citizen clicks 'New Complaint'. - System displays the complaint form. - Citizen enters: complaint category, description, date of the incident, and optionally the relevant officer or department. - Citizen submits the form. - System validates all required fields. - System saves the complaint and routes it to the Administrator. - System confirms with a success message and provides a complaint reference number.
Description of Alternative Sequence	- Missing required fields: System highlights empty fields. - System save error: System retains the form and prompts retry.
Non-Functional Requirements	- Complaints must be saved within 3 seconds. - Citizens must receive a reference number. - All complaints must be routed to the Administrator for review.
Postconditions	- Complaint is saved and routed to the Administrator. - Citizen receives a reference number.

- Complaint is accessible in the citizen's portal for status tracking.

UC32: Report a Missing Person

UC Name	UC32: Report a Missing Person
Summary	A Citizen submits a missing person report through the public portal.
Dependency	UC1 (Citizen logs in)
Actors	Primary: Citizen
Preconditions	• Citizen is logged in to their portal account. • A person has gone missing.
Description of Main Sequence	• Citizen navigates to the 'Report Missing Person' section. • Citizen clicks 'File Report'. • System displays the missing person report form. • Citizen enters: missing person's name, date of birth, physical description, last known location, date last seen, and citizen's contact information. • Citizen optionally uploads a recent photo. • Citizen submits the form. • System validates all required fields. • System saves the report and assigns it to the relevant police department. • System confirms with a success message and provides a reference number.
Description of Alternative Sequence	• Missing required fields: System highlights empty fields. • Photo upload failure: System retains the form and prompts retry.
Non-Functional Requirements	• Missing person reports must be saved within 3 seconds. • Citizens must receive a reference number. • Reports must be immediately accessible to the assigned department.
Postconditions	• Missing person report is saved and submitted to the police department. • Citizen receives a reference number. • Report is accessible in the citizen's portal for status tracking.

UC33: Check Case Status

UC Name	UC33: Check Case Status
Summary	A Citizen checks the current status of an incident report, complaint, or missing person report they have submitted.
Dependency	UC1 (Citizen logs in), UC30/UC31/UC32
Actors	Primary: Citizen

Preconditions	• Citizen is logged in. • The citizen has at least one submitted report in the system.
Description of Main Sequence	• Citizen navigates to 'My Reports' section. • System retrieves all reports submitted by the citizen. • System displays the list with: report type, reference number, date submitted, and current status. • Citizen selects a specific report. • System displays the full report details and current status. • Citizen reviews the status update.
Description of Alternative Sequence	• No reports found: System displays a message 'You have not submitted any reports'. • Report status not yet updated: System displays the last known status with a 'Pending Review' label.
Non-Functional Requirements	• Report list must load within 2 seconds. • Citizens can only view their own submitted reports. • Status information must reflect the most recent update.
Postconditions	• Citizen can view the current status of all their submitted reports. • Citizen is informed of any updates or next steps.

UC34: View Traffic Fines

UC Name	UC34: View Traffic Fines
Summary	A Citizen views any traffic fines issued to them by a Traffic Officer through the citizen portal.
Dependency	UC1 (Citizen logs in)
Actors	Primary: Citizen
Preconditions	• Citizen is logged in. • At least one fine has been issued to the citizen by a Traffic Officer.
Description of Main Sequence	• Citizen navigates to the 'My Fines' section. • System retrieves all fines linked to the citizen's account. • System displays the list with: fine reference, violation type, amount, date issued, and payment status. • Citizen selects a specific fine. • System displays the full fine details: issuing officer, location, violation description, and deadline.
Description of Alternative Sequence	• No fines found: System displays a message 'No fines have been issued to your account'. • Fine data loading error: System shows an error and provides a refresh option.

Non-Functional Requirements	- Fine list must load within 2 seconds. - Citizens can only view their own fines. - Fine data must be accurate and reflect the most recent status.
Postconditions	- Citizen can view all traffic fines issued to them. - Citizen is informed of the violation type, amount, and payment status.

7. Administrator

UC35: Manage Users and Roles	
UC Name	UC35: Manage Users and Roles
Summary	The Administrator creates, updates, deactivates, and manages user accounts and role assignments for all staff roles.
Dependency	UC1 (Administrator logs in)
Actors	Primary: Administrator
Preconditions	- Administrator is logged in. - The system is operational.
Description of Main Sequence	- Administrator navigates to the User Management section. - To add a user: Administrator clicks 'Add User', fills in Full Name, Username, Password, Role, and Department. Clicks Save. System validates and creates the account. - To edit a user: Administrator selects a user, clicks 'Edit', modifies the required fields, and saves. - To deactivate a user: Administrator selects a user and clicks 'Deactivate'. System prompts for confirmation. Administrator confirms. System deactivates the account. - System confirms each action with a success message.
Description of Alternative Sequence	- Duplicate username: System prevents creation and displays an error. - Missing required fields: System highlights missing fields. - Attempt to deactivate an active user with ongoing cases: System warns the administrator and requests confirmation.
Non-Functional Requirements	- User management operations must complete within 3 seconds. - Passwords must be hashed before storage. - All user management actions must be audit-logged.
Postconditions	- New user account is created with the correct role and department. - Updated user information is reflected in the system. - Deactivated user can no longer log in.

UC36: Manage Departments

UC Name	UC36: Manage Departments
Summary	The Administrator adds, edits, and manages police department records to organize staff by their respective units.
Dependency	UC1 (Administrator logs in)
Actors	Primary: Administrator
Preconditions	- Administrator is logged in. - System is operational.
Description of Main Sequence	- Administrator navigates to the Departments section. - To add: Administrator clicks 'Add', enters the Department Name, and saves. - To edit: Administrator selects a department, clicks 'Edit', modifies the name, and saves. - To delete: Administrator selects a department and clicks 'Delete'. System checks if staff are assigned. If no staff assigned, department is deleted. If staff are assigned, system warns the administrator. - System confirms each action with a success message.
Description of Alternative Sequence	- Department name already exists: System displays a duplicate error. - Attempt to delete department with assigned staff: System warns and prevents deletion until staff are reassigned.
Non-Functional Requirements	- Department operations should complete within 2 seconds. - Department names must be unique within the system. - Changes must be immediately visible to all relevant users.
Postconditions	- New department is saved and available for staff assignment. - Updated department information is reflected in the system. - Deleted department is removed from all dropdown lists.

UC37: Assign Officer to Case

UC Name	UC37: Assign Officer to Case
Summary	The Administrator assigns a Police Officer to a specific case to ensure proper investigative coverage.
Dependency	UC1 (Administrator logs in), UC4 (Create Incident Report)
Actors	Primary: Administrator
Preconditions	- Administrator is logged in. - An active case exists in the system. - One or more Police Officers are registered in the system.
Description of Main Sequence	- Administrator navigates to the active case. - Administrator selects 'Assign Officer'.

	• System displays a list of available Police Officers. • Administrator selects the appropriate officer. • Administrator confirms the assignment. • System validates the selection. • System saves the assignment and notifies the assigned officer. • System confirms with a success message.
Description of Alternative Sequence	• No available officers: System displays a message indicating all officers are currently assigned. • Case already has an assigned officer: System asks the Administrator to confirm the reassignment.
Non-Functional Requirements	• Assignment must be saved and reflected within 2 seconds. • The assigned officer must receive a notification. • Officer assignment history must be logged.
Postconditions	• The selected officer is assigned to the case. • The case details are updated with the assigned officer's information. • The officer can now access and manage the case.

UC38: Monitor System Statistics

UC Name	UC38: Monitor System Statistics
Summary	The Administrator monitors real-time analytics and statistics across the entire system including cases, users, departments, forensic activity, and traffic data.
Dependency	UC1 (Administrator logs in), UC2 (View Dashboard)
Actors	Primary: Administrator
Preconditions	• Administrator is logged in. • Data has been entered into the system by users across all roles.
Description of Main Sequence	• Administrator navigates to the Dashboard. • System retrieves all system-wide statistics. • System displays: number of active departments, users per role, open cases, cases under investigation, recent incidents, pending forensic analyses, active traffic violations, and recent fines. • Administrator reviews the data. • Administrator can navigate to specific sections for more detailed information.
Description of Alternative Sequence	• No data available: System displays zero values and a note indicating the system has no recorded data. • Data loading failure: System shows an error and provides a refresh option.
Non-Functional Requirements	• Dashboard statistics must reflect real-time data updated within 1 minute. • Statistics page must load within 5 seconds. • Data displayed must be accurate and sourced from the live database.

Postconditions	- Administrator has a comprehensive, real-time overview of all system operations. - Administrator can take corrective action based on the statistics displayed.
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UC39: Reset User Password

UC Name	UC39: Reset User Password
Summary	The Administrator resets the password of a staff user account when requested or necessary.
Dependency	UC1 (Administrator logs in), UC35 (Manage Users and Roles)
Actors	Primary: Administrator
Preconditions	- Administrator is logged in. - The target user account exists in the system.
Description of Main Sequence	- Administrator navigates to the Users section. - Administrator selects the target user. - Administrator clicks 'Reset Password'. - System prompts for confirmation. - Administrator confirms. - System generates a temporary password and updates the user's account. - System notifies the user via the system or email with the temporary credentials. - System confirms with a success message.
Description of Alternative Sequence	- User account not found: System displays an error. - Notification delivery failure: System logs the failure and alerts the administrator.
Non-Functional Requirements	- Temporary password must meet minimum security requirements. - Password reset must be logged in the audit trail. - User must be required to change the temporary password on next login.
Postconditions	- User's password is reset to a temporary value. - User is notified of the reset. - The reset action is recorded in the audit log.

UC40: View System Audit Log

UC Name	UC40: View System Audit Log
Summary	The Administrator views a log of all significant actions performed in the system for accountability and security monitoring.
Dependency	UC1 (Administrator logs in)

Actors	Primary: Administrator
Preconditions	- Administrator is logged in. - The system has recorded at least one auditable action.
Description of Main Sequence	- Administrator navigates to the Audit Log section. - System retrieves all logged actions from the database. - System displays the log: action type, user role, username, timestamp, and affected record. - Administrator can filter logs by user, date range, role, or action type. - Administrator reviews the audit entries.
Description of Alternative Sequence	- No logs available: System displays a message 'No audit records found'. - Filter returns no results: System displays 'No results match the selected filters'.
Non-Functional Requirements	- Audit log must be read-only -- no entries can be edited or deleted. - Log entries must include user identity, timestamp, and action description. - Audit log must load within 3 seconds.
Postconditions	- Administrator has visibility into all audited system actions. - Administrator can identify any unauthorized or suspicious activity.

UC41: Search Officers by Department

UC Name	UC41: Search Officers by Department
Summary	The Administrator searches for and filters staff members based on their department assignment or role.
Dependency	UC1 (Administrator logs in), UC36 (Manage Departments)
Actors	Primary: Administrator
Preconditions	- Administrator is logged in. - At least one department and one staff member exist in the system.
Description of Main Sequence	- Administrator navigates to the Users section. - Administrator selects a department from the filter dropdown or enters a role filter. - System queries the database for staff in the selected department or role. - System displays the filtered list of staff members with their ID, name, role, and department. - Administrator can select a staff member to view or edit their profile.
Description of Alternative Sequence	- No staff in selected department: System displays 'No staff found in this department'. - Invalid filter selection: System resets to show all staff.
Non-Functional Requirements	- Search results must be returned within 2 seconds. - Results must reflect the current state of the database.

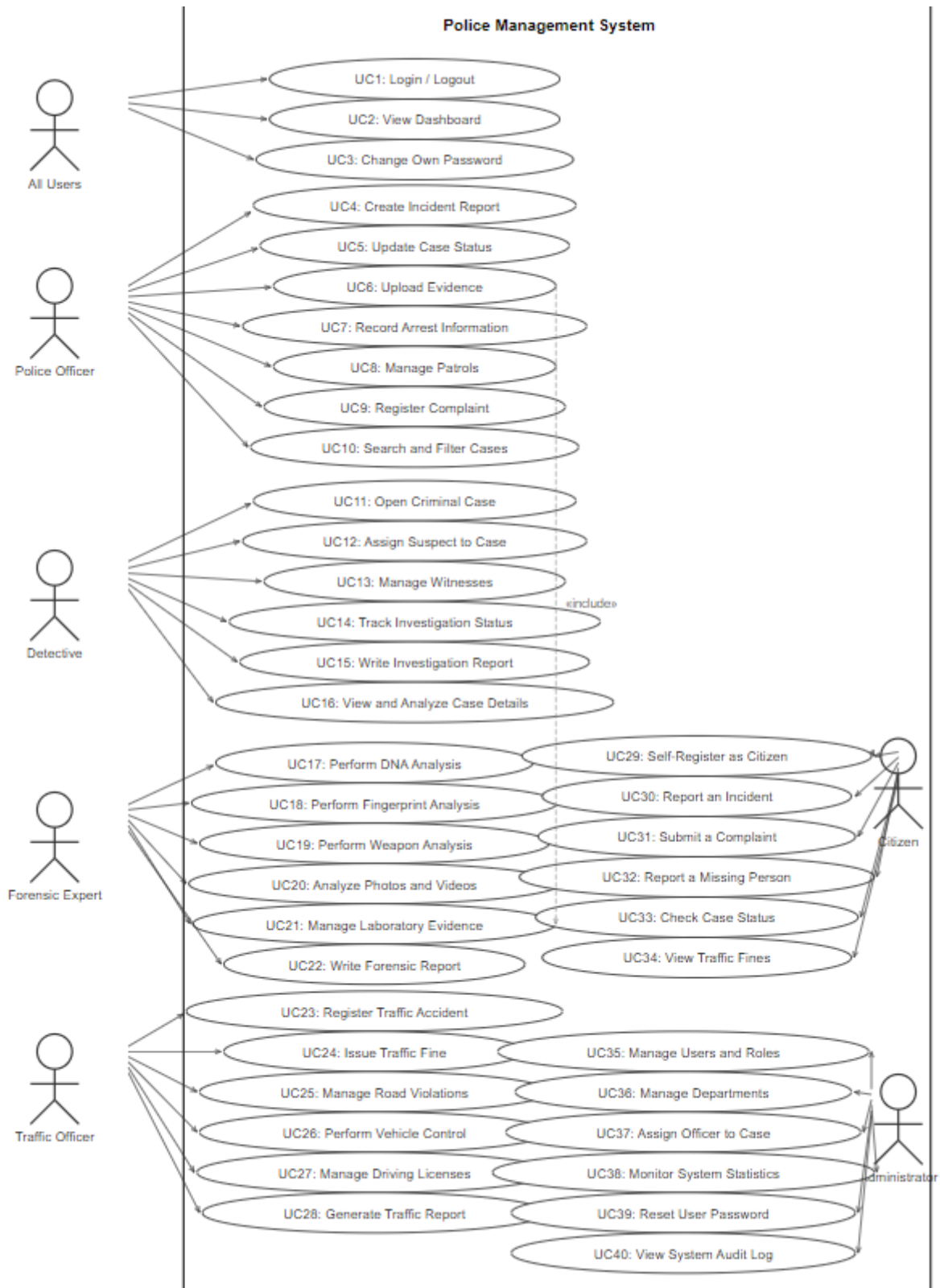
	Only authorized administrators can access staff data.
Postconditions	- A filtered list of staff matching the selected criteria is displayed. - Administrator can take action on any staff member from the results.

UC42: Deactivate User Account

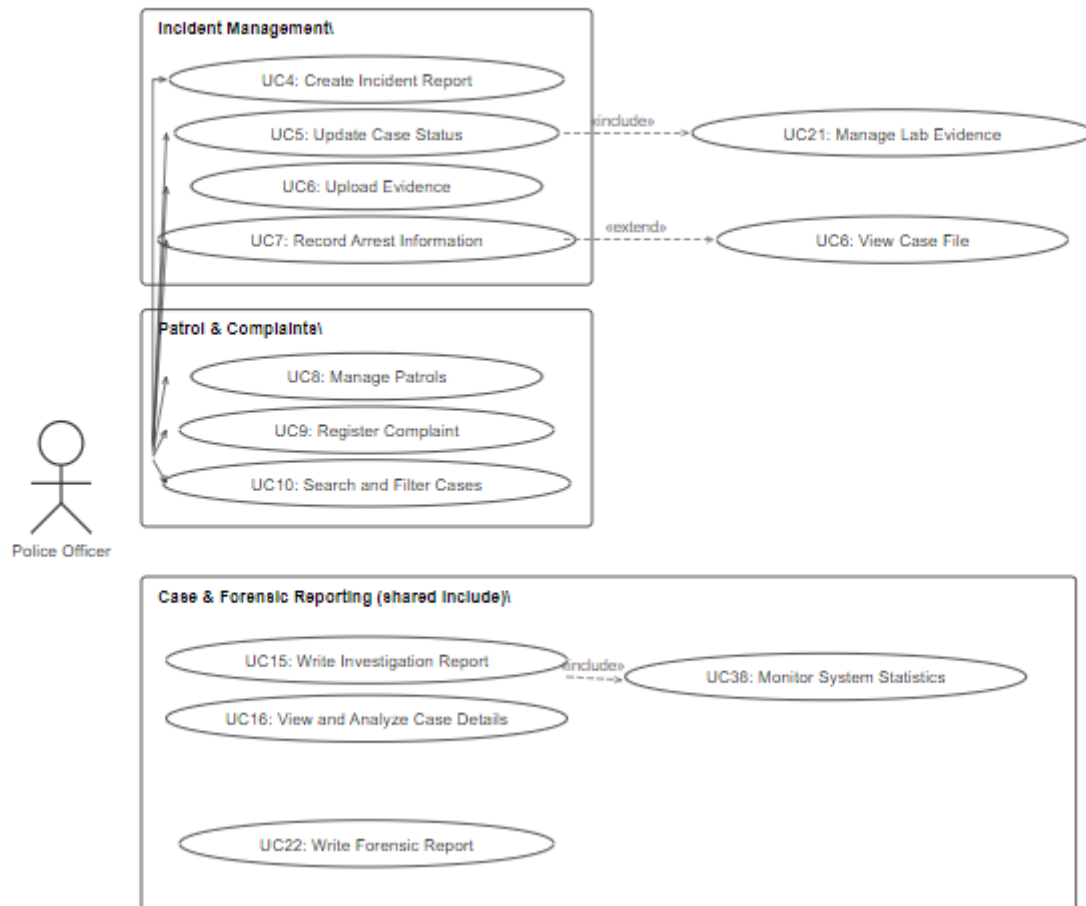
UC Name	UC42: Deactivate User Account
Summary	The Administrator deactivates a staff user account to revoke system access without permanently deleting the account and its data.
Dependency	UC1 (Administrator logs in), UC35 (Manage Users and Roles)
Actors	Primary: Administrator
Preconditions	- Administrator is logged in. - The target user account exists and is currently active.
Description of Main Sequence	- Administrator navigates to the Users section. - Administrator selects the target user. - Administrator clicks 'Deactivate Account'. - System prompts for confirmation and optionally a reason. - Administrator enters the reason and confirms. - System sets the account status to Inactive. - System terminates any active session for the deactivated user. - System logs the deactivation with the reason and timestamp. - System confirms with a success message.
Description of Alternative Sequence	- User has ongoing open cases: System warns the administrator and requests confirmation before proceeding. - Administrator cancels: Account remains active.
Non-Functional Requirements	- Deactivation must take effect immediately and terminate the user's active session. - Deactivation reason must be stored in the audit log. - Deactivated account data must be preserved for historical integrity.
Postconditions	- The user account is set to Inactive and access is revoked. - The user can no longer log in. - All account data is preserved and remains accessible to administrators. - Deactivation is recorded in the audit log with reason and timestamp.

4.2 Behavioral Diagrams

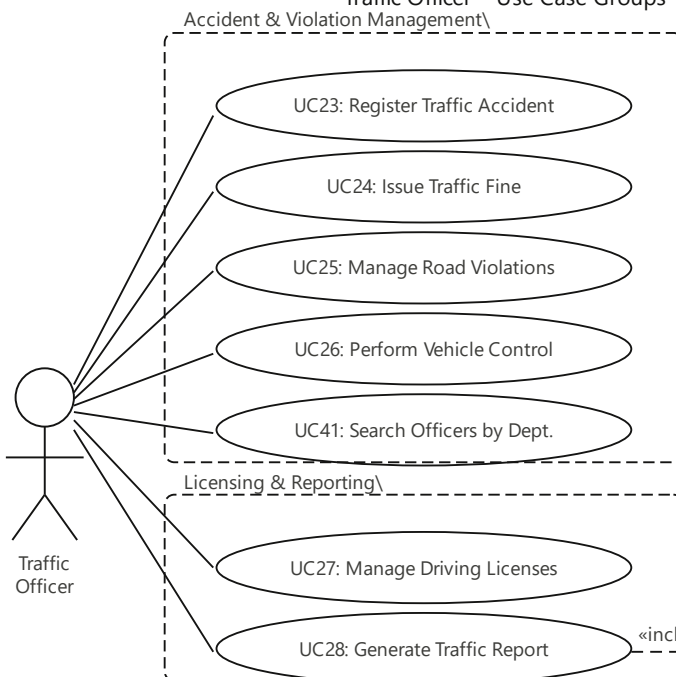
4.2.1 Use Case Diagrams



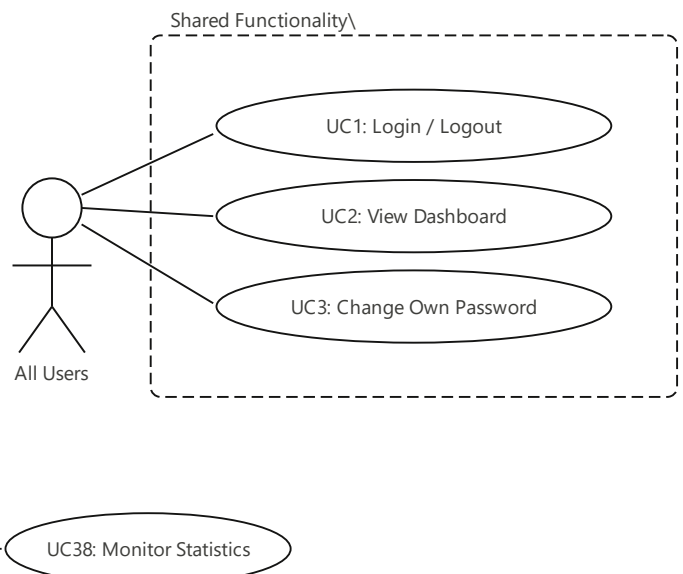
Police Officer – Use Case Groups



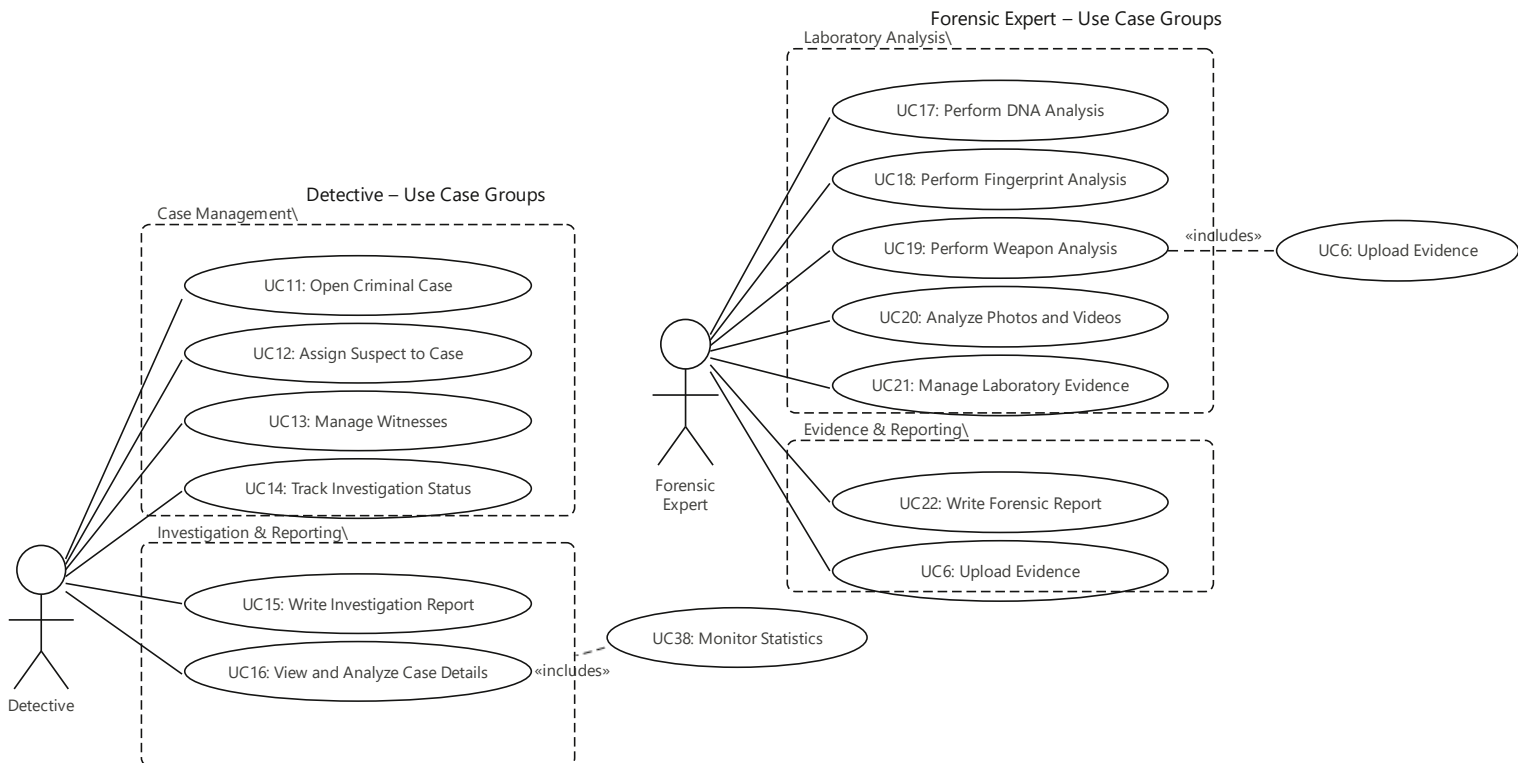
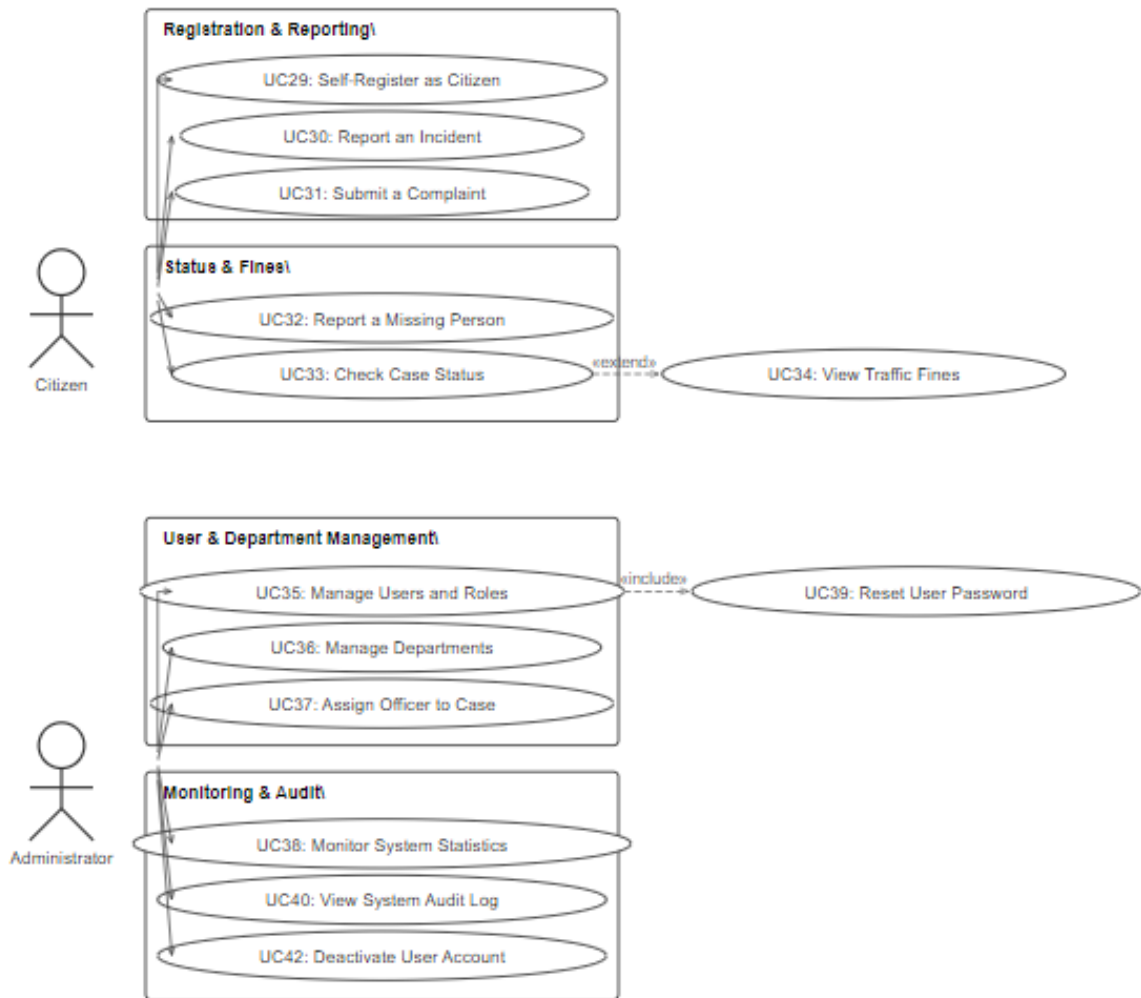
Traffic Officer – Use Case Groups



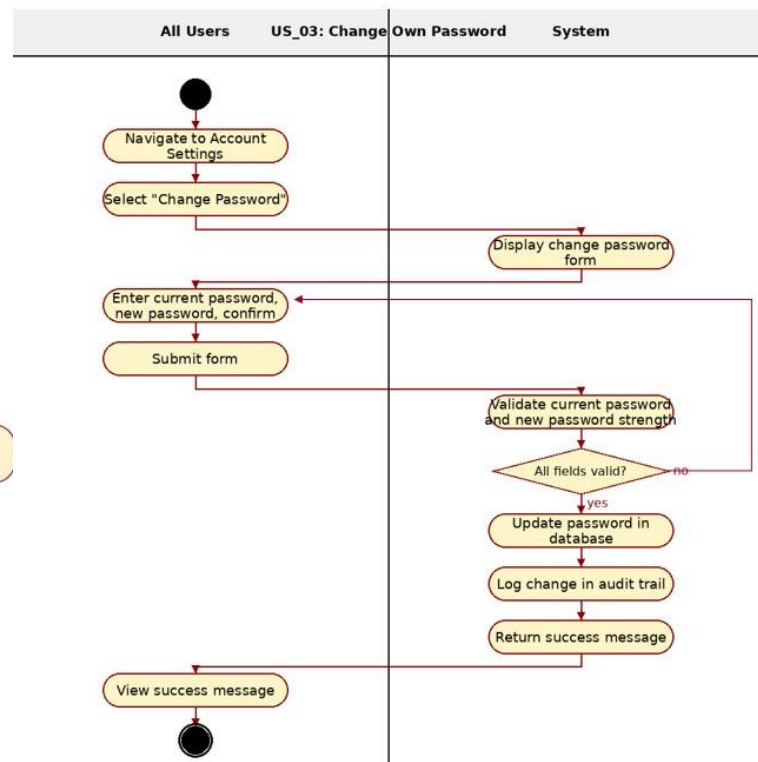
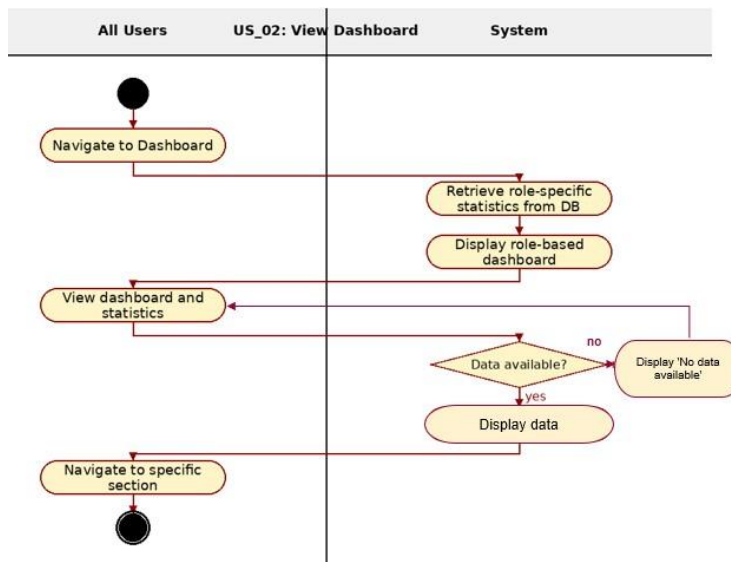
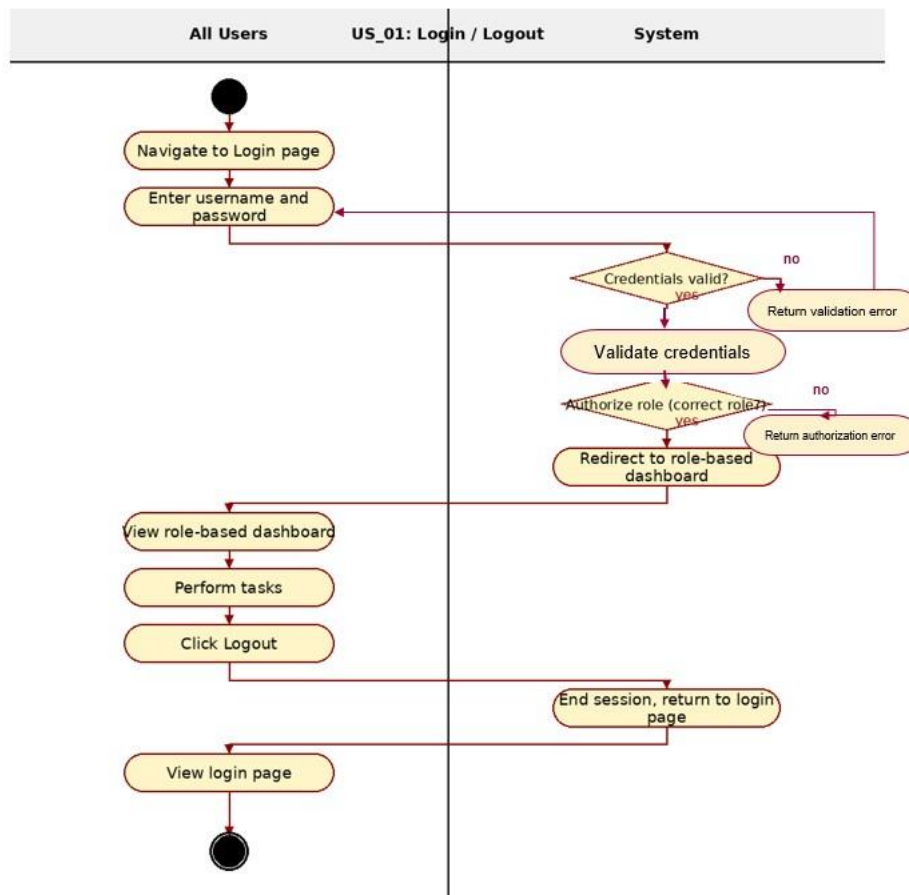
All Users – Use Case Groups

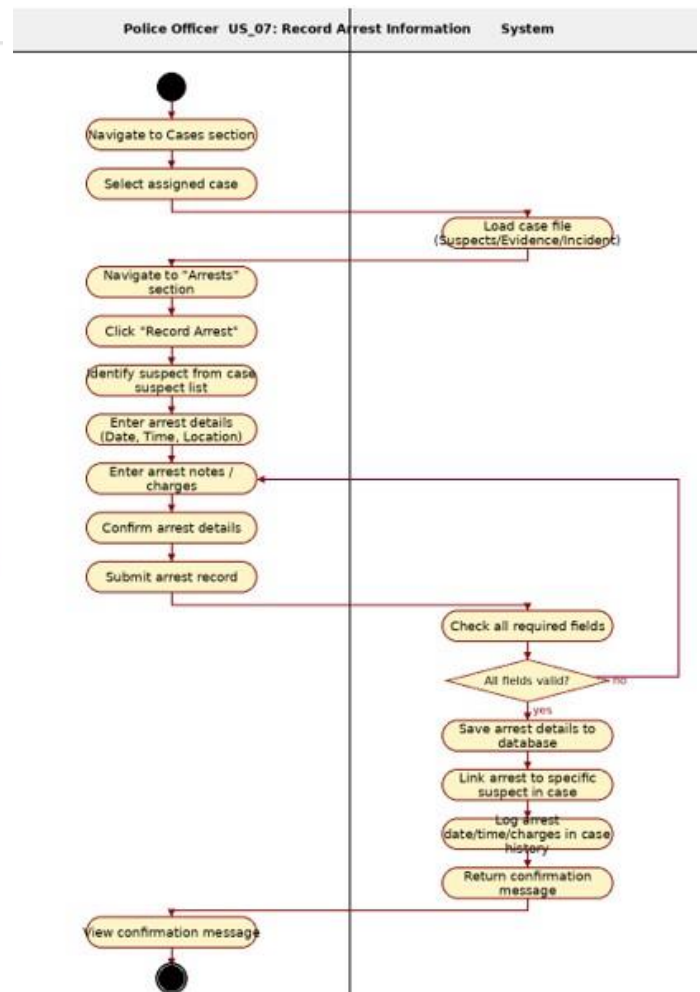
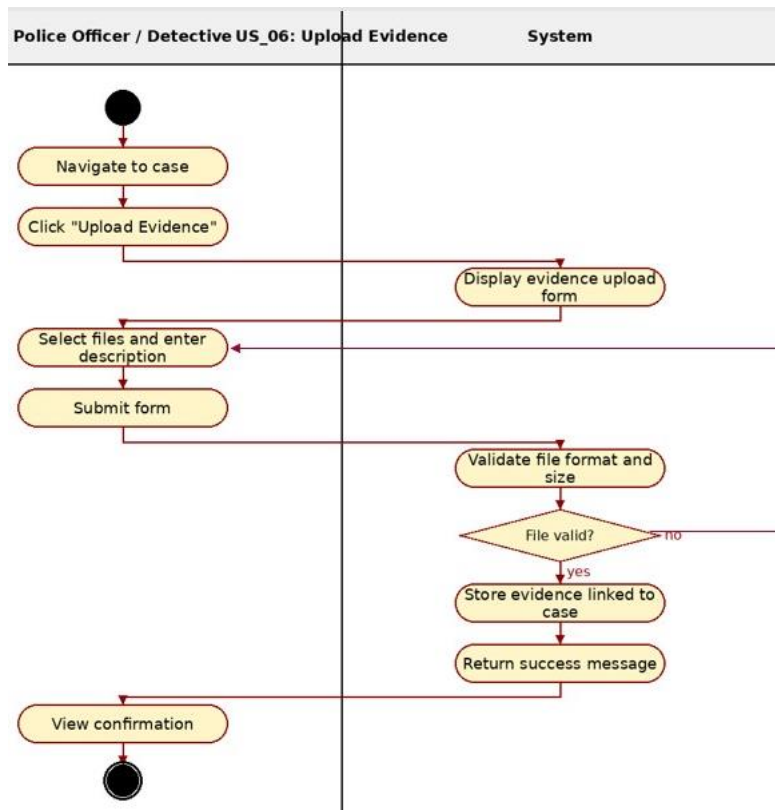
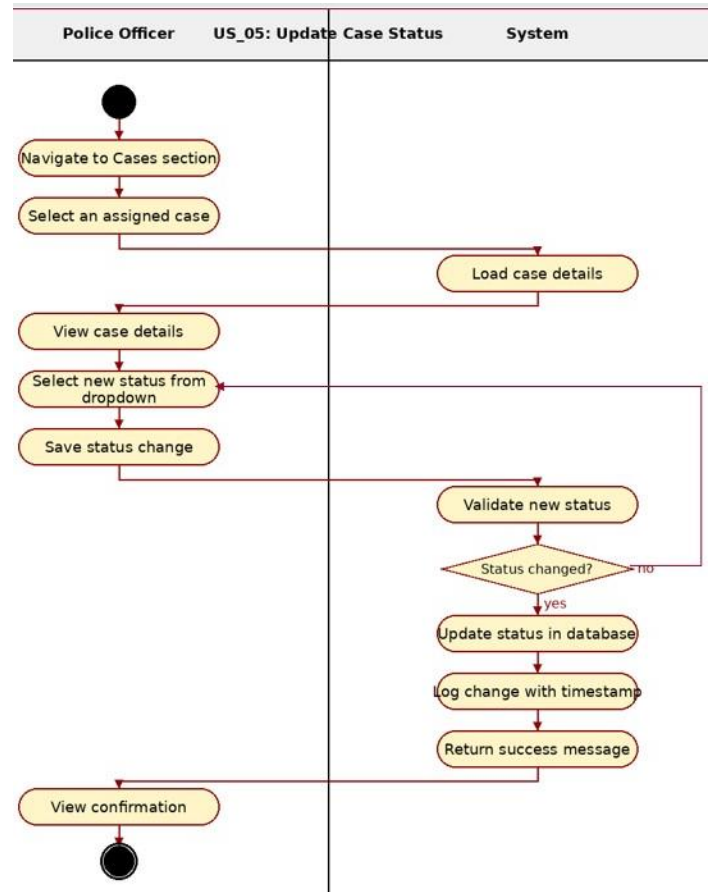
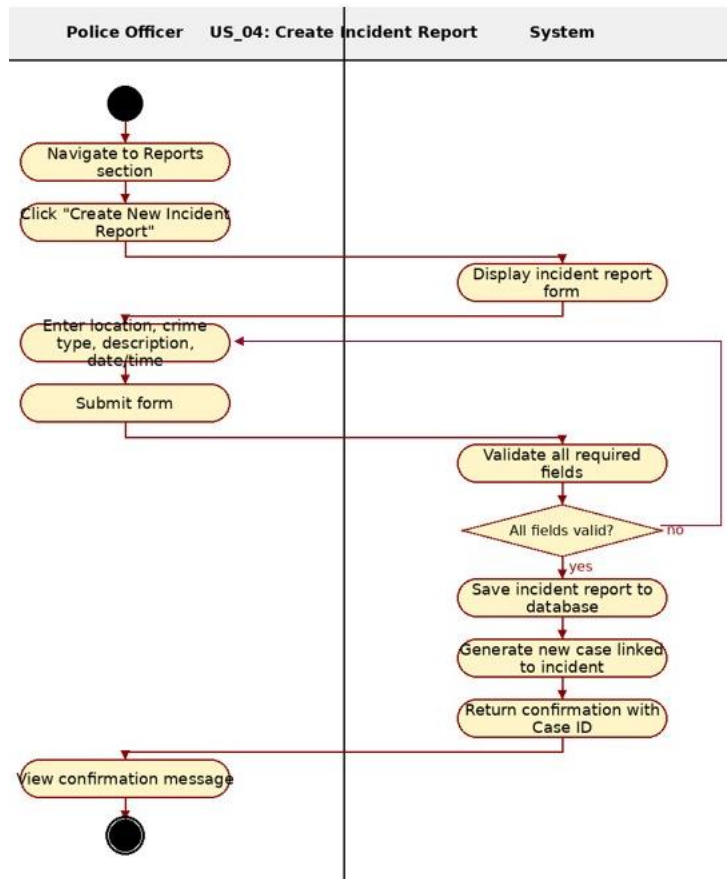


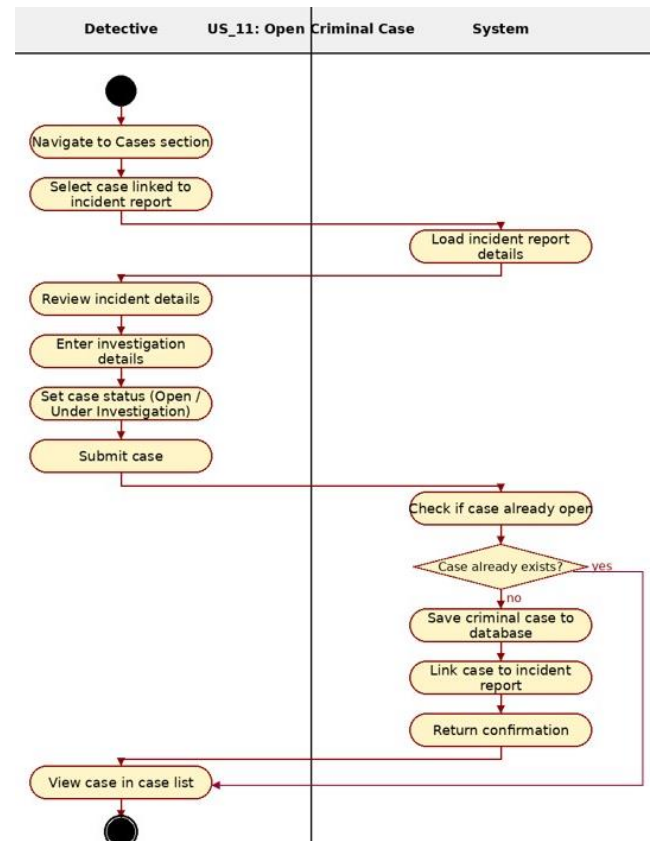
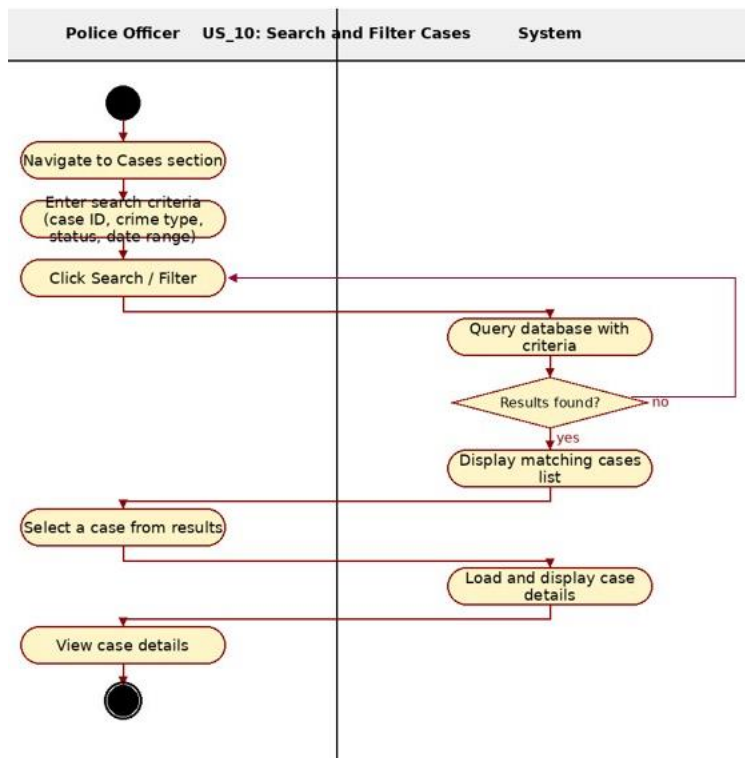
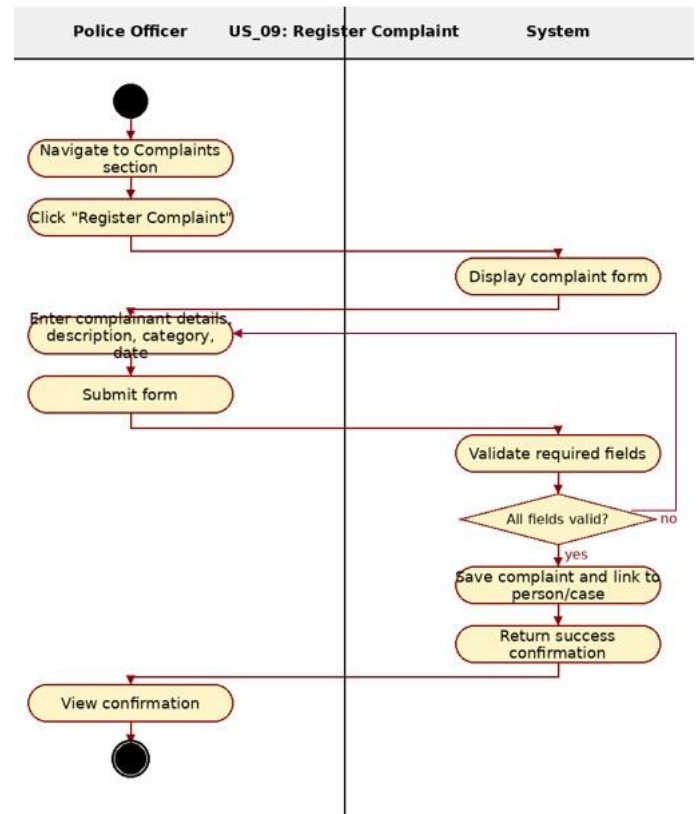
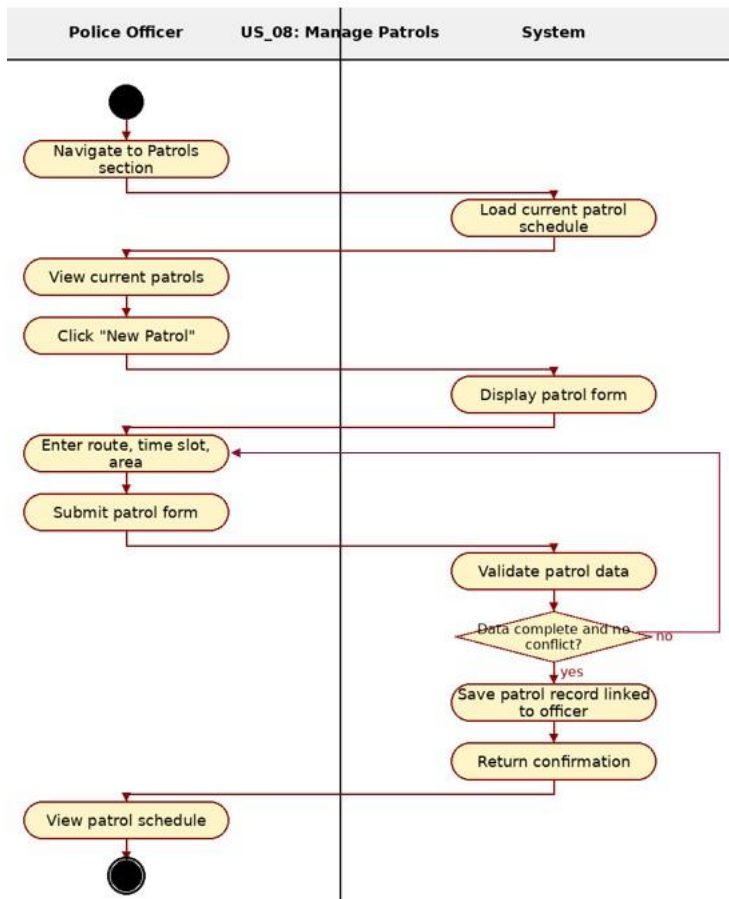
Citizen & Administrator – Use Case Groups

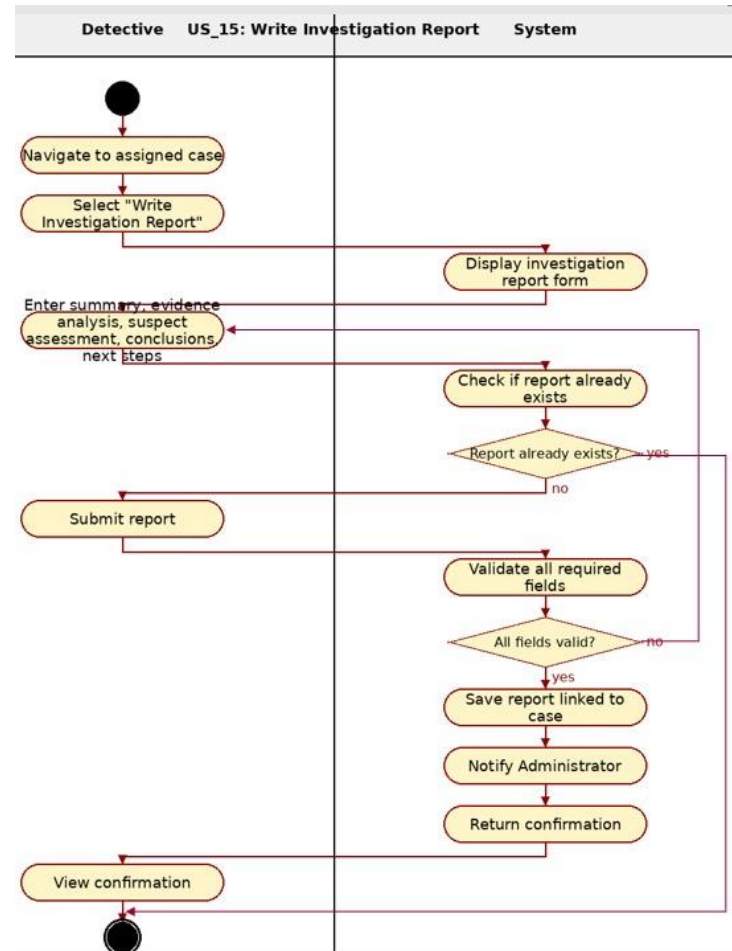
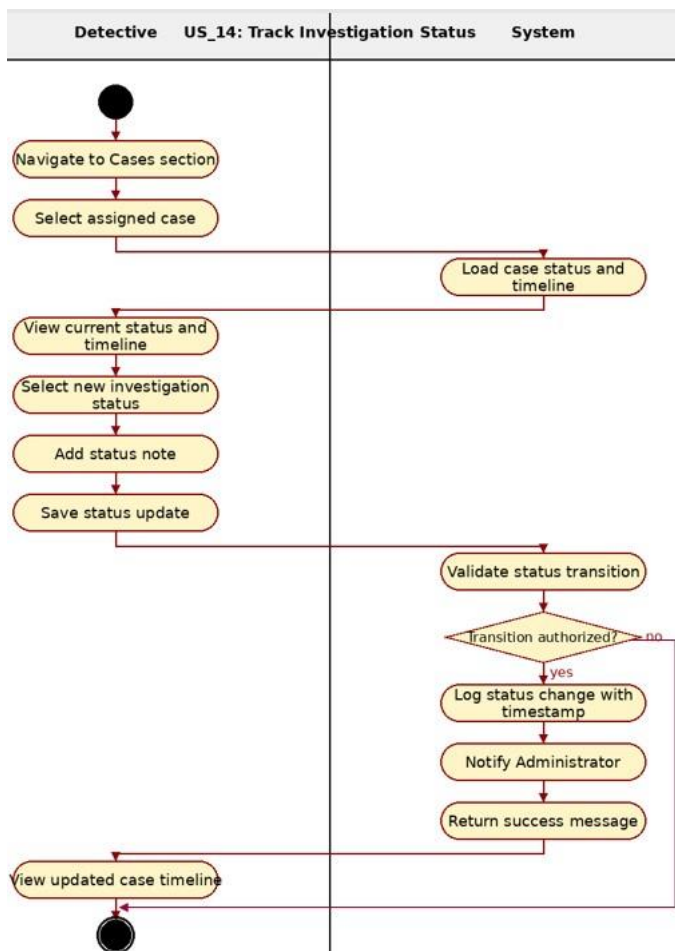
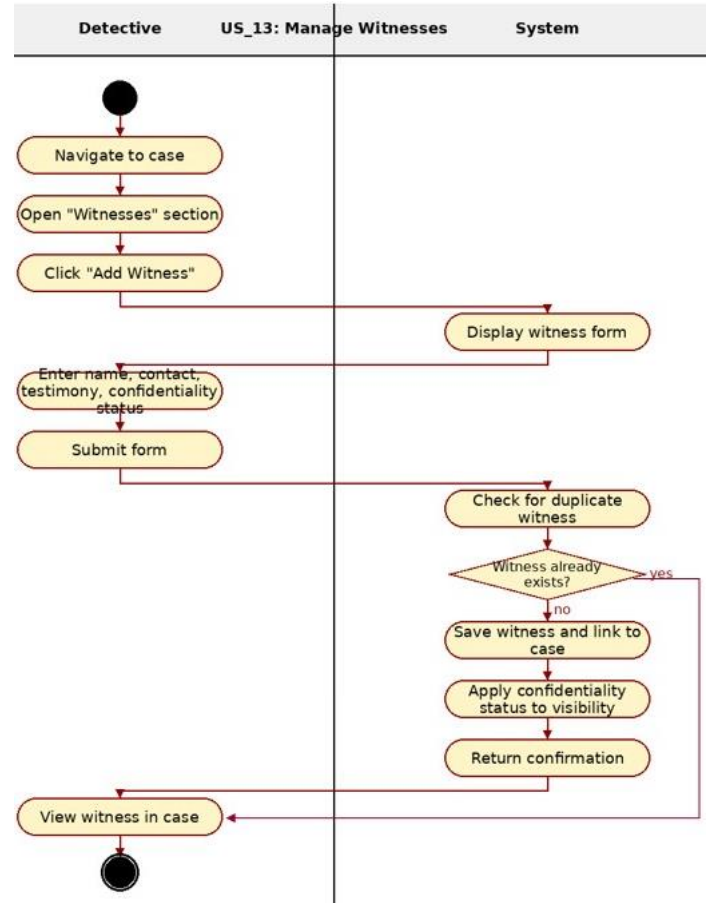
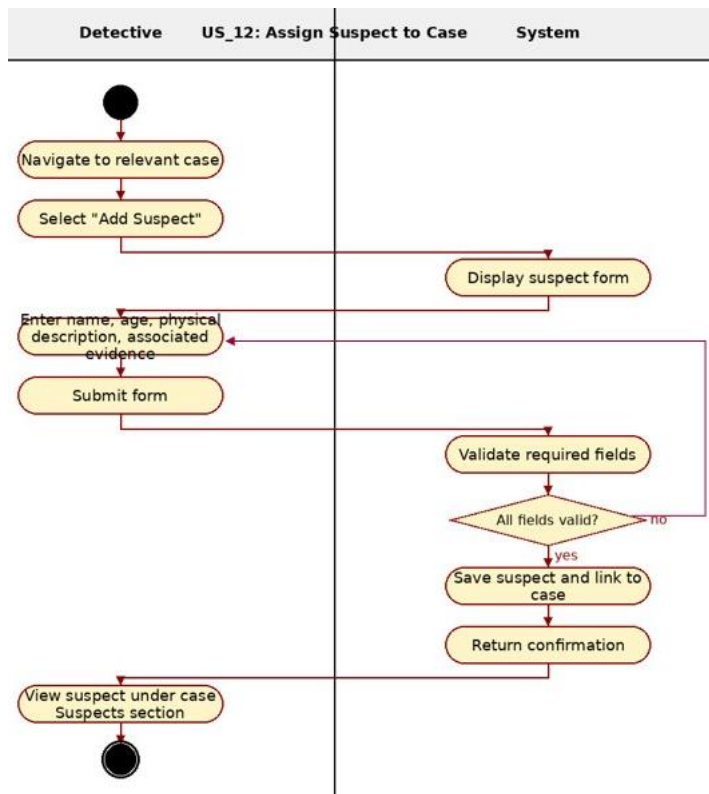


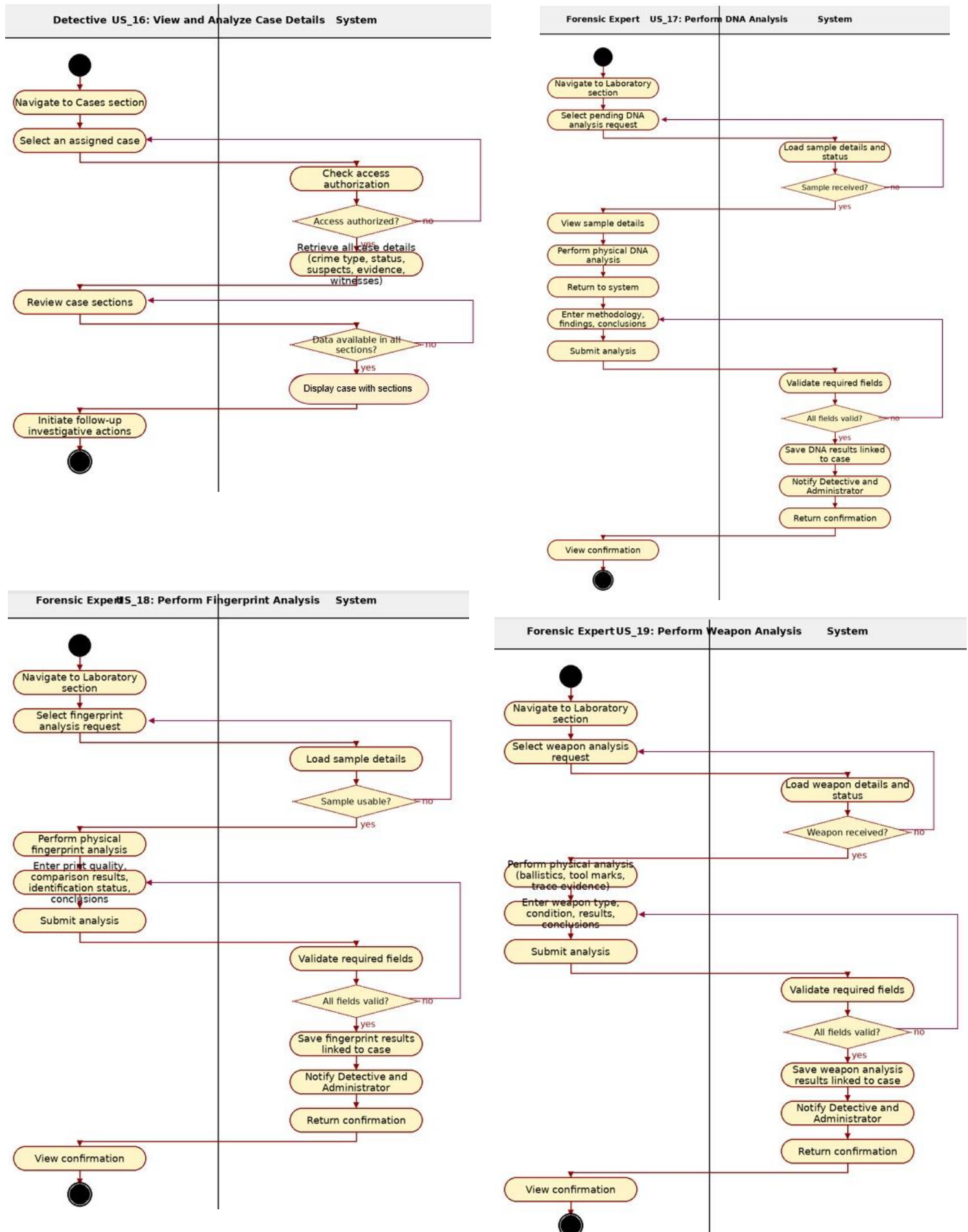
4.2.2 Use Case Diagrams

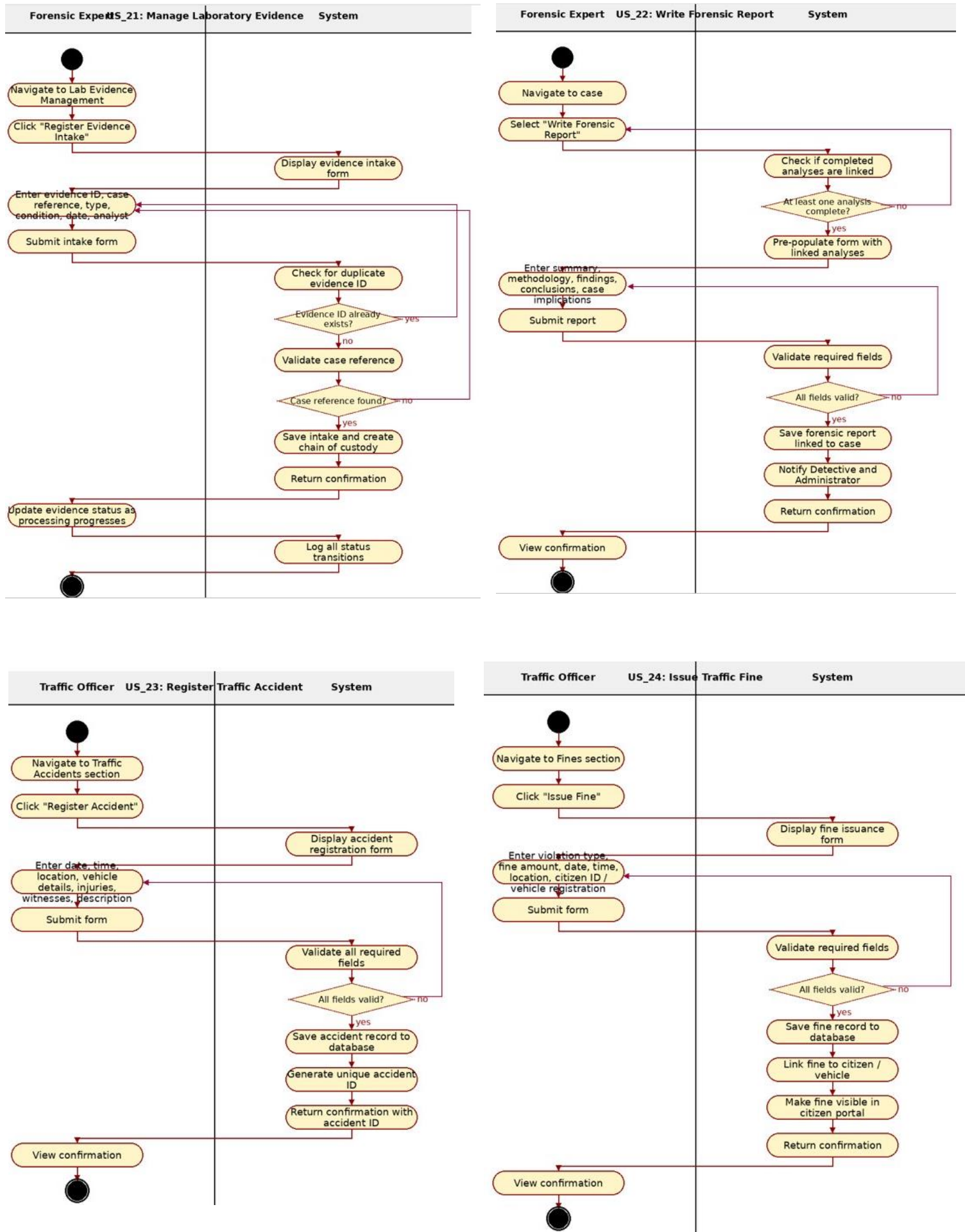


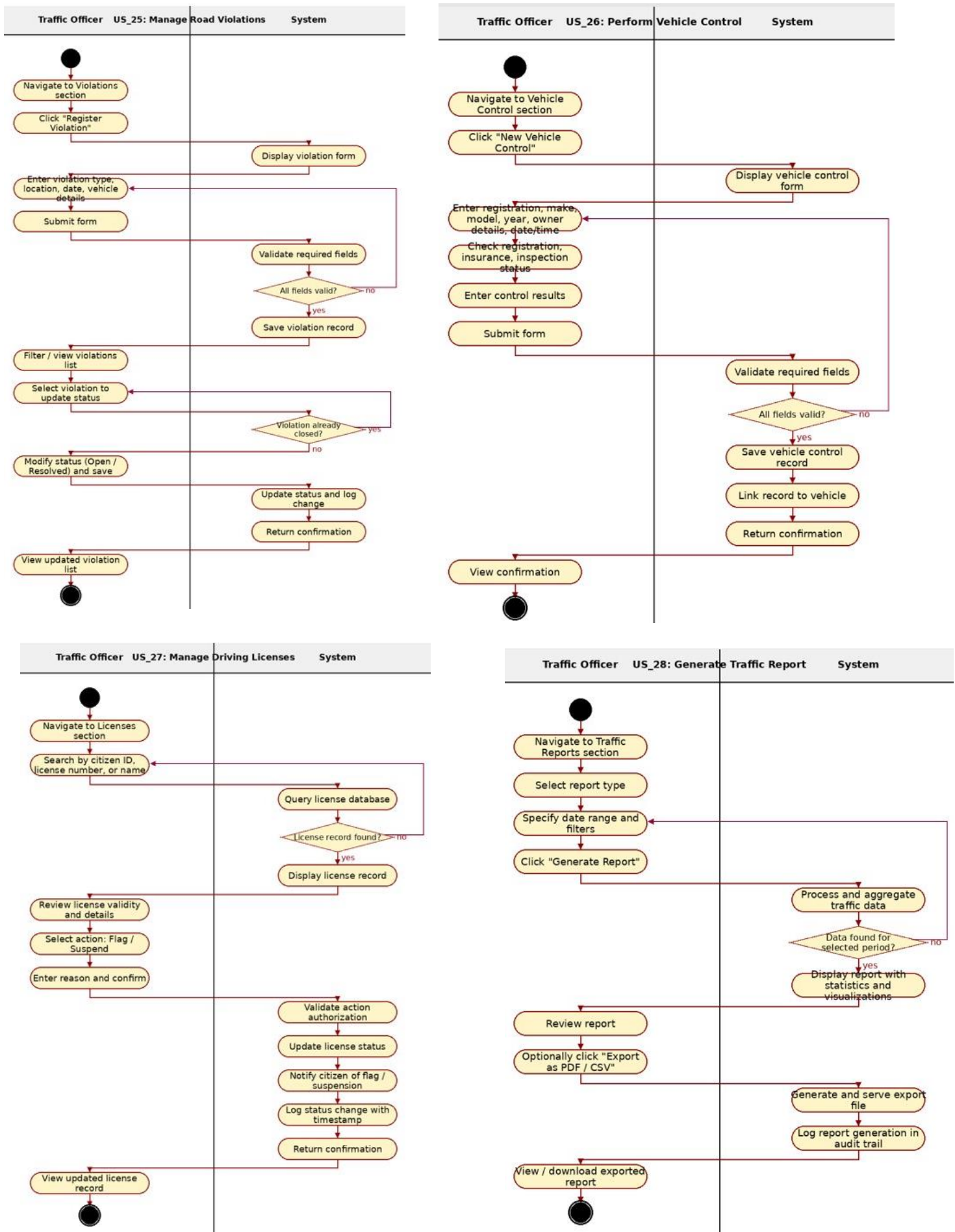


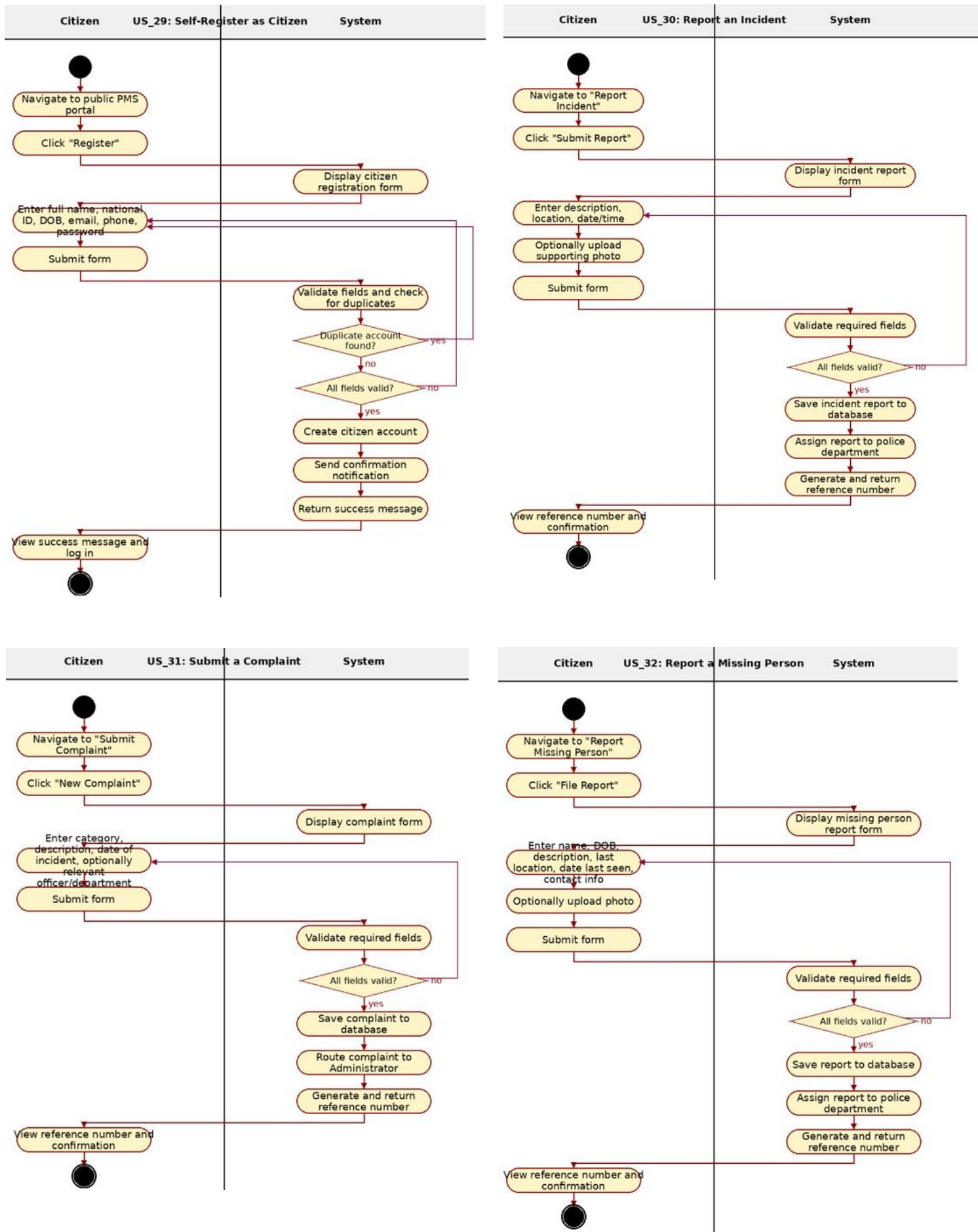


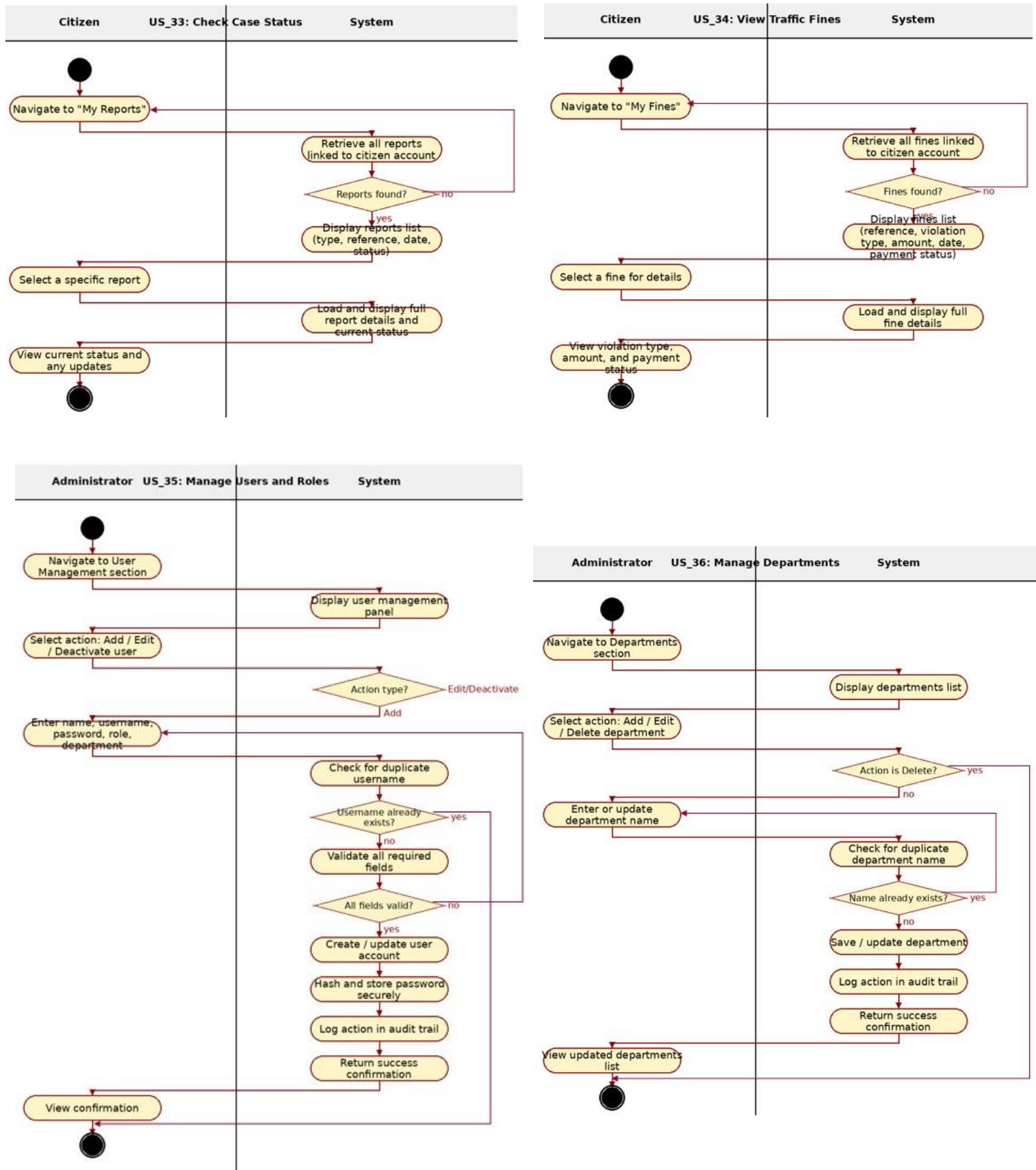


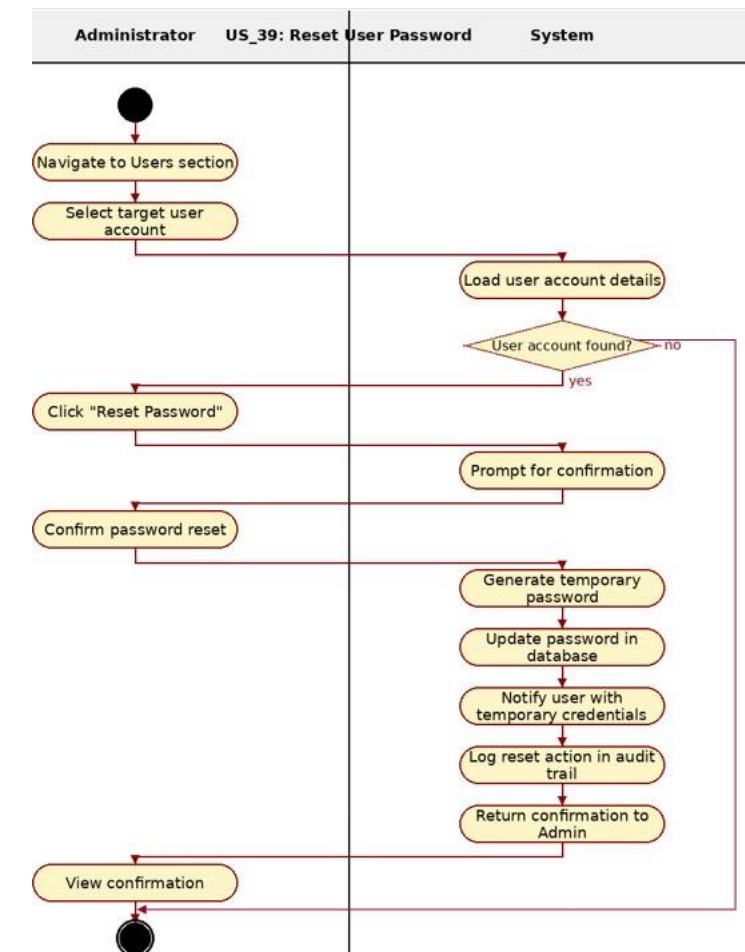
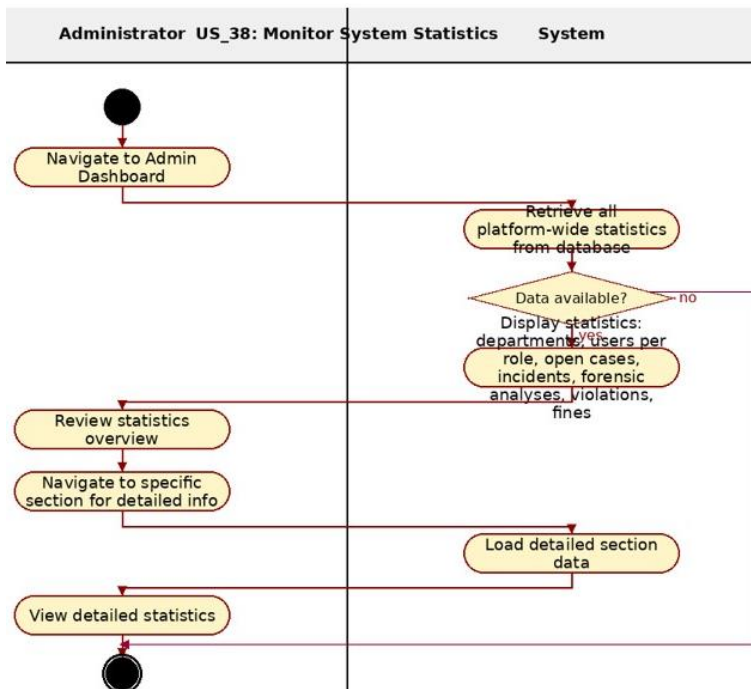
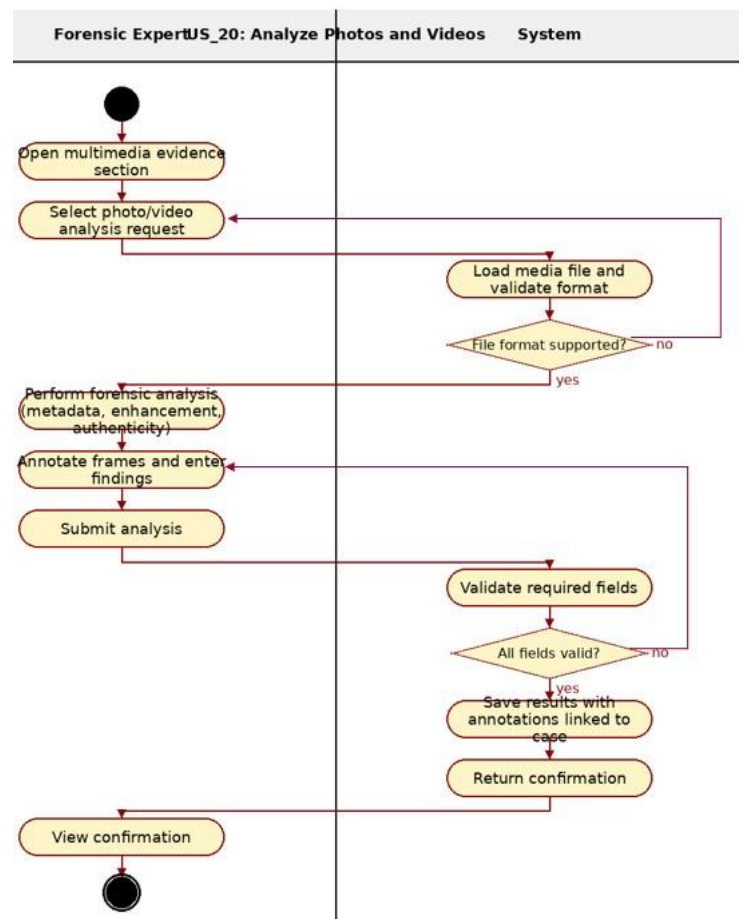
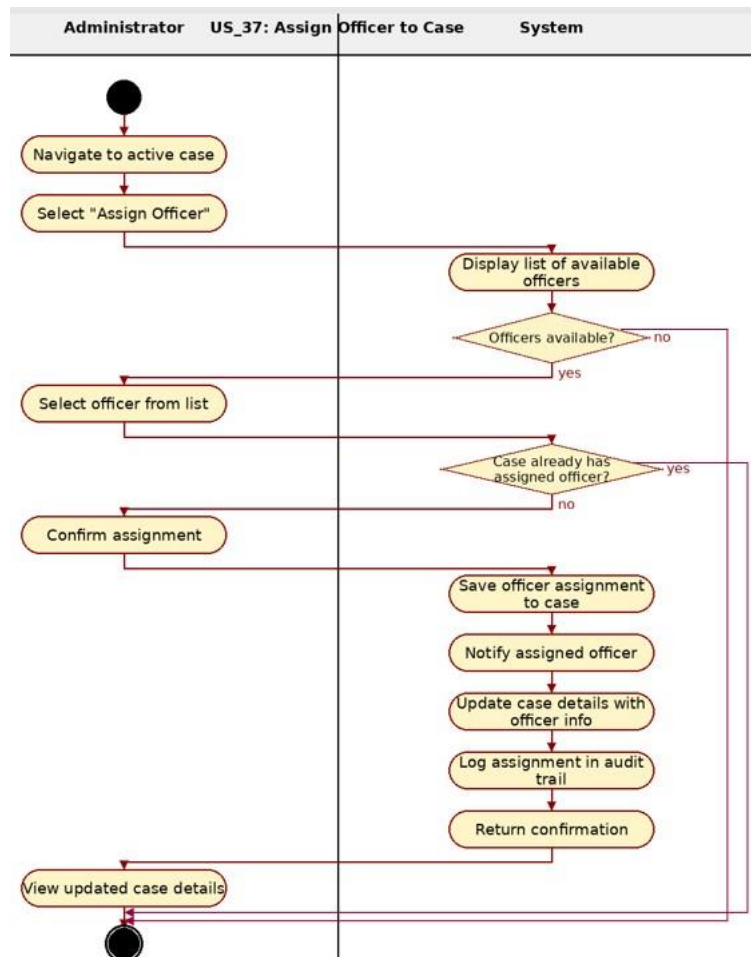


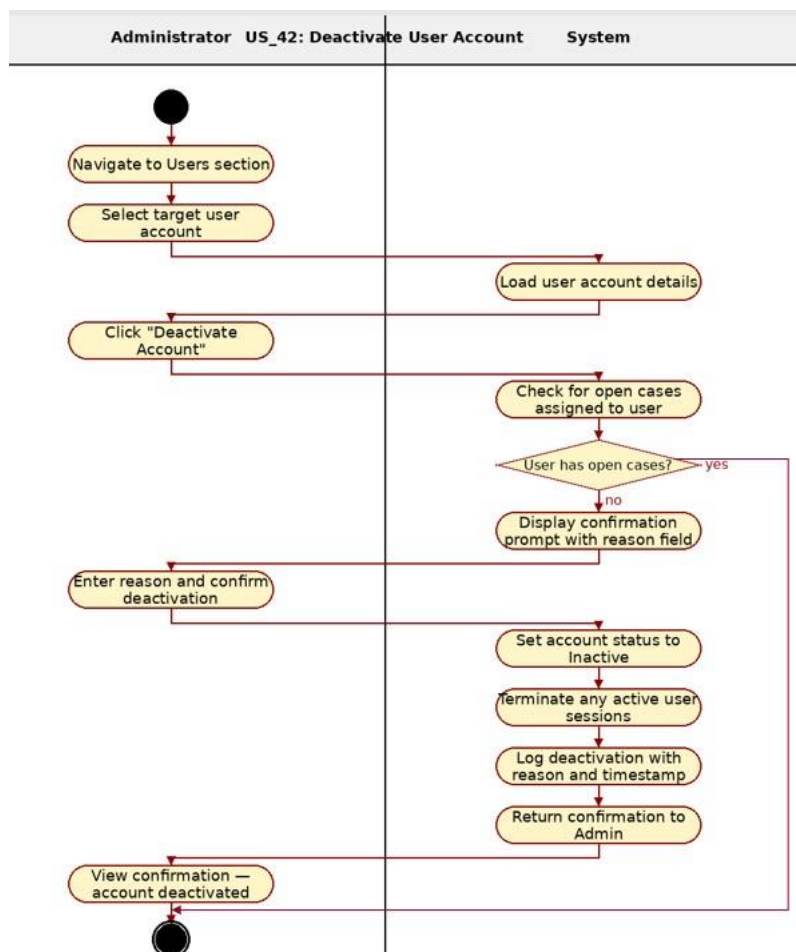
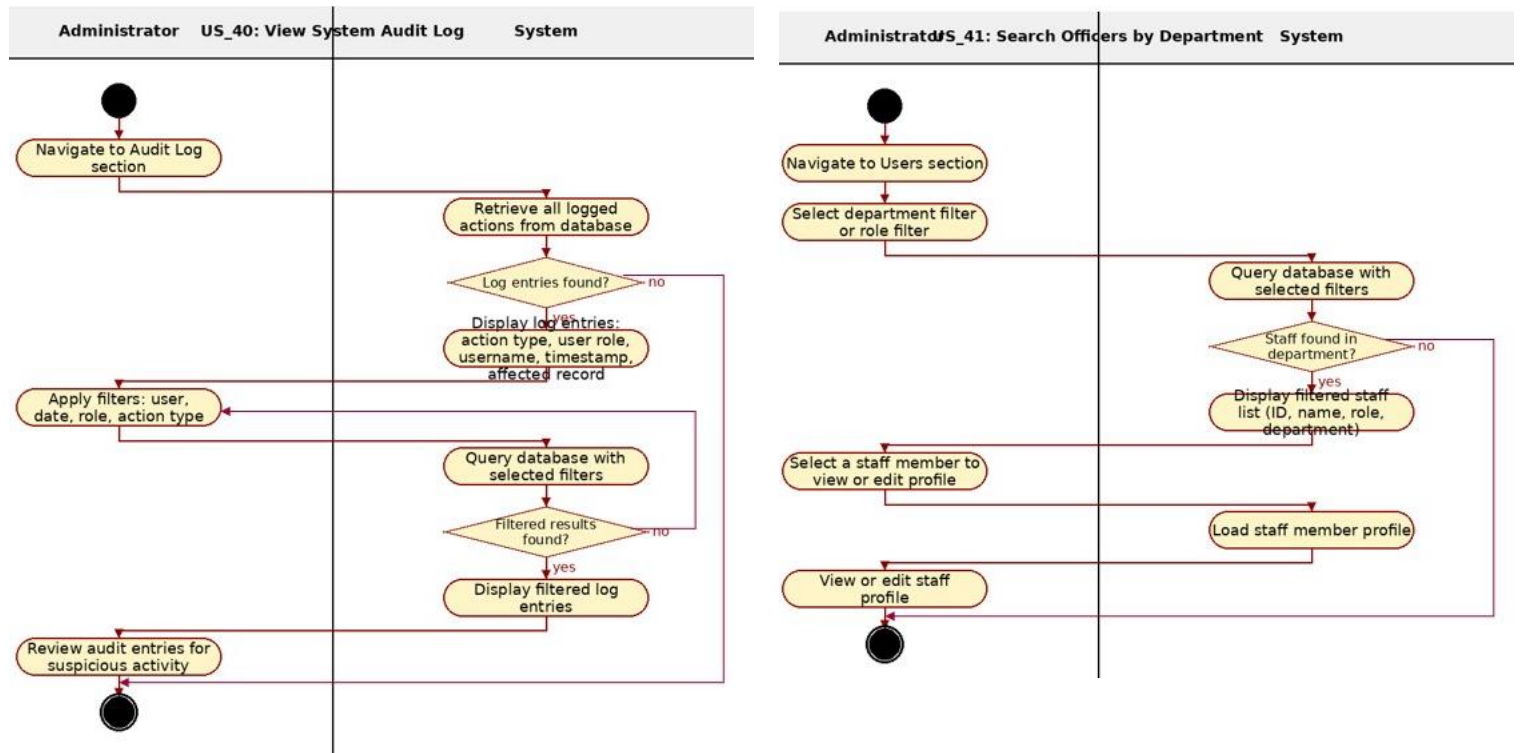








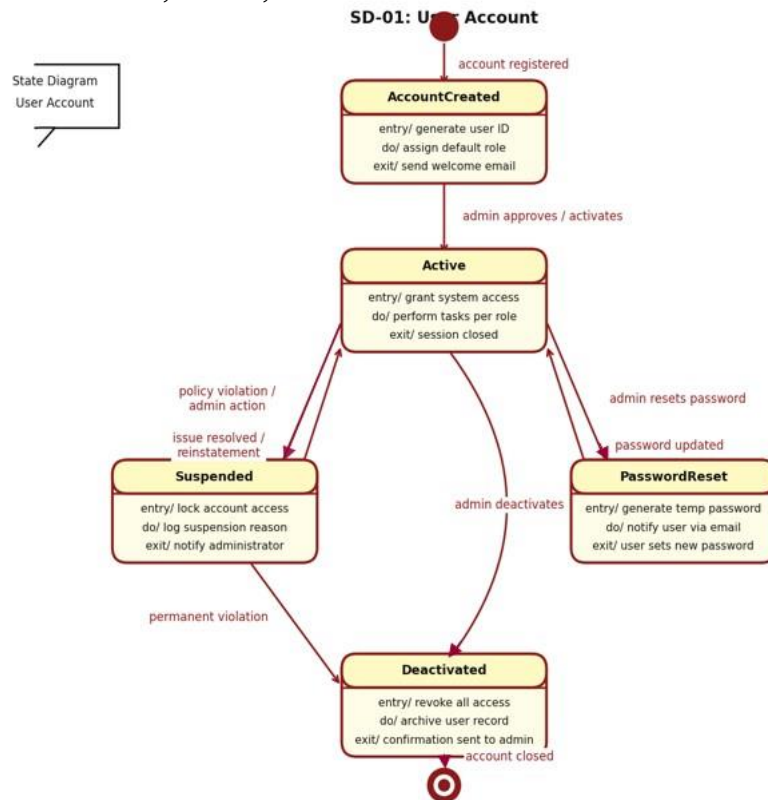




4.2.3 State Diagrams

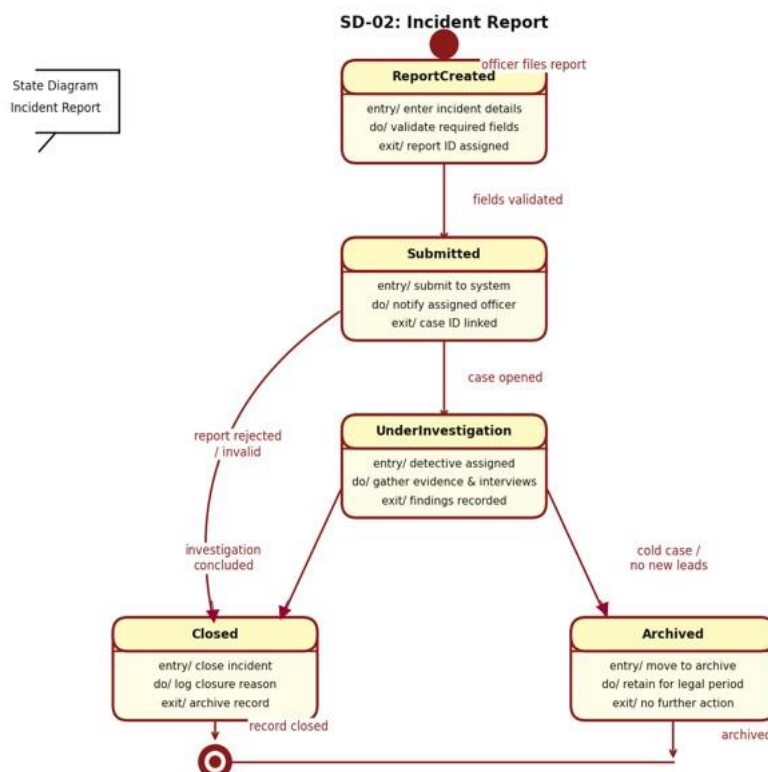
SD-01: User Account

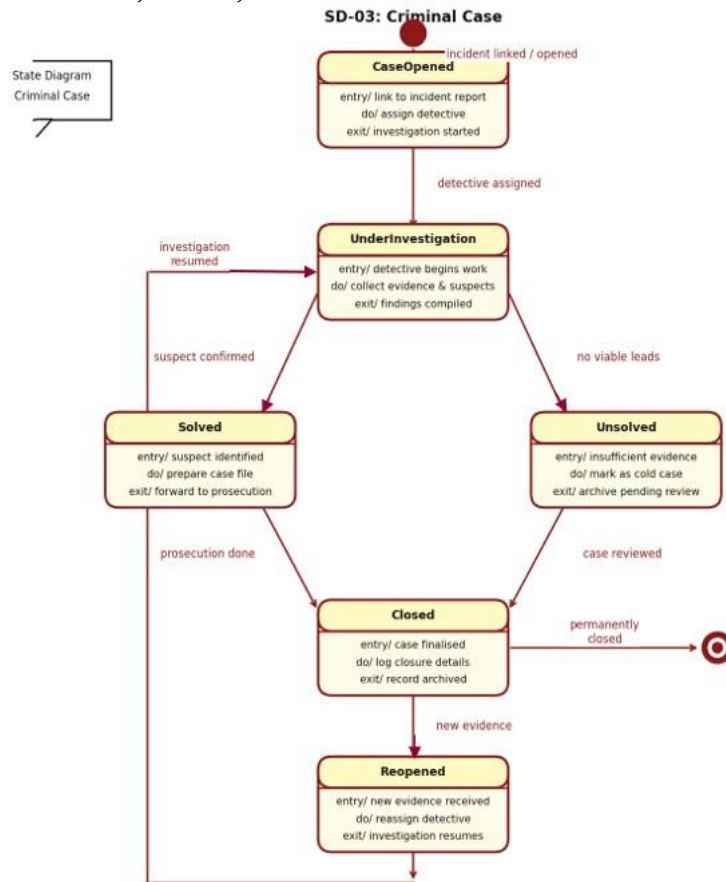
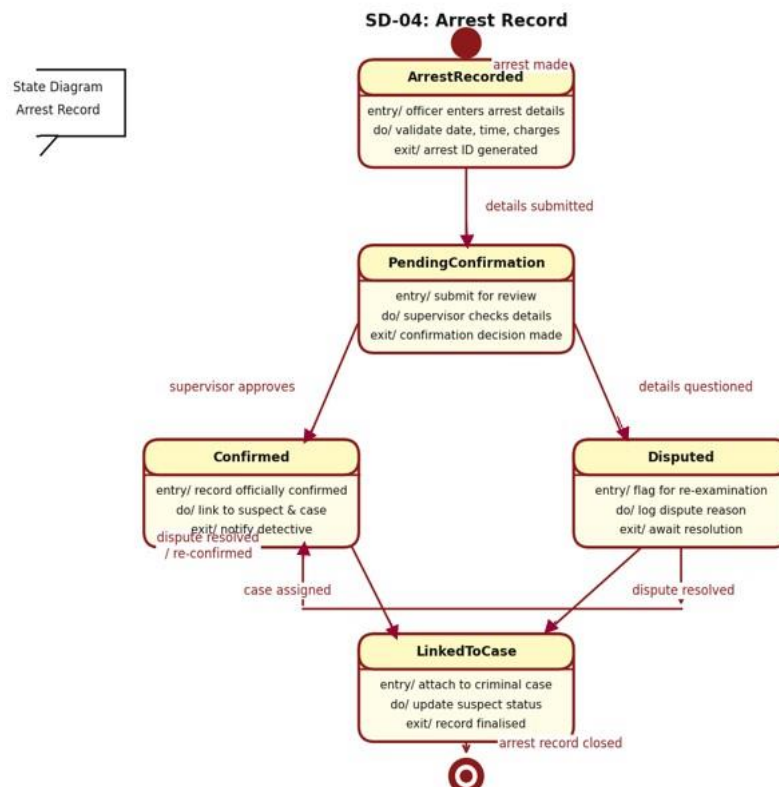
Related User Scenarios: US-35, US-39, US-42

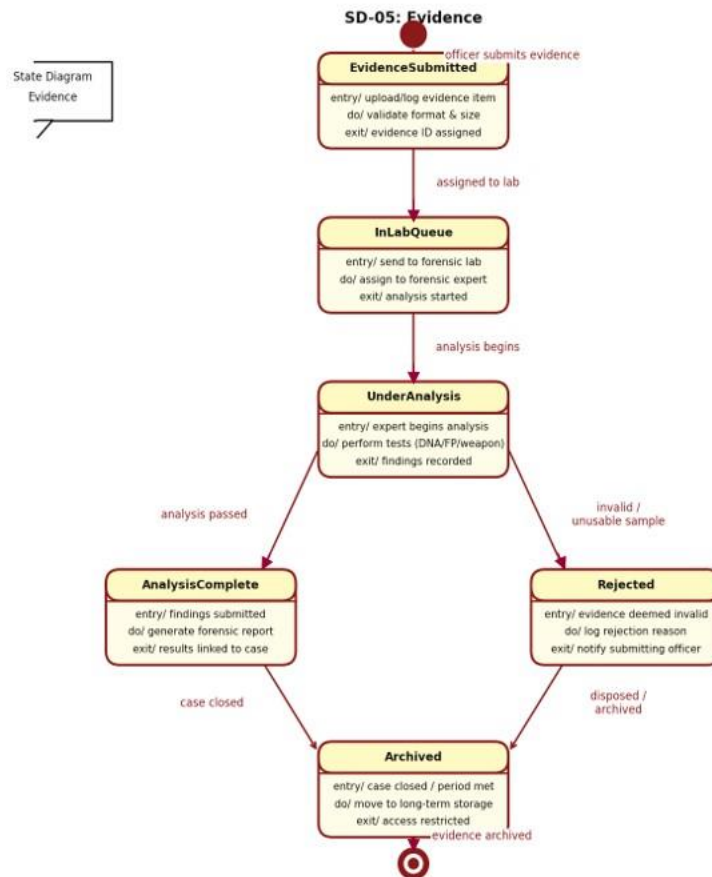
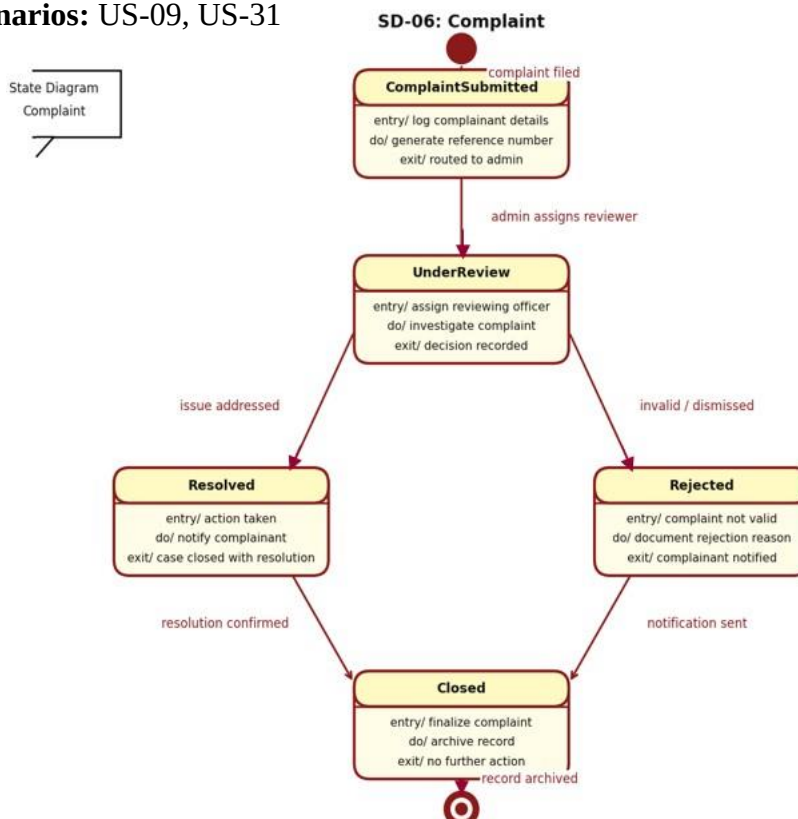


SD-02: Incident Report

Related User Scenarios: US-04, US-30

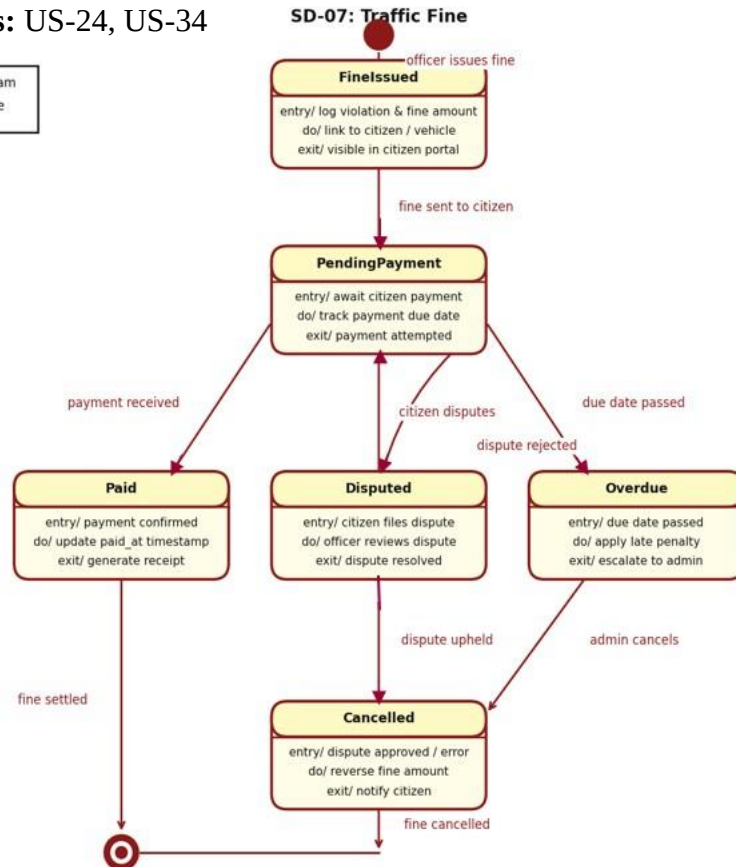


SD-03: Criminal Case**Related User Scenarios:** US-11, US-14, US-16**SD-04: Arrest Record****Related User Scenarios:** US-07

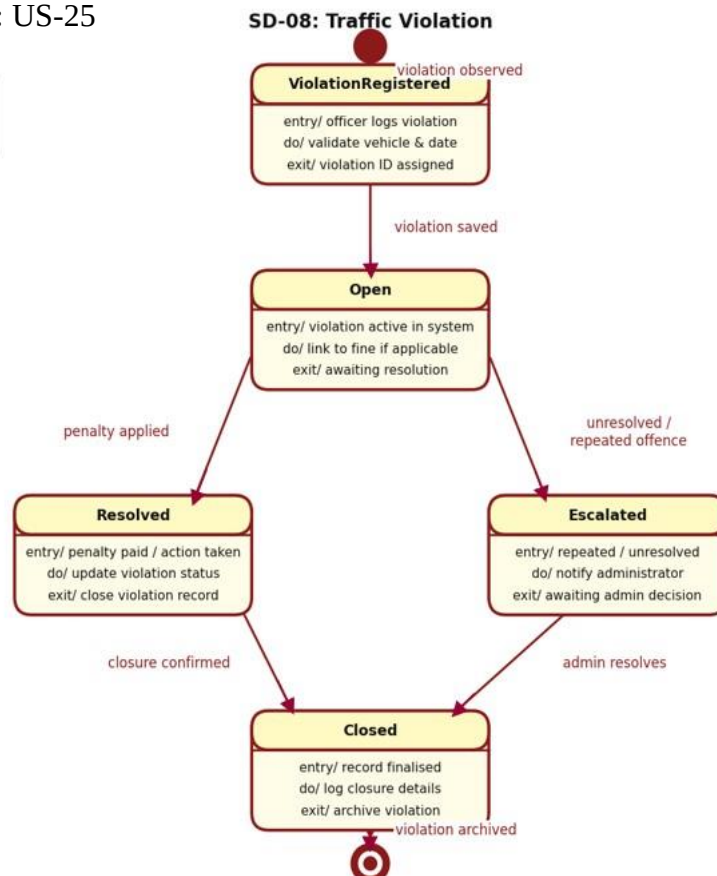
SD-05: Evidence**Related User Scenarios:** US-06, US-17–US-22**SD-06: Complaint****Related User Scenarios:** US-09, US-31

SD-07: Traffic Fine**Related User Scenarios:** US-24, US-34

State Diagram
Traffic Fine

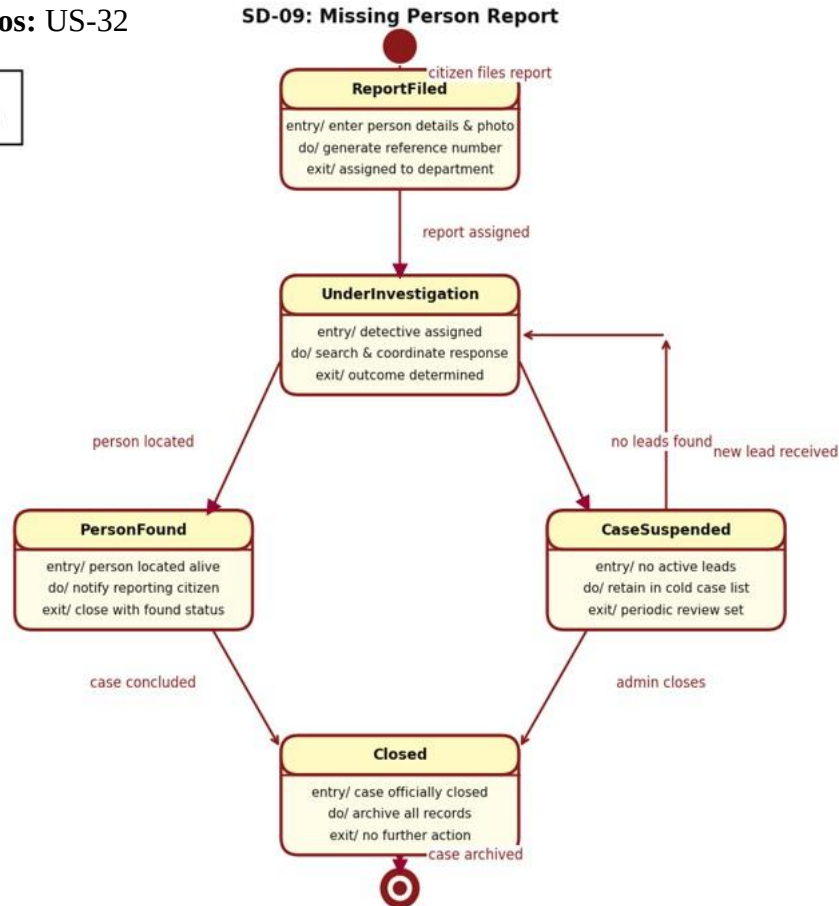
**SD-08: Traffic Violation****Related User Scenarios:** US-25

State Diagram
Traffic Violation

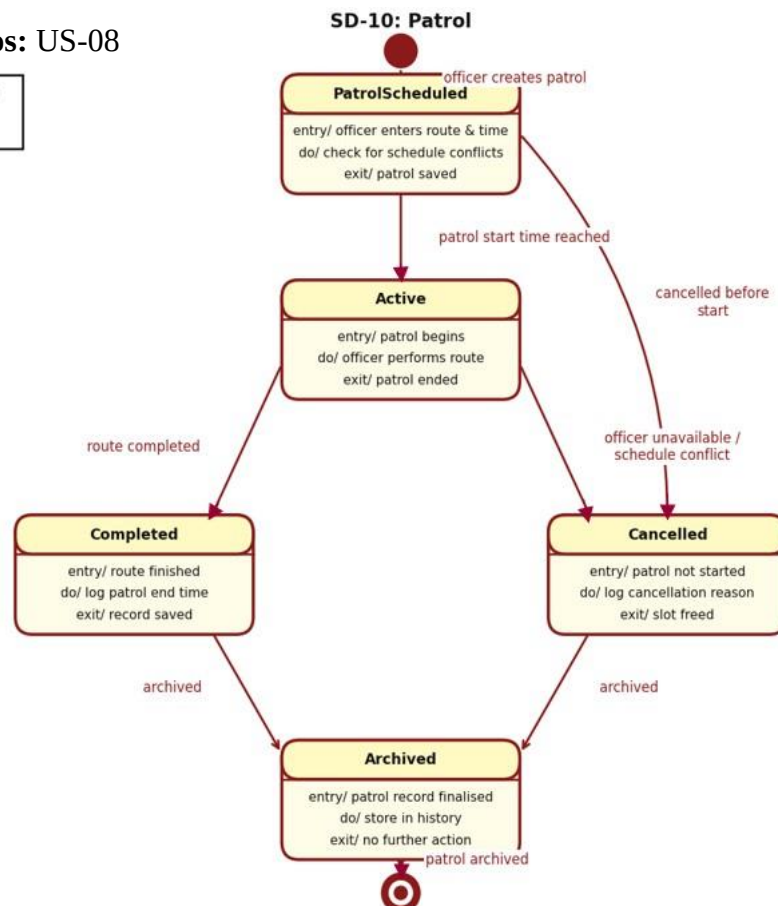


SD-09: Missing Person Report**Related User Scenarios: US-32**

State Diagram
Missing Person
Report

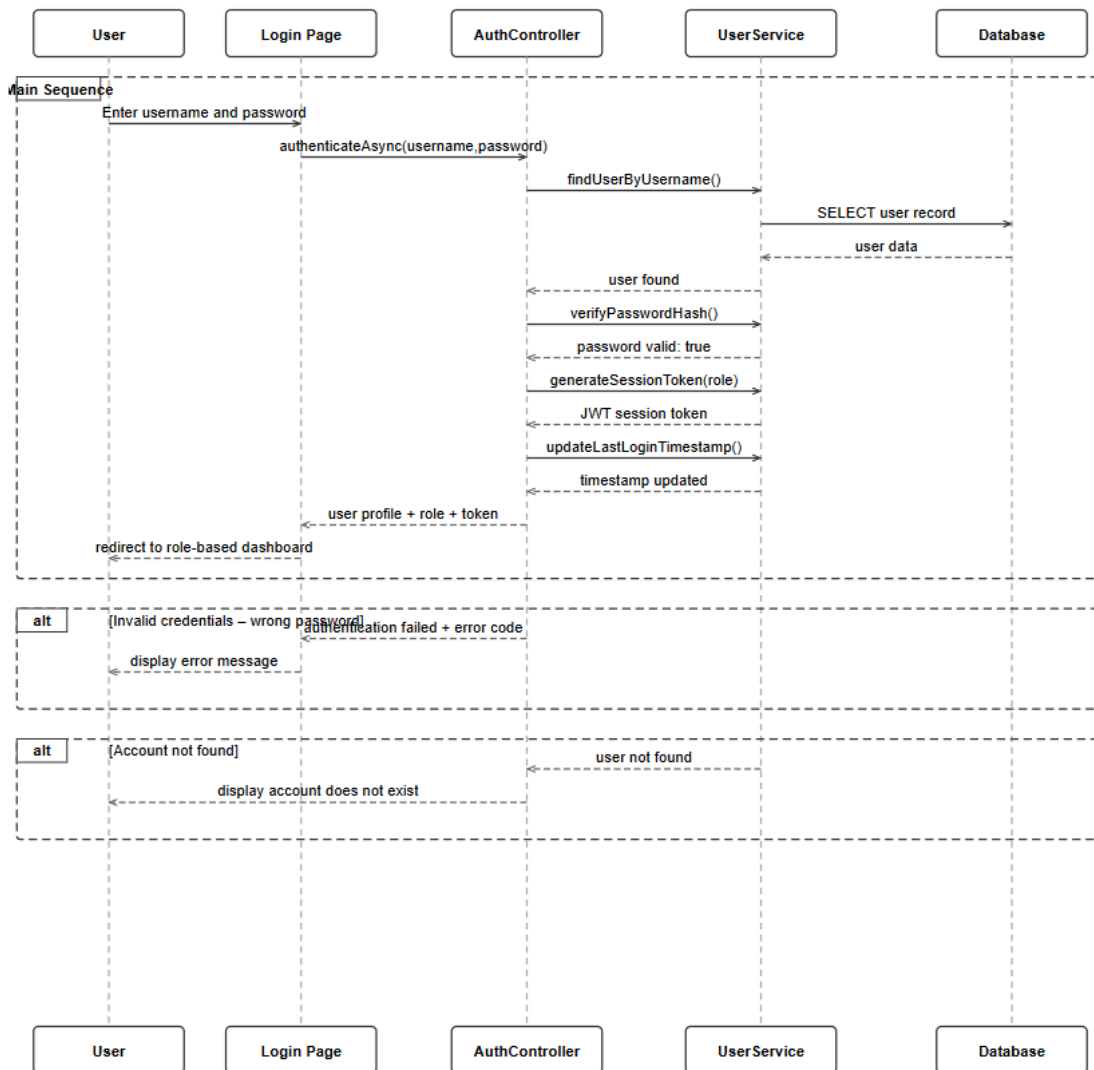
**SD-10: Patrol****Related User Scenarios: US-08**

State Diagram
Patrol

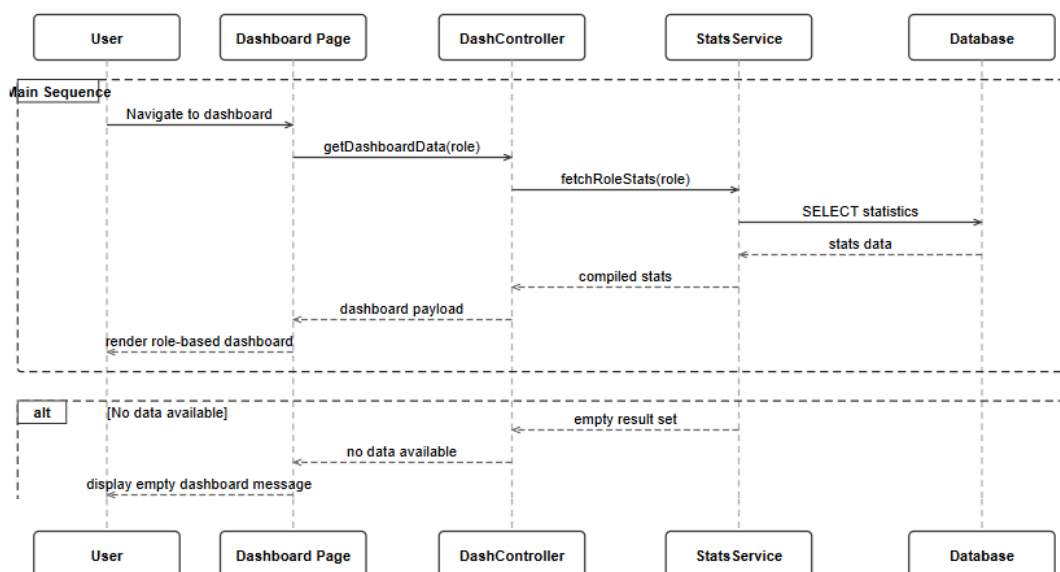


4.2.4 Sequence Diagrams

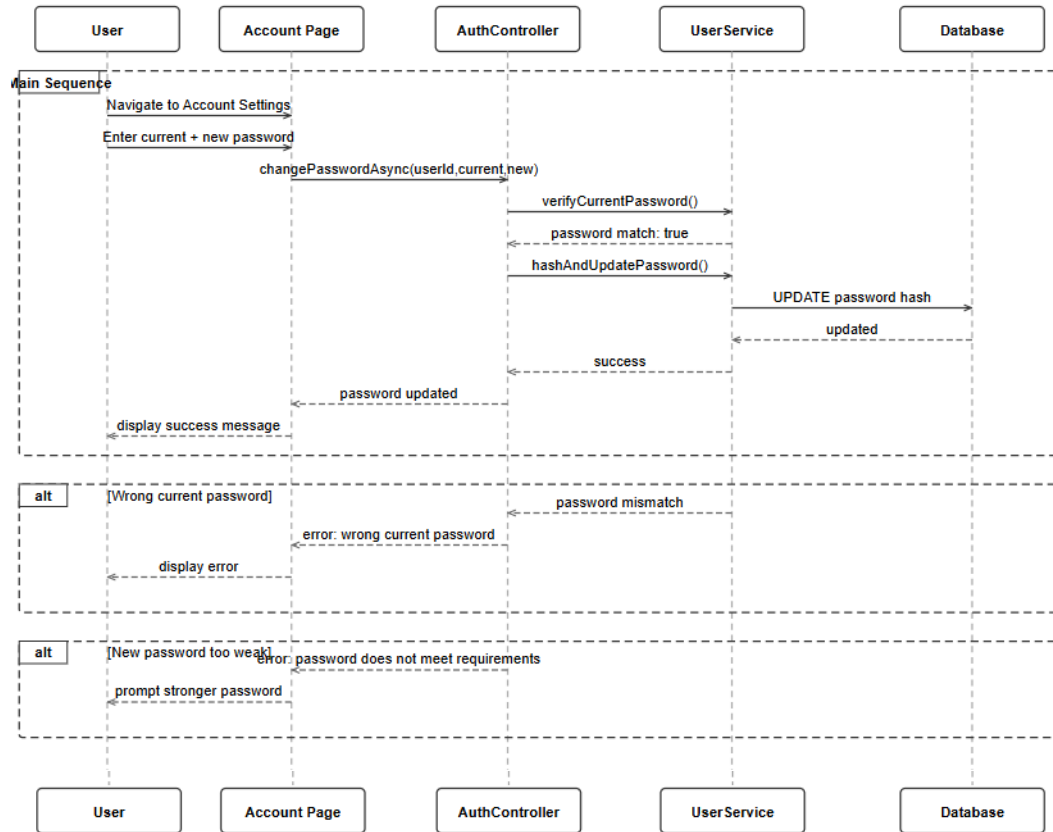
SeqD_01 – User Authentication (Login)



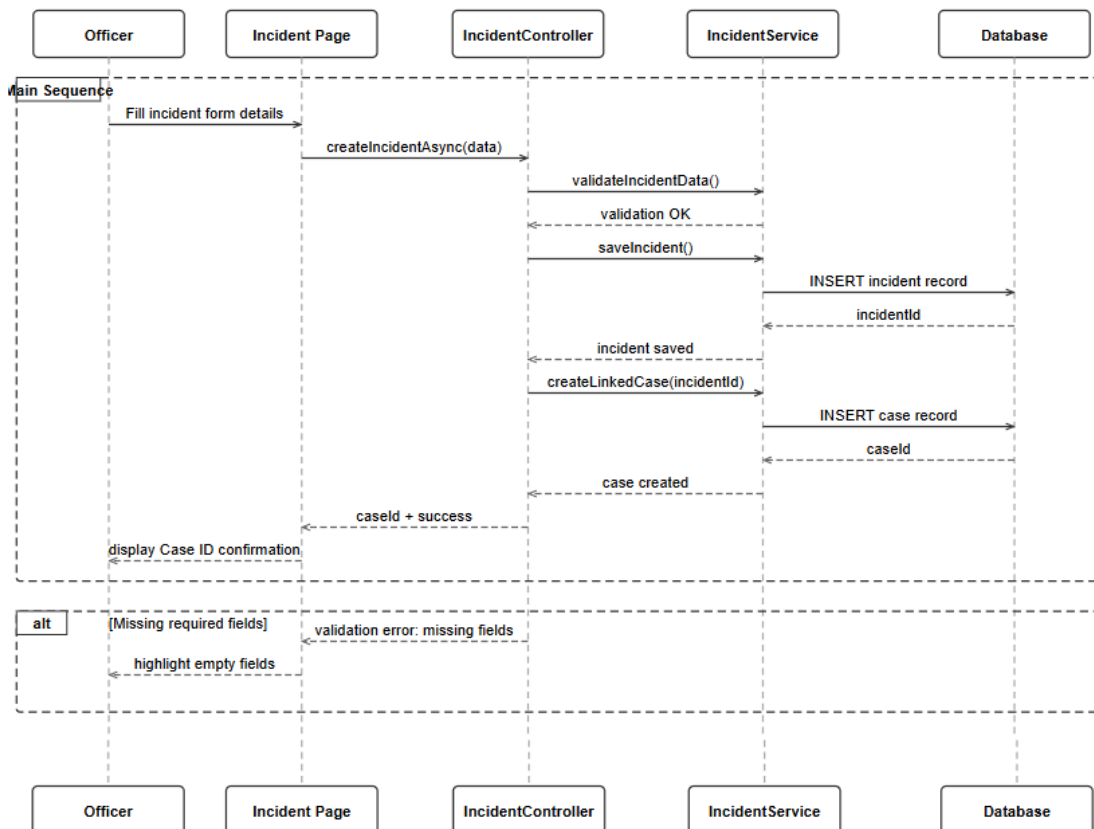
SeqD_02 – View Dashboard



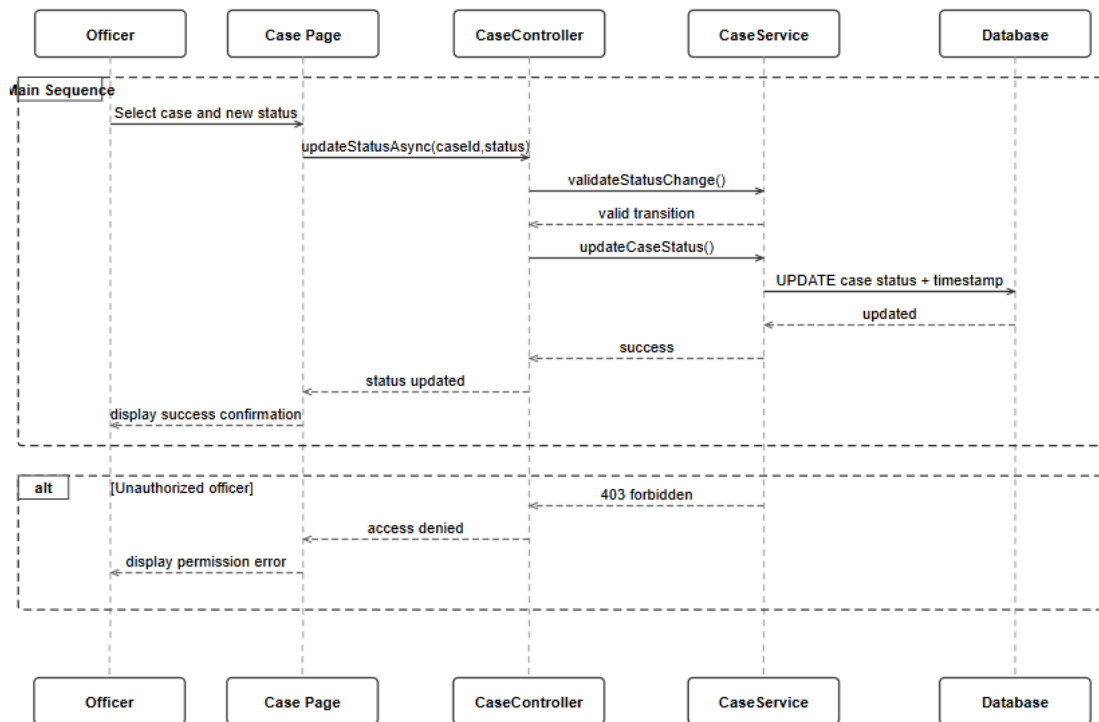
SeqD_03 – Change Own Password



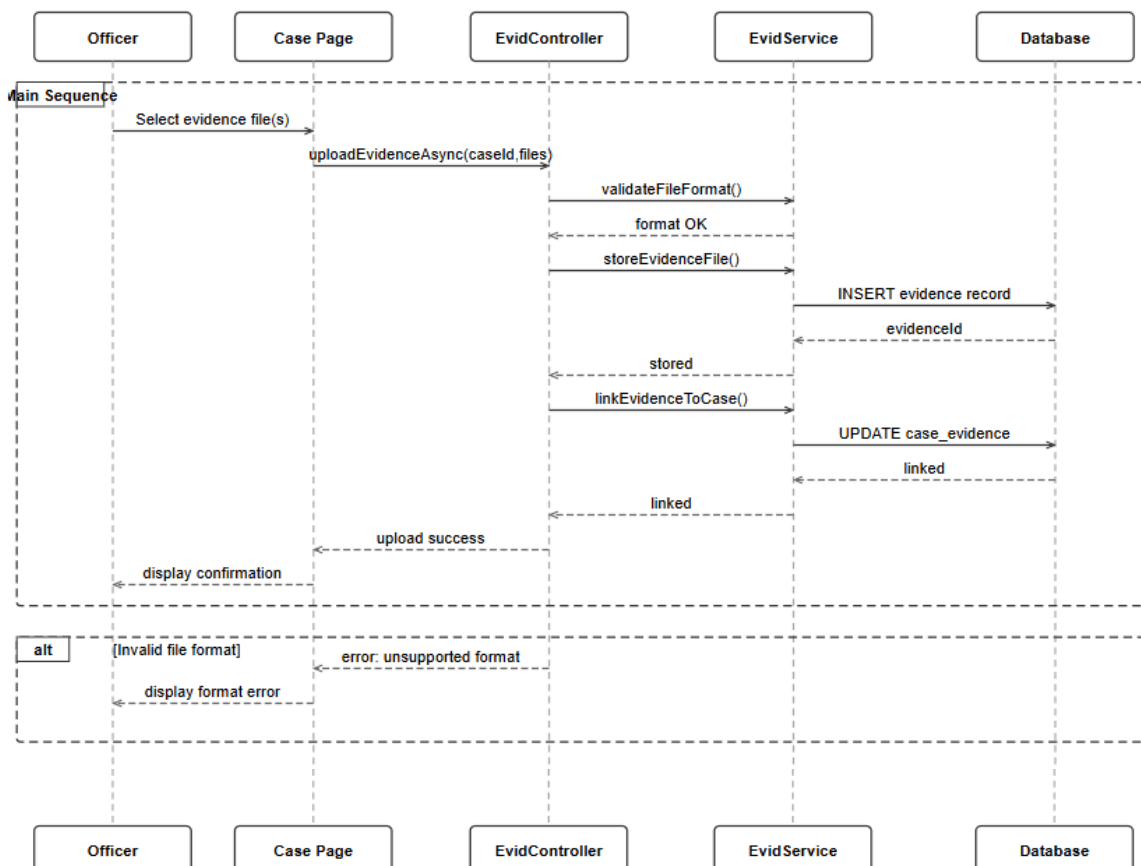
SeqD_04 – Create Incident Report



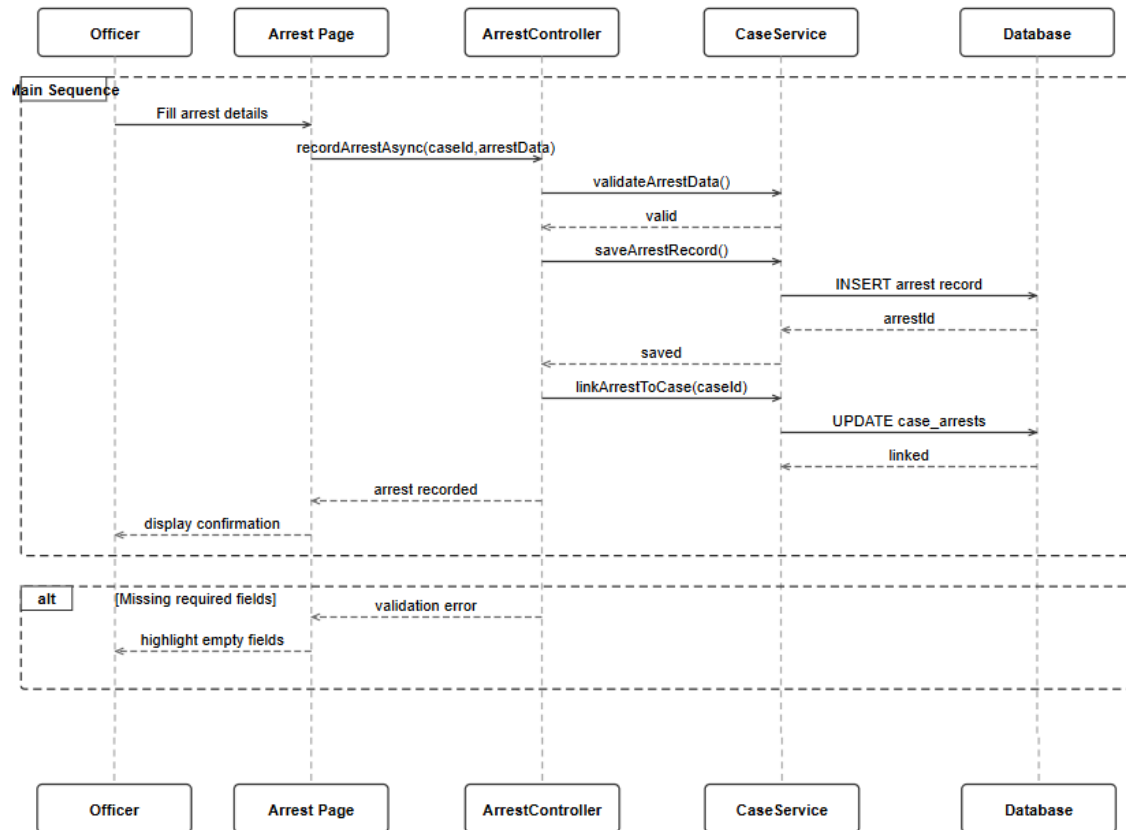
SeqD_05 – Update Case Status



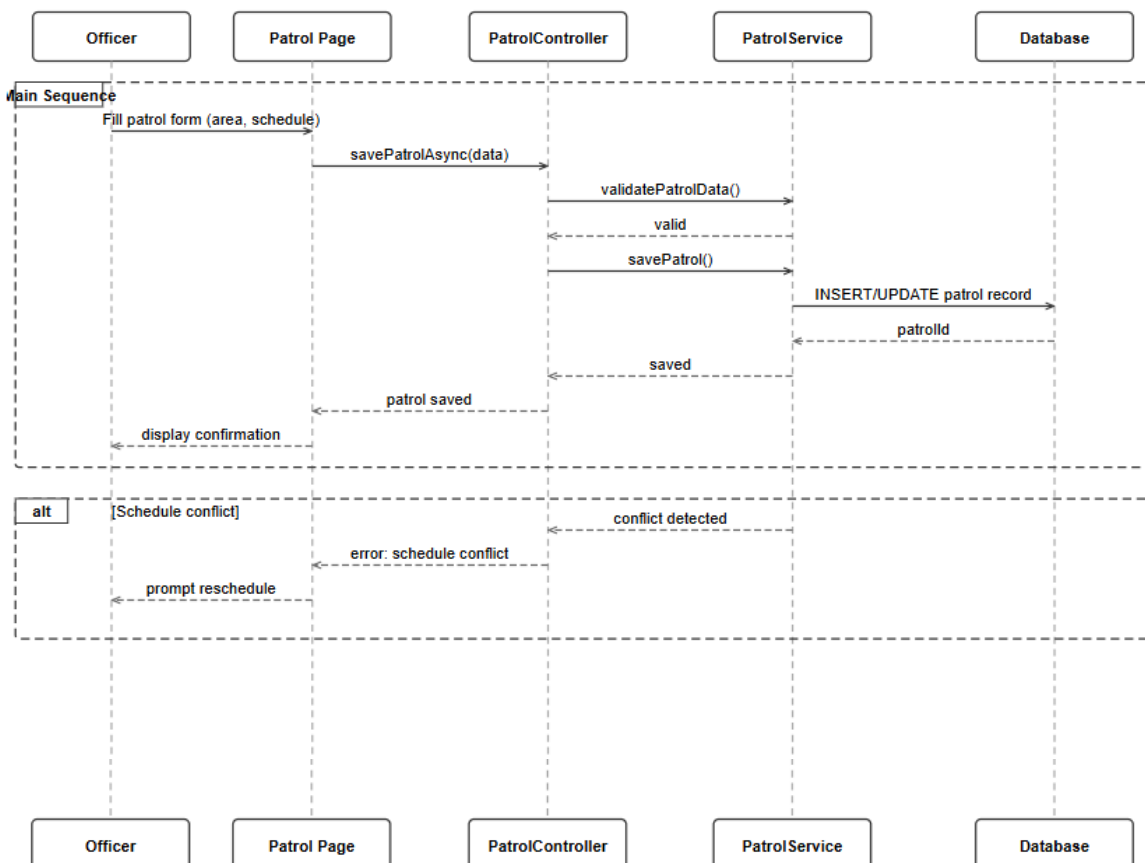
SeqD_06 – Upload Evidence



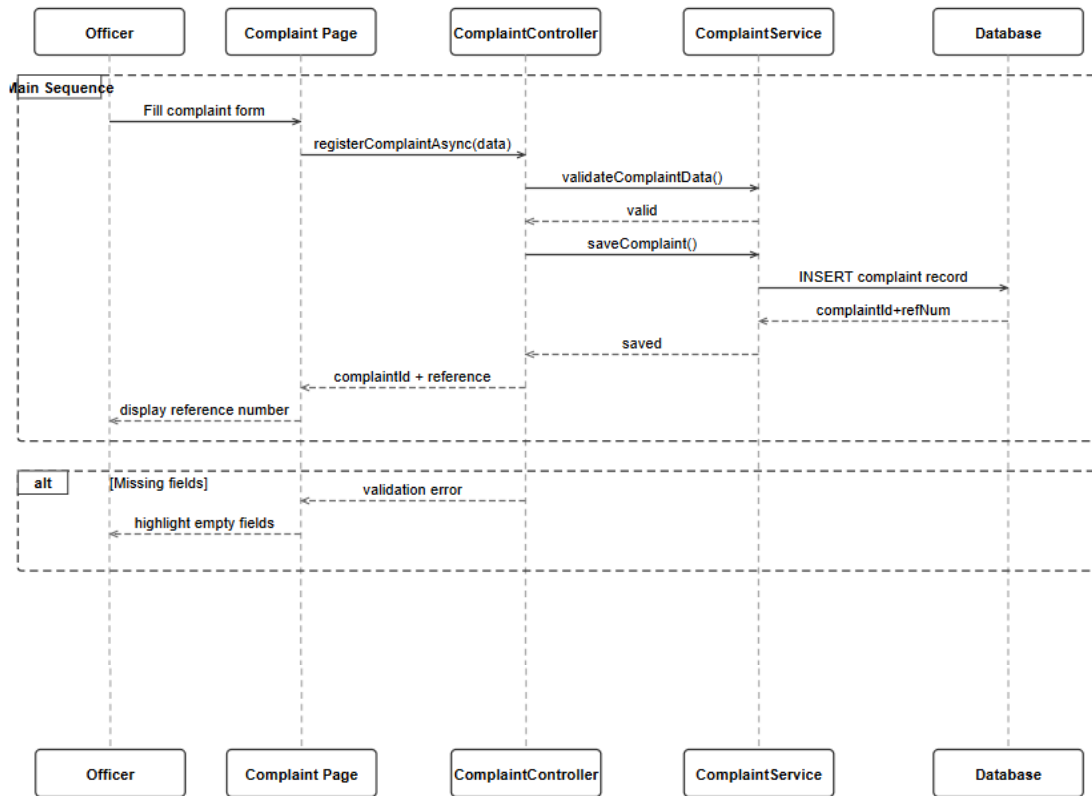
SeqD_07 – Record Arrest Information



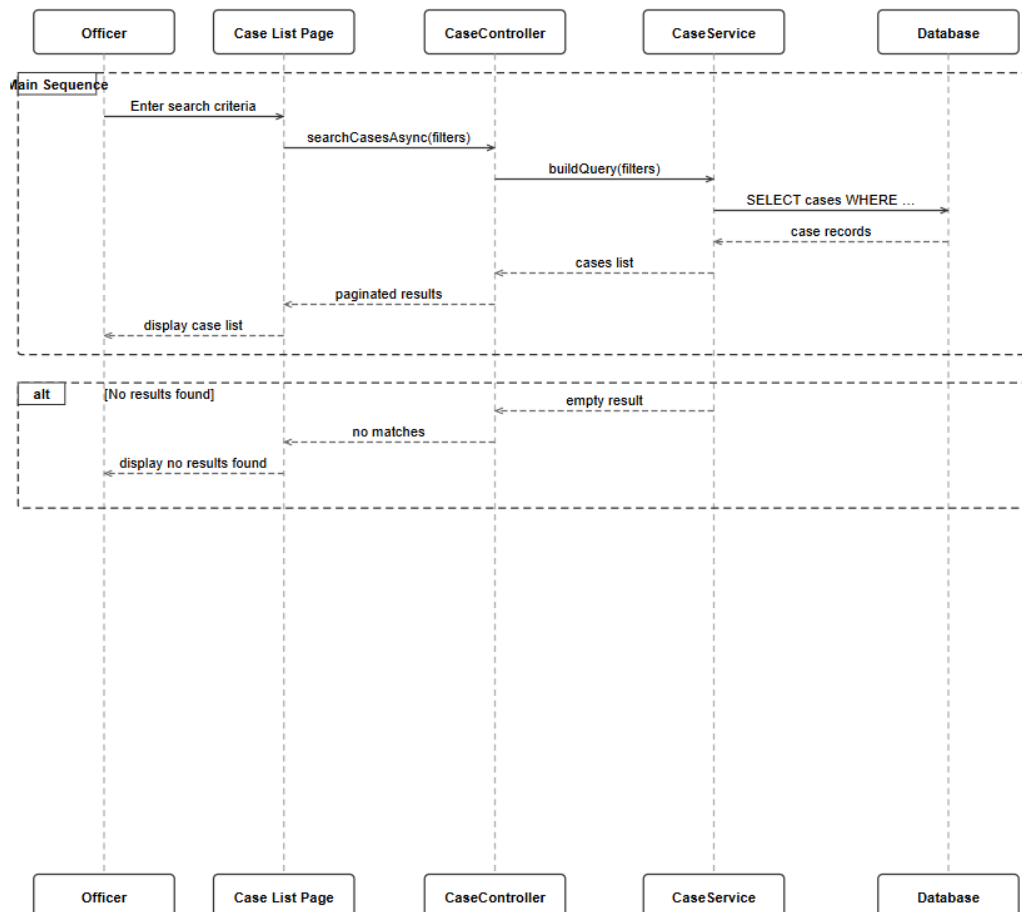
SeqD_08 – Manage Patrols



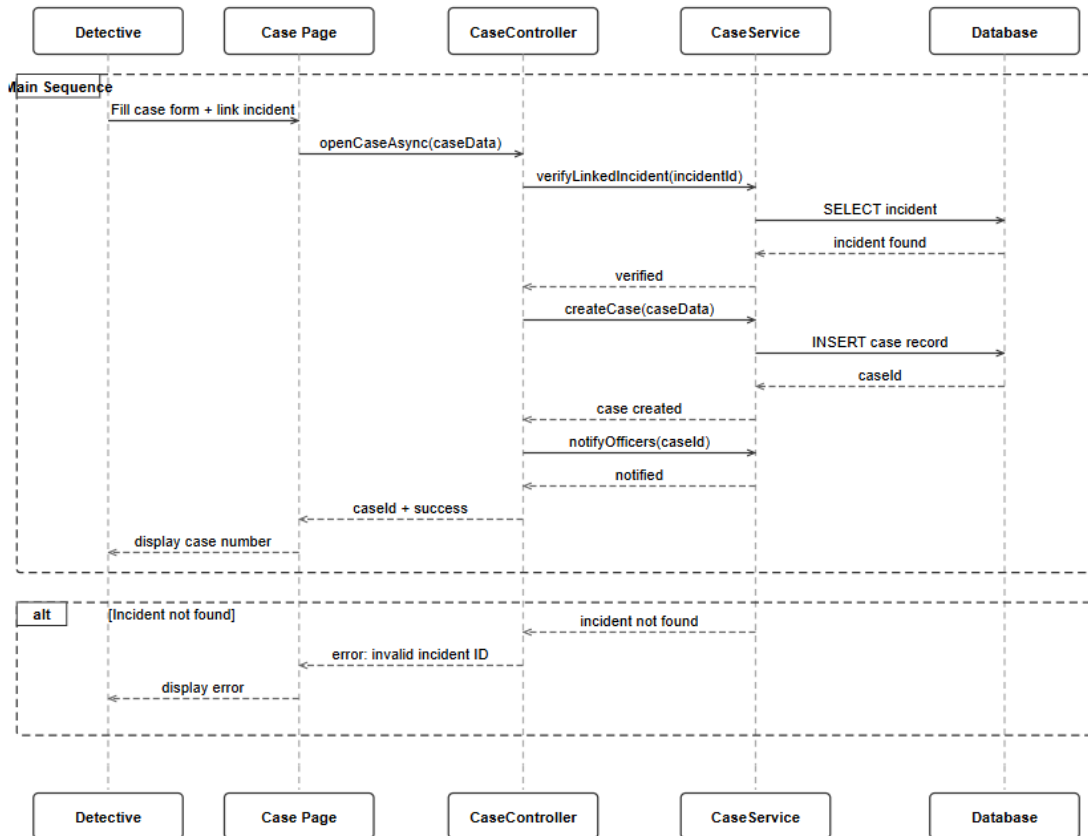
SeqD_09 – Register Complaint



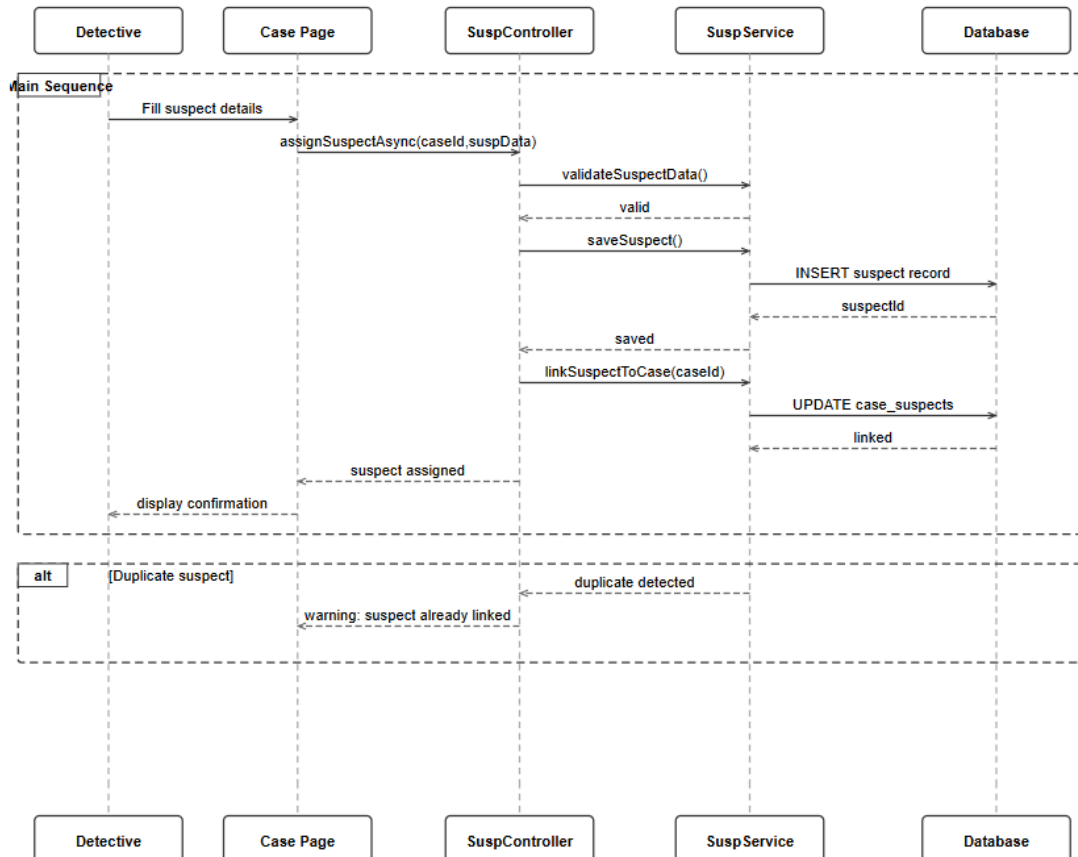
SeqD_10 – Search and Filter Cases



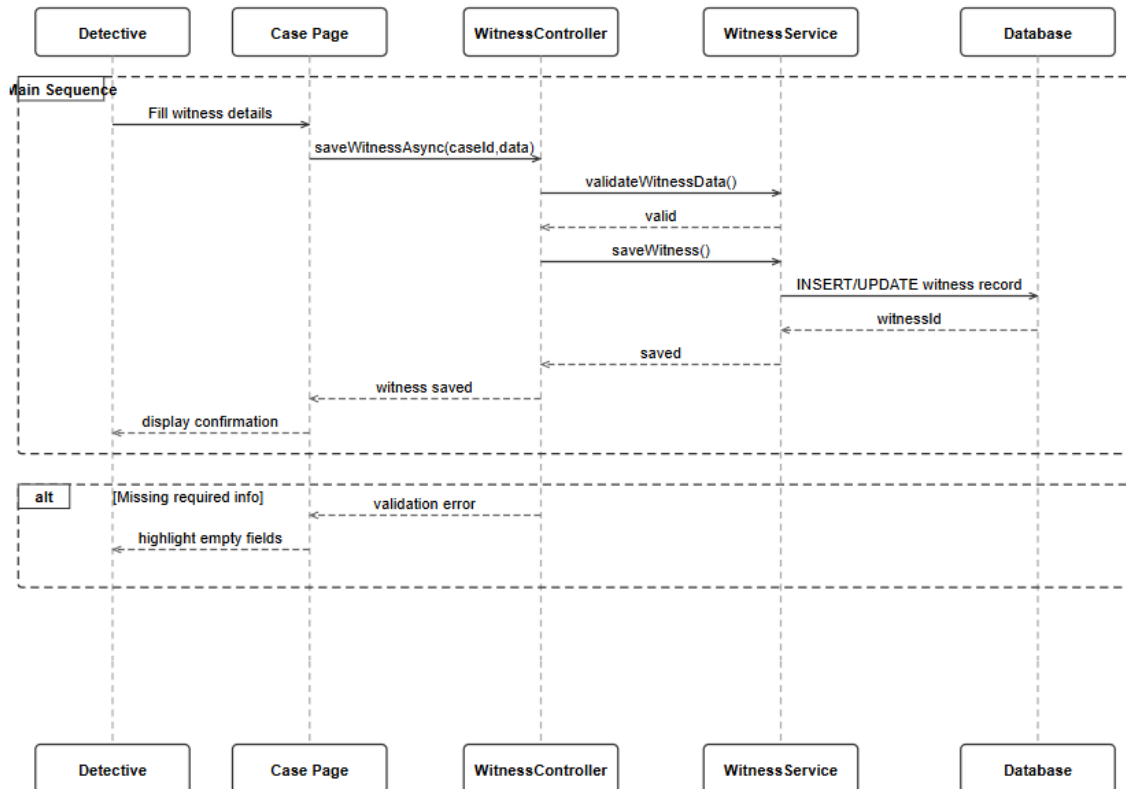
SeqD_11 – Open Criminal Case



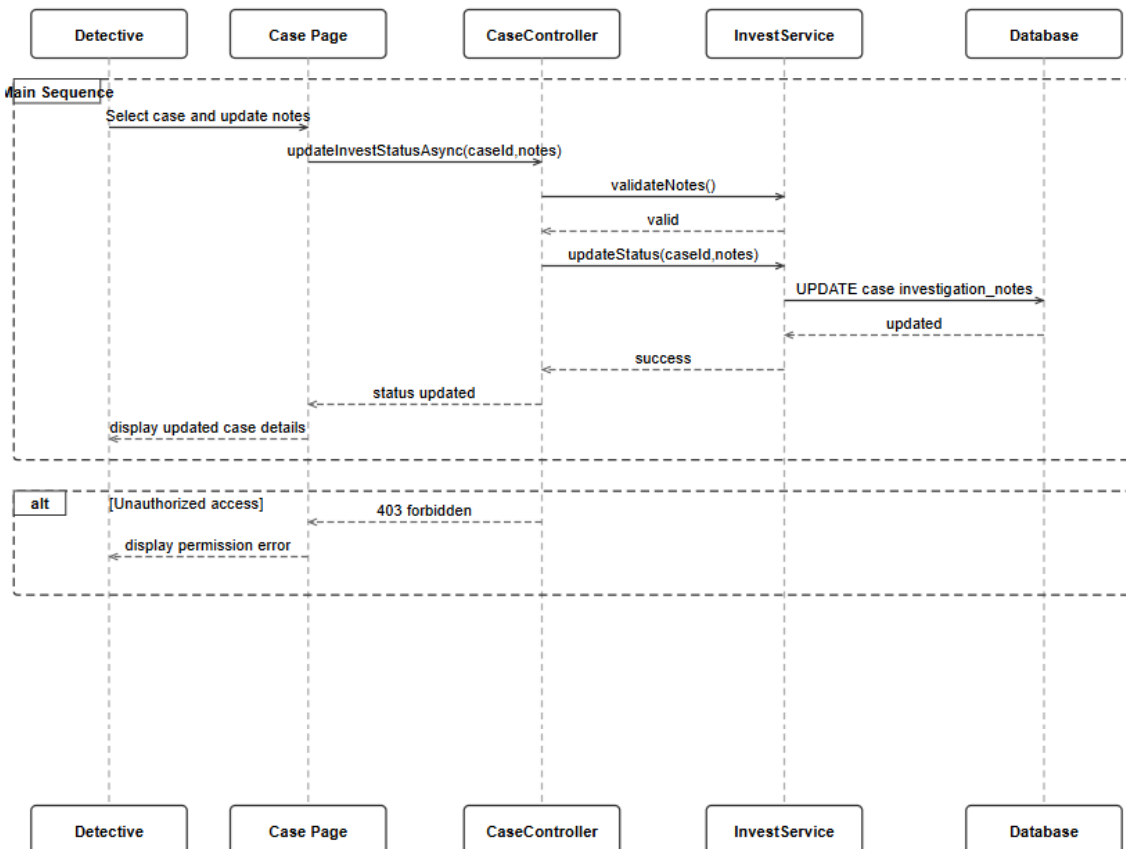
SeqD_12 – Assign Suspect to Case



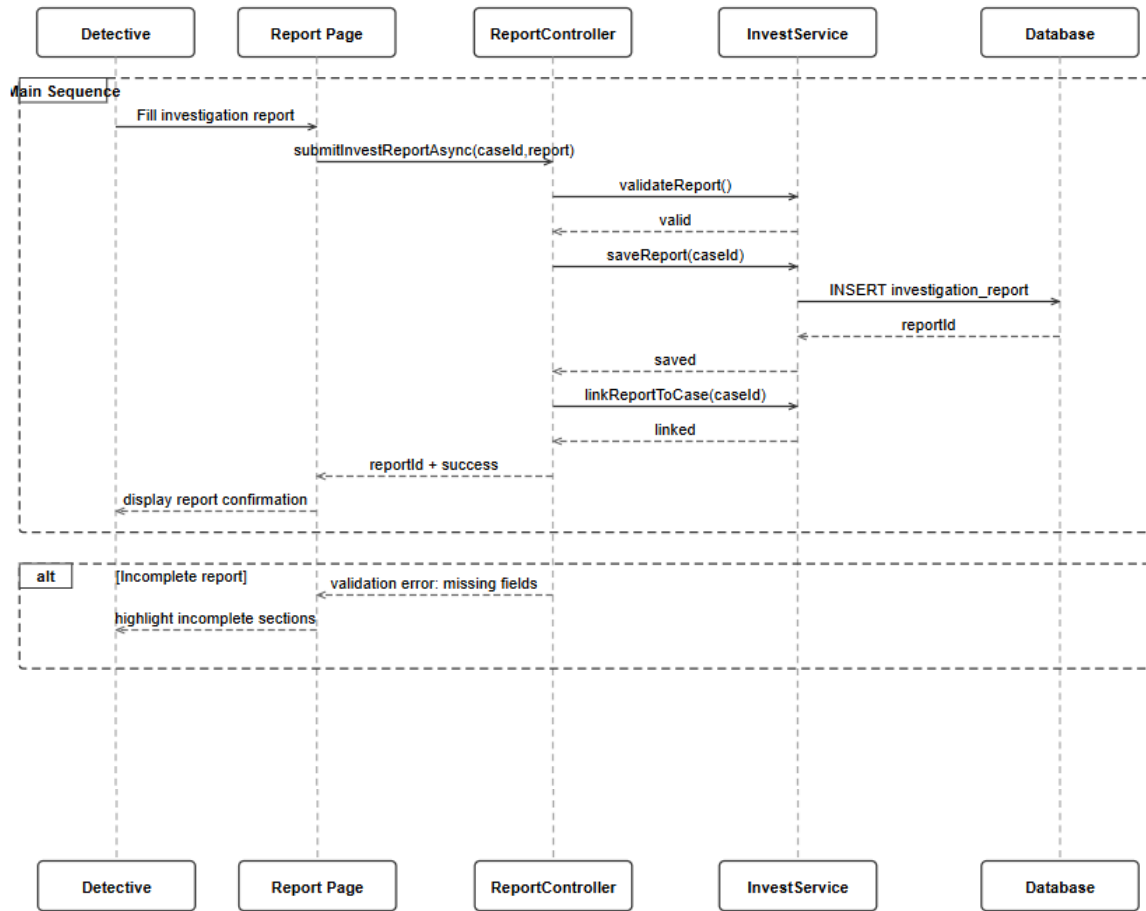
SeqD_13 – Manage Witnesses



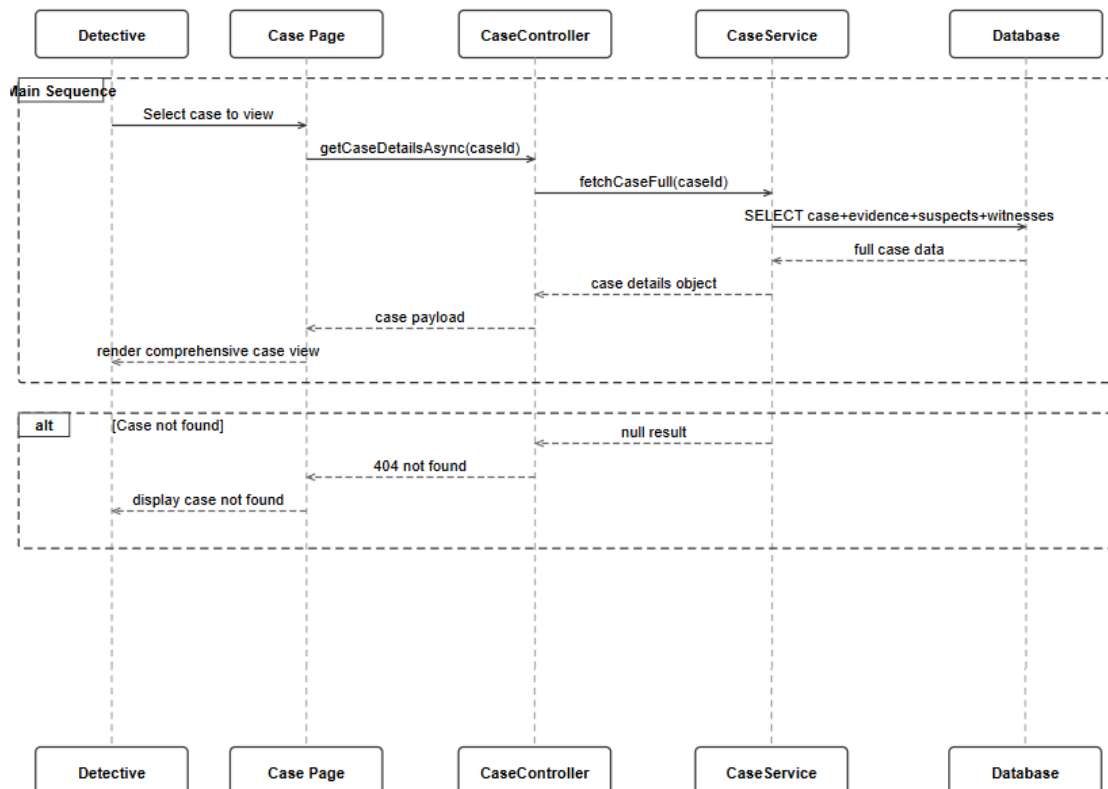
SeqD_14 – Track Investigation Status



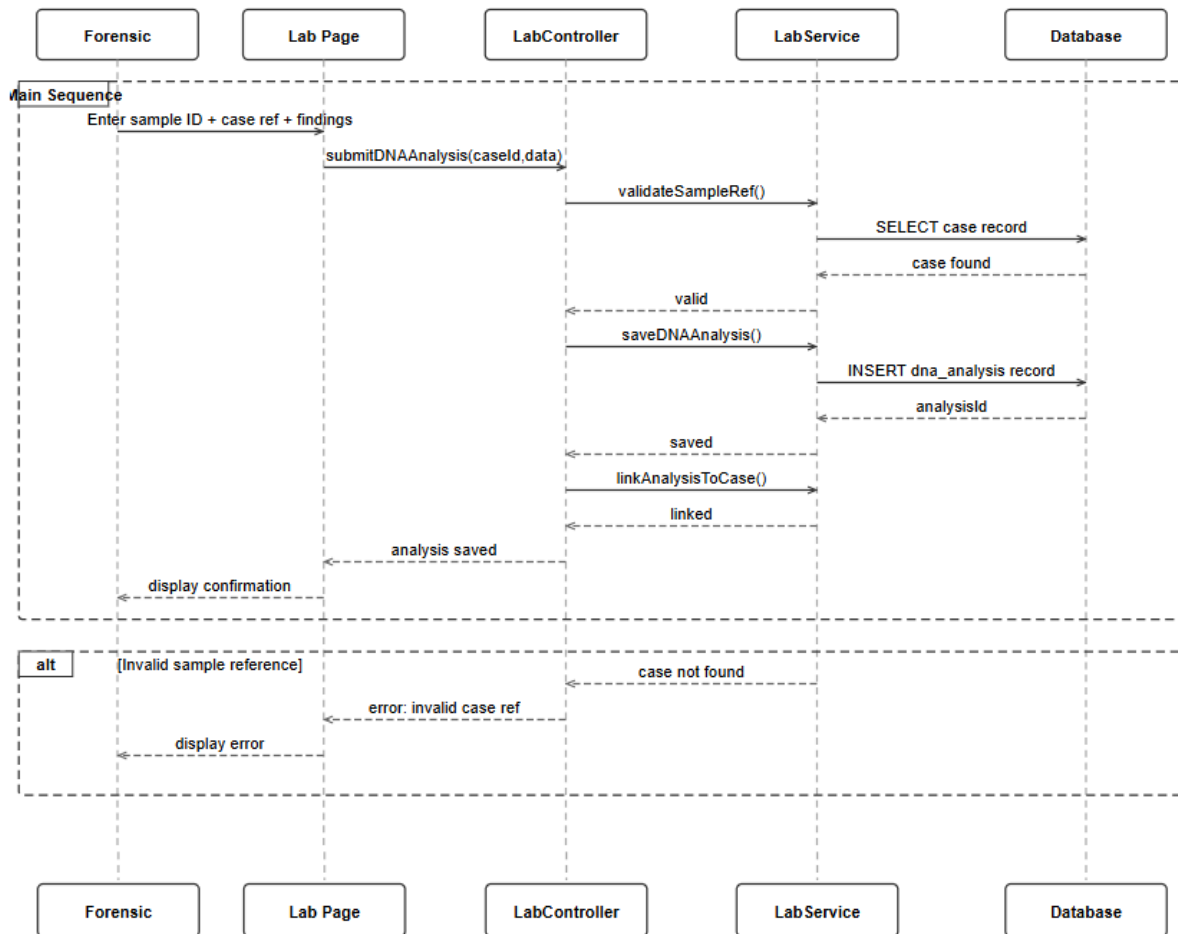
SeqD_15 – Write Investigation Report



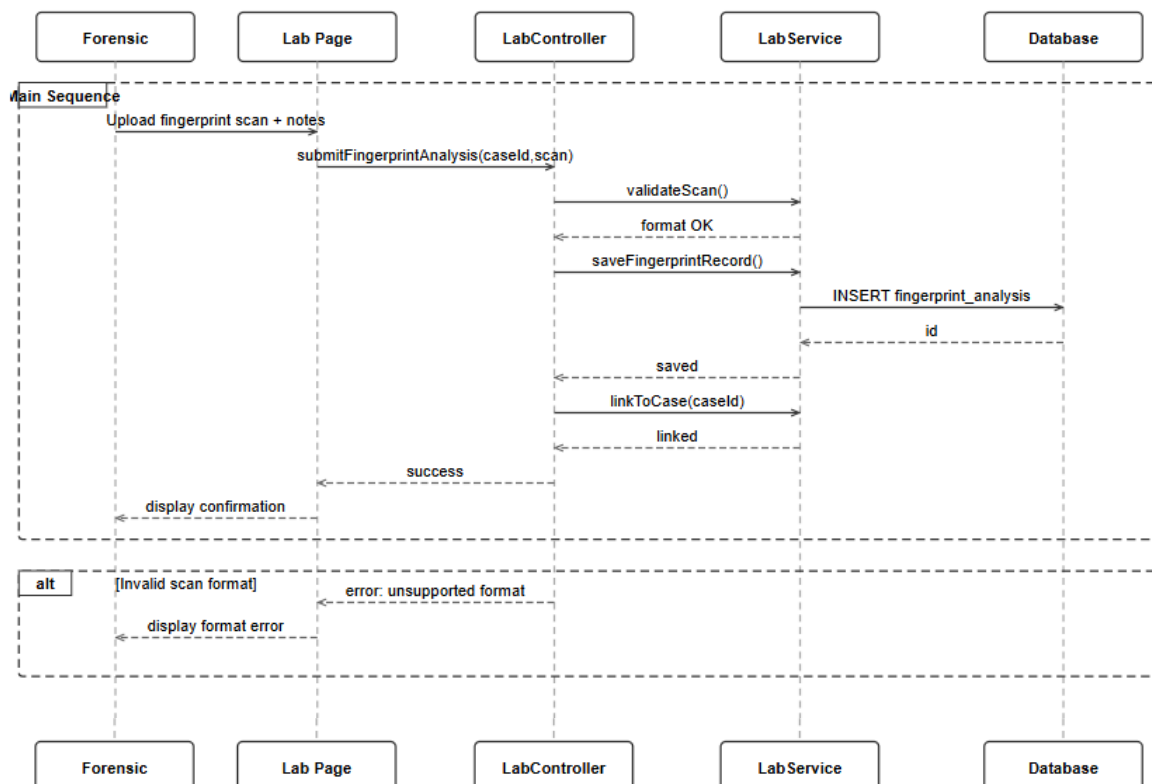
SeqD_16 – View and Analyze Case Details



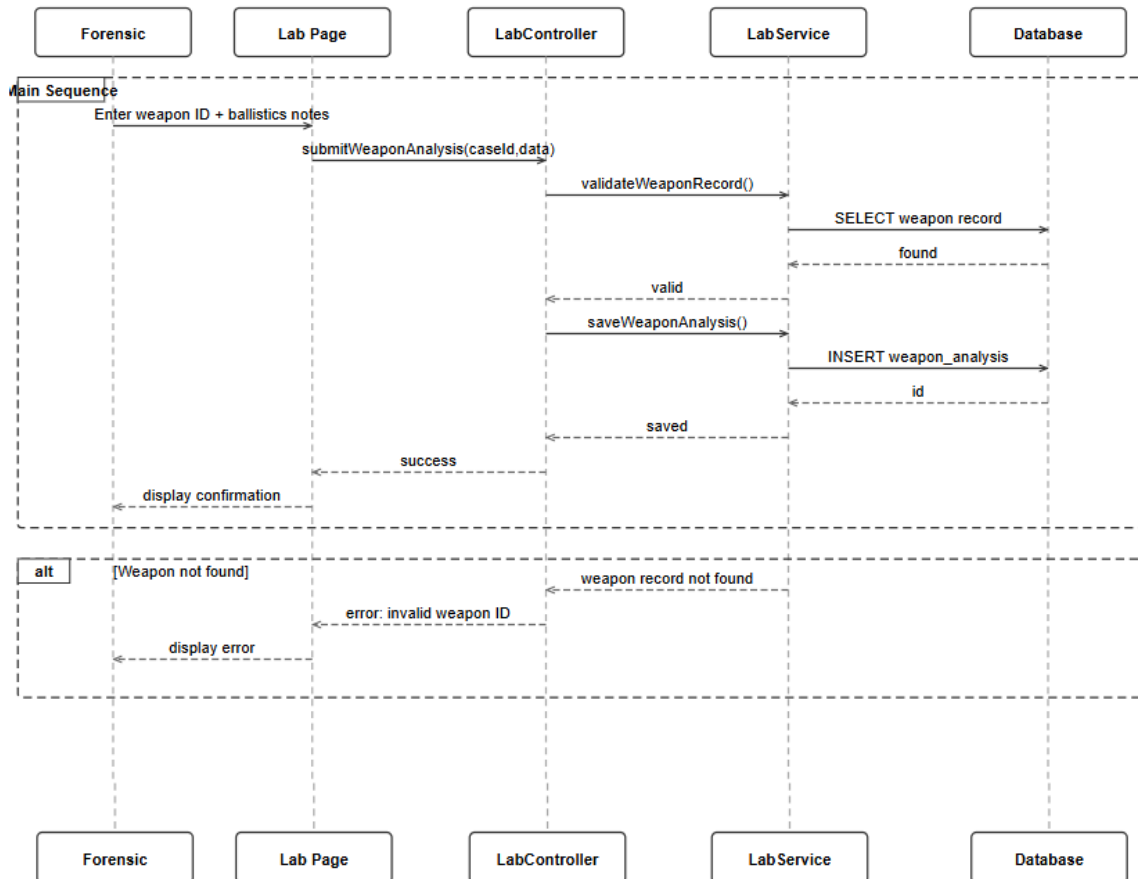
SeqD_17 – Perform DNA Analysis



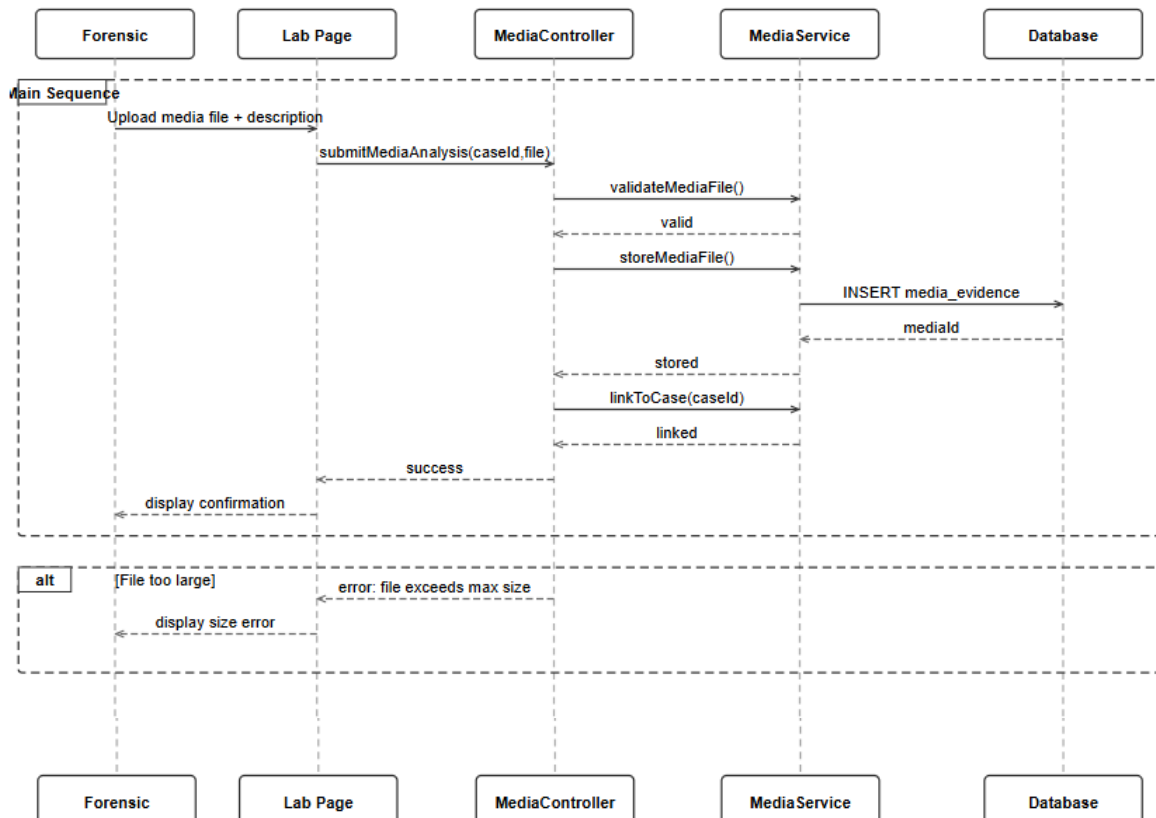
SeqD_18 – Perform Fingerprint Analysis



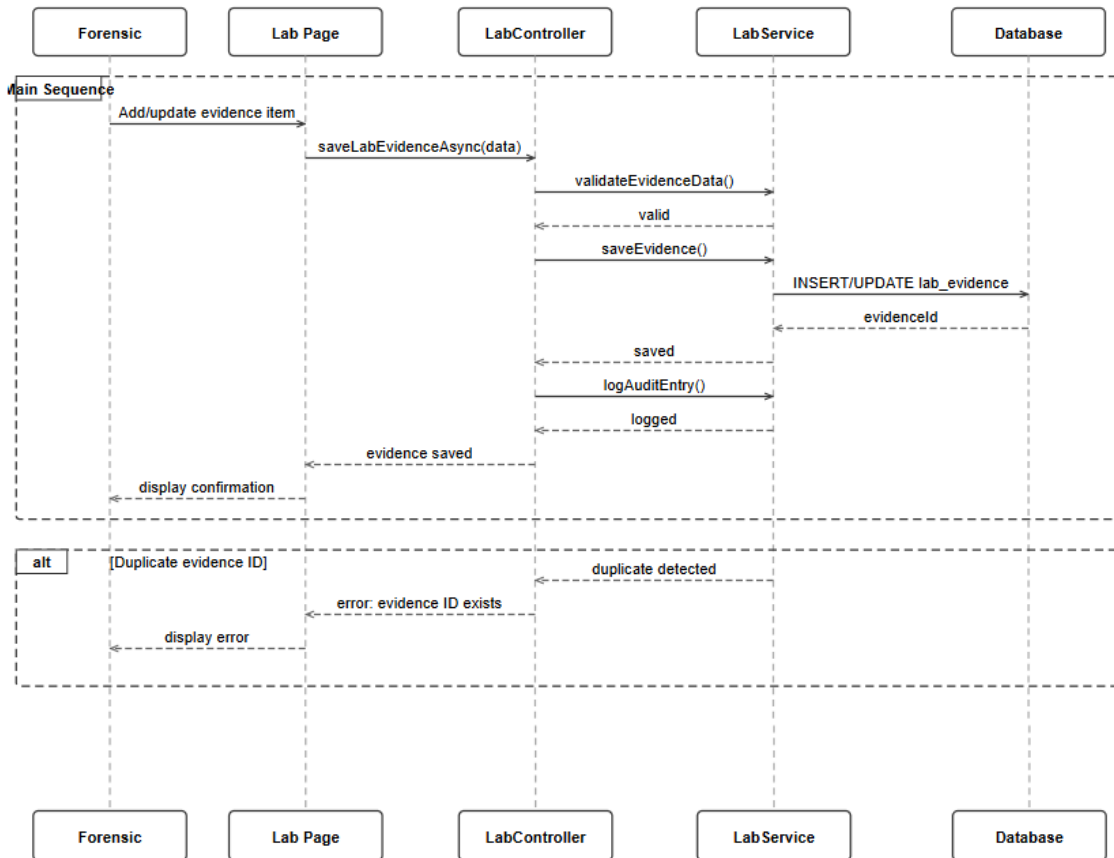
SeqD_19 – Perform Weapon Analysis



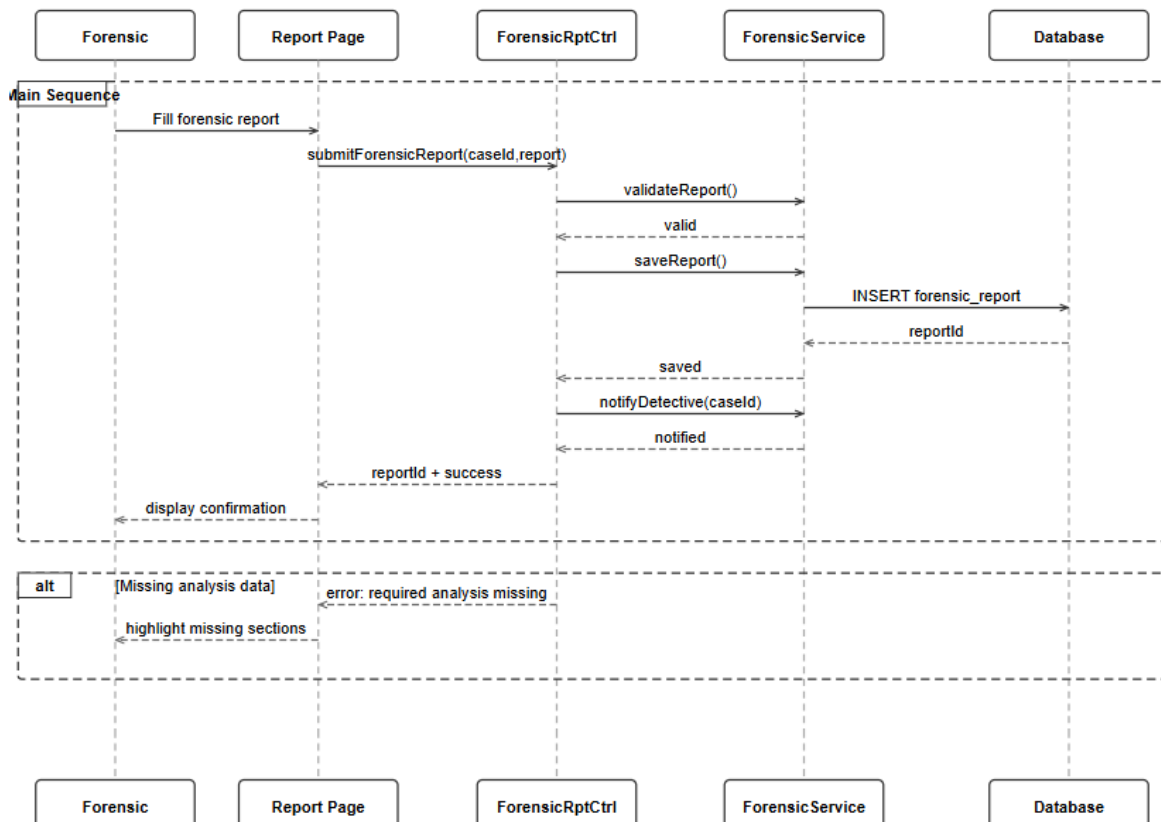
SeqD_20 – Analyze Photos and Videos



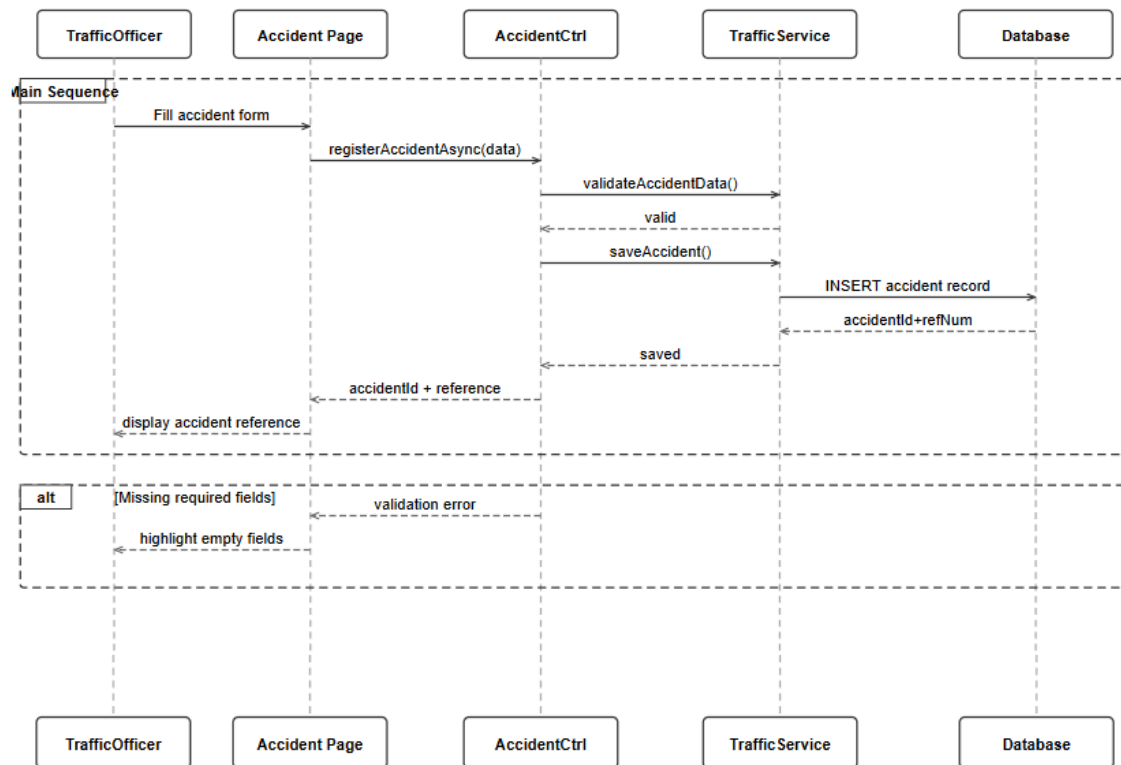
SeqD_21 – Manage Laboratory Evidence



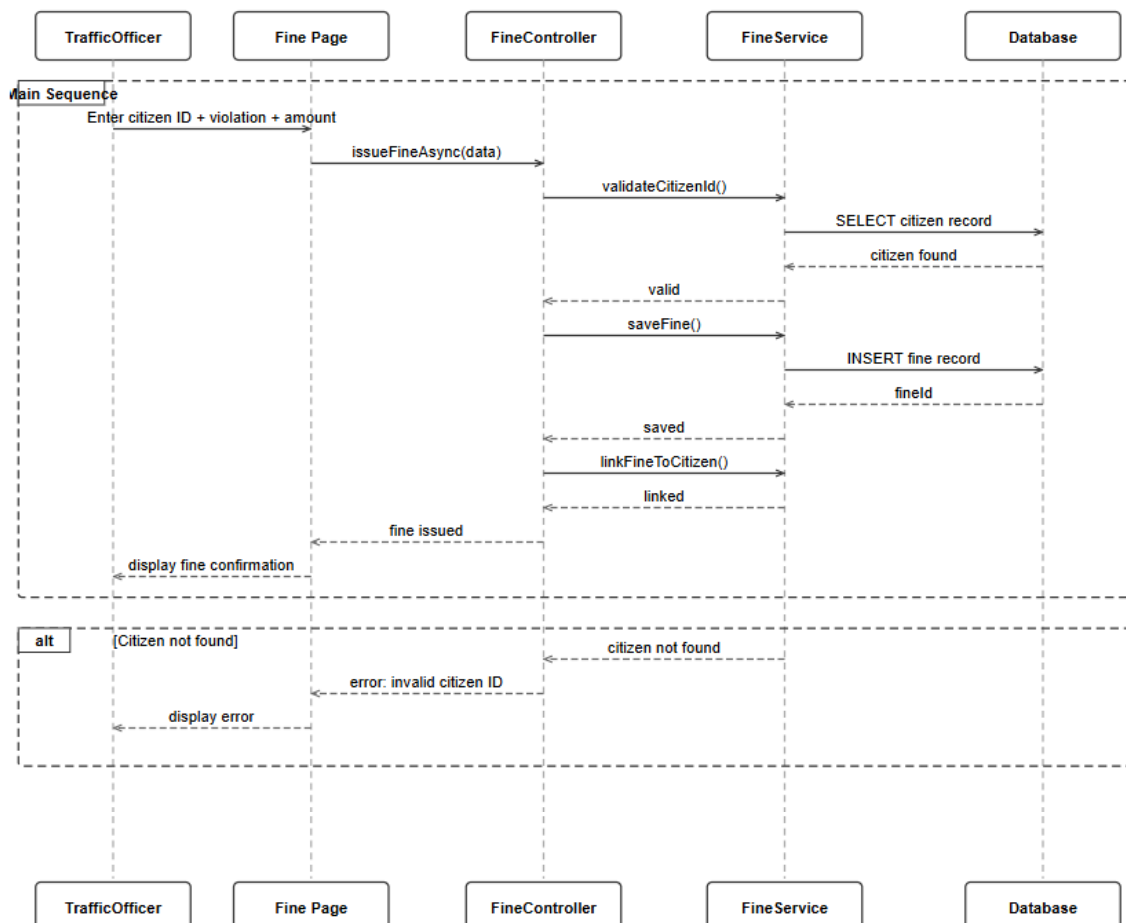
SeqD_22 – Write Forensic Report



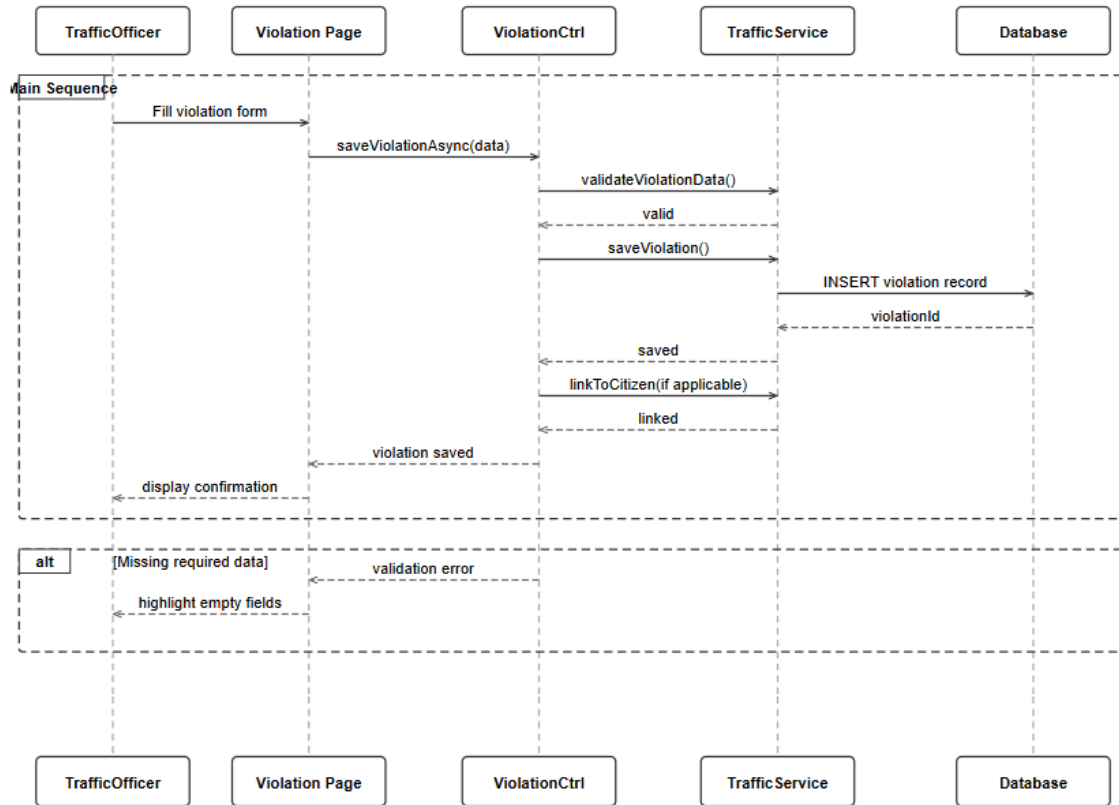
SeqD_23 – Register Traffic Accident



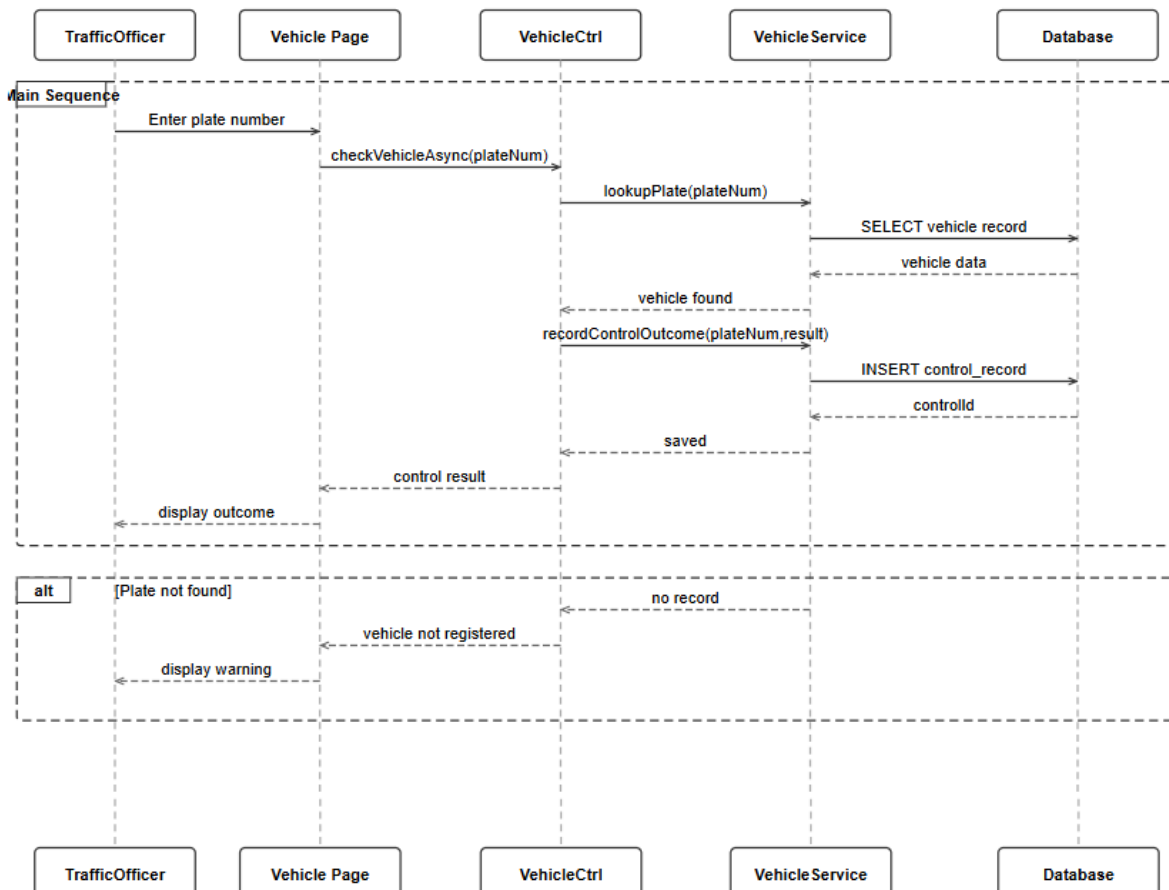
SeqD_24 – Issue Traffic Fine



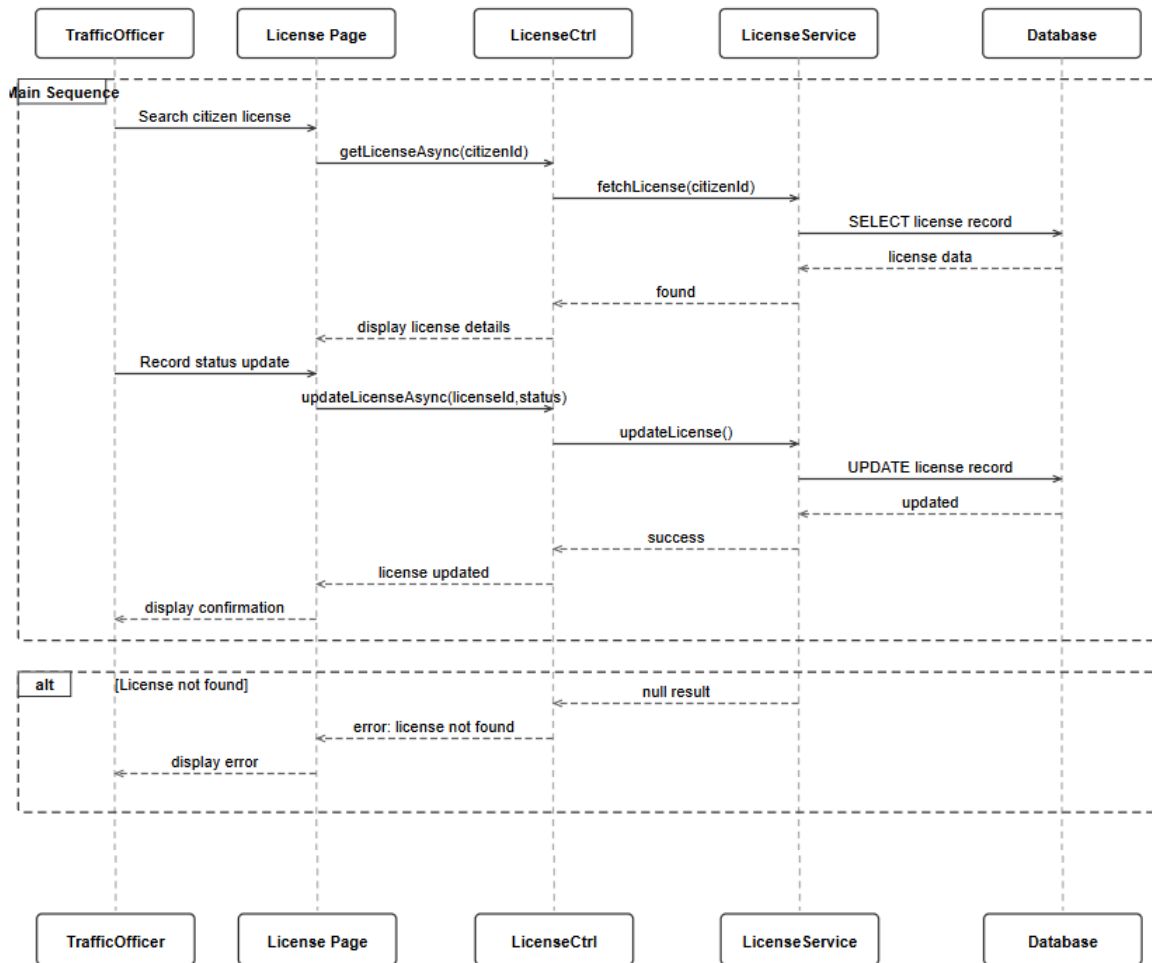
SeqD_25 – Manage Road Violations



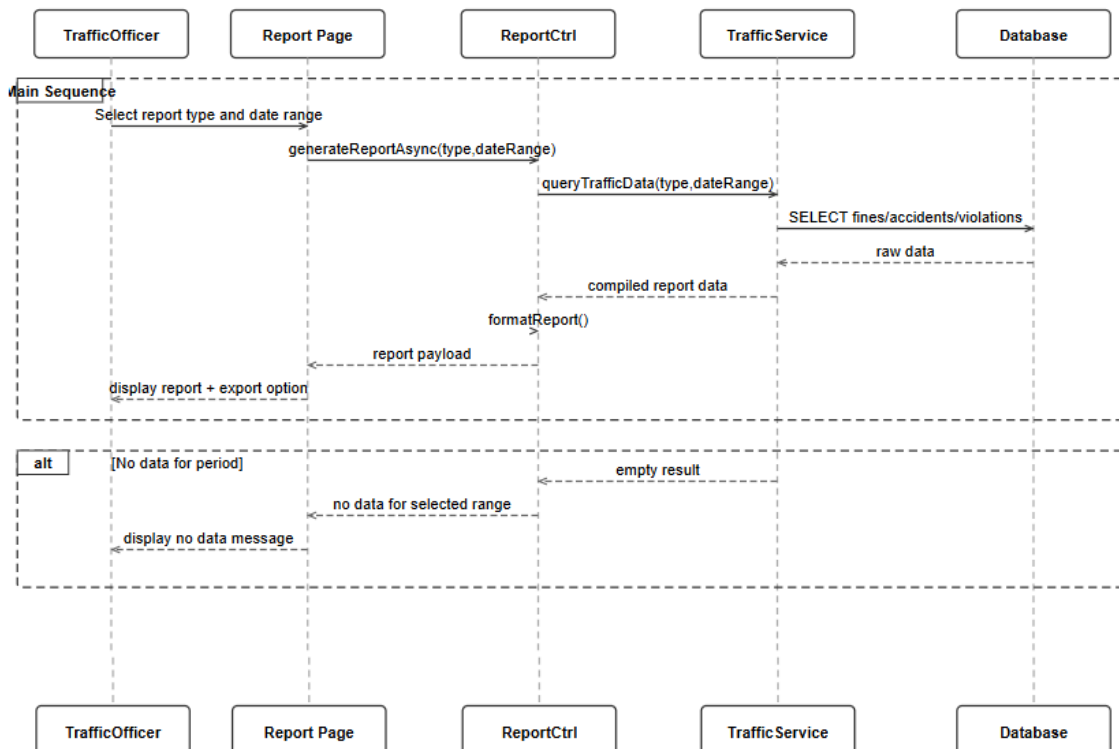
SeqD_26 – Perform Vehicle Control



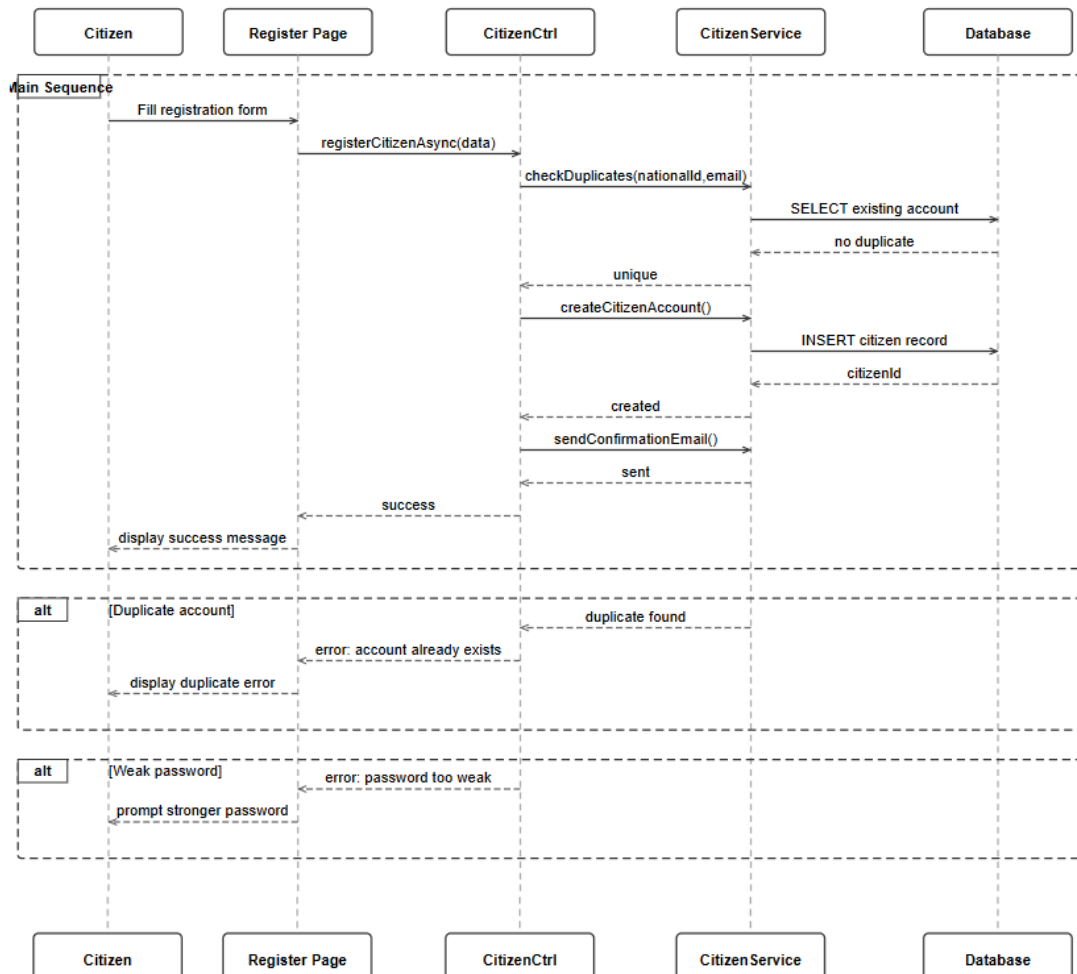
SeqD_27 – Manage Driving Licenses



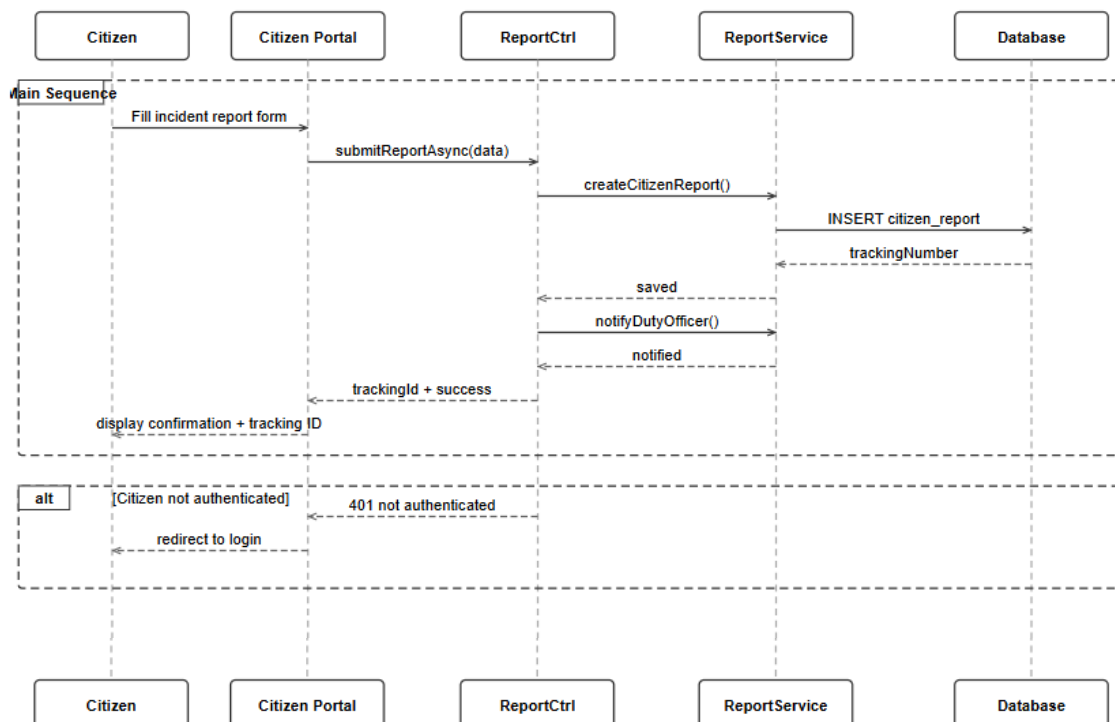
SeqD_28 – Generate Traffic Report



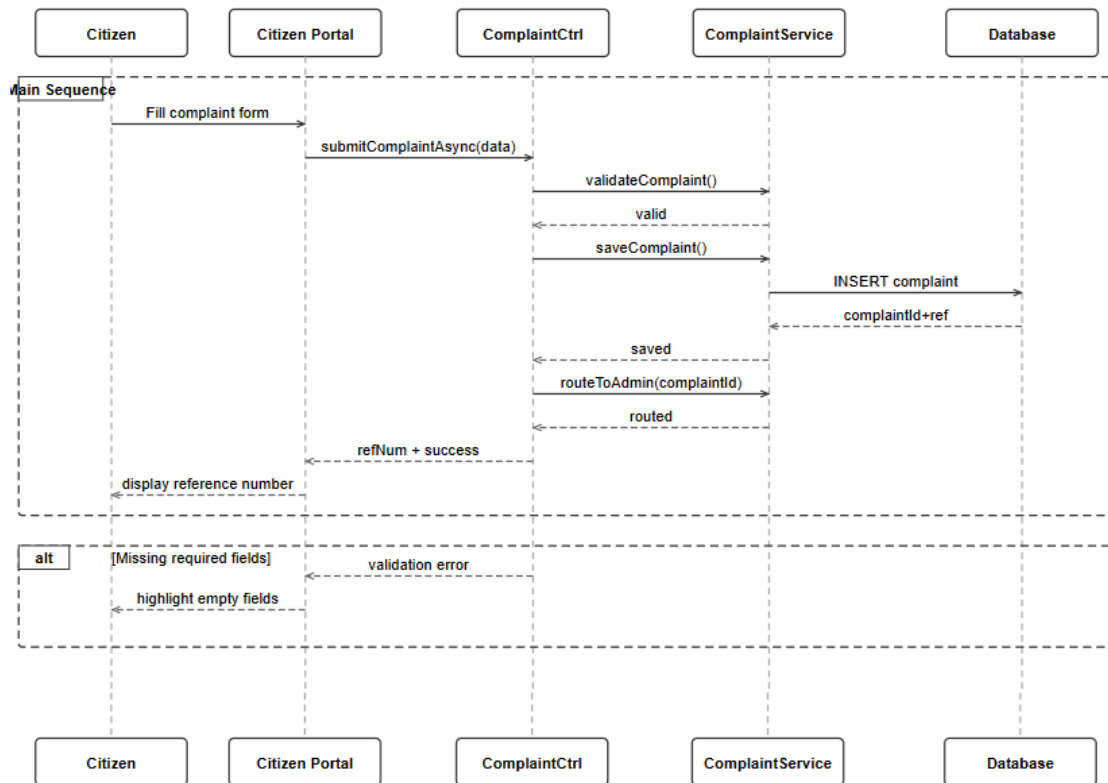
SeqD_29 – Self-Register as Citizen



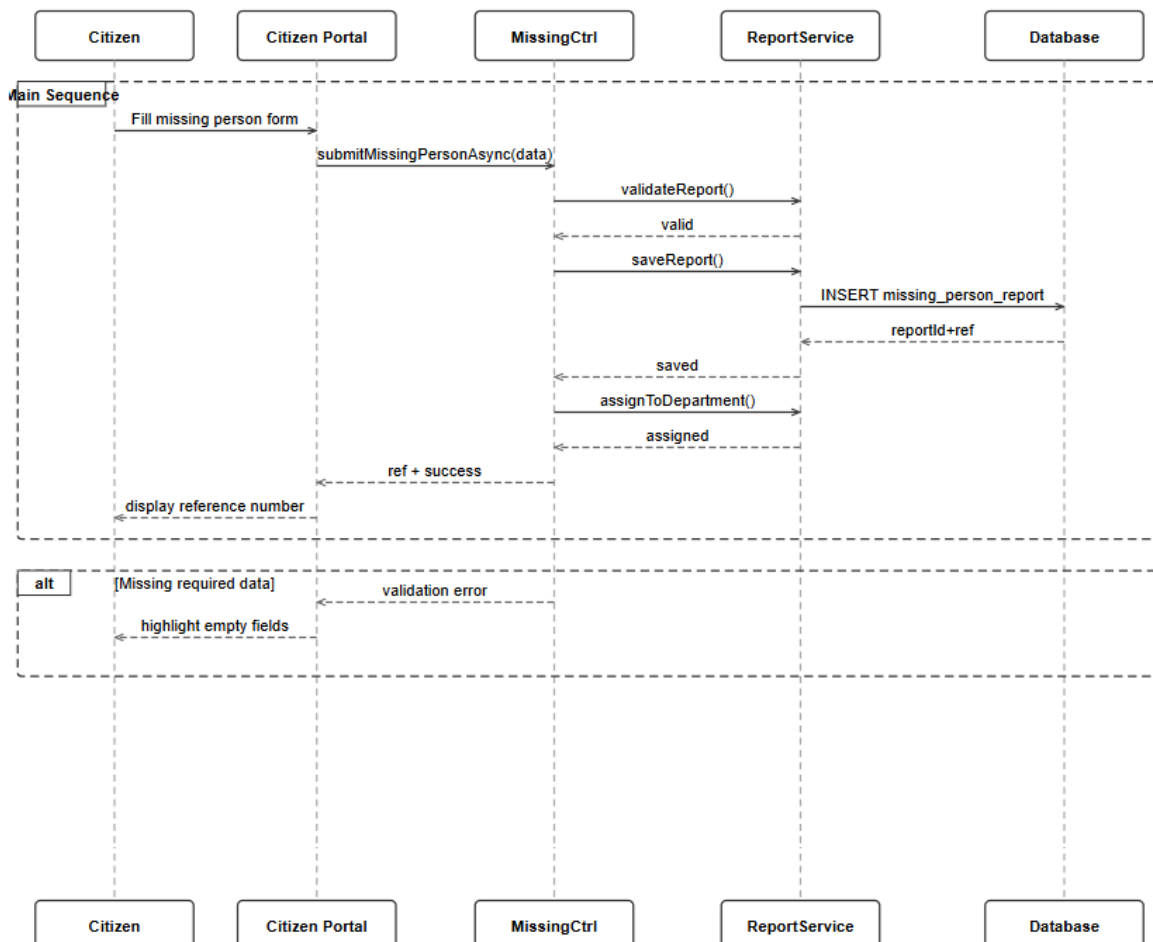
SeqD_30 – Report an Incident (Citizen)



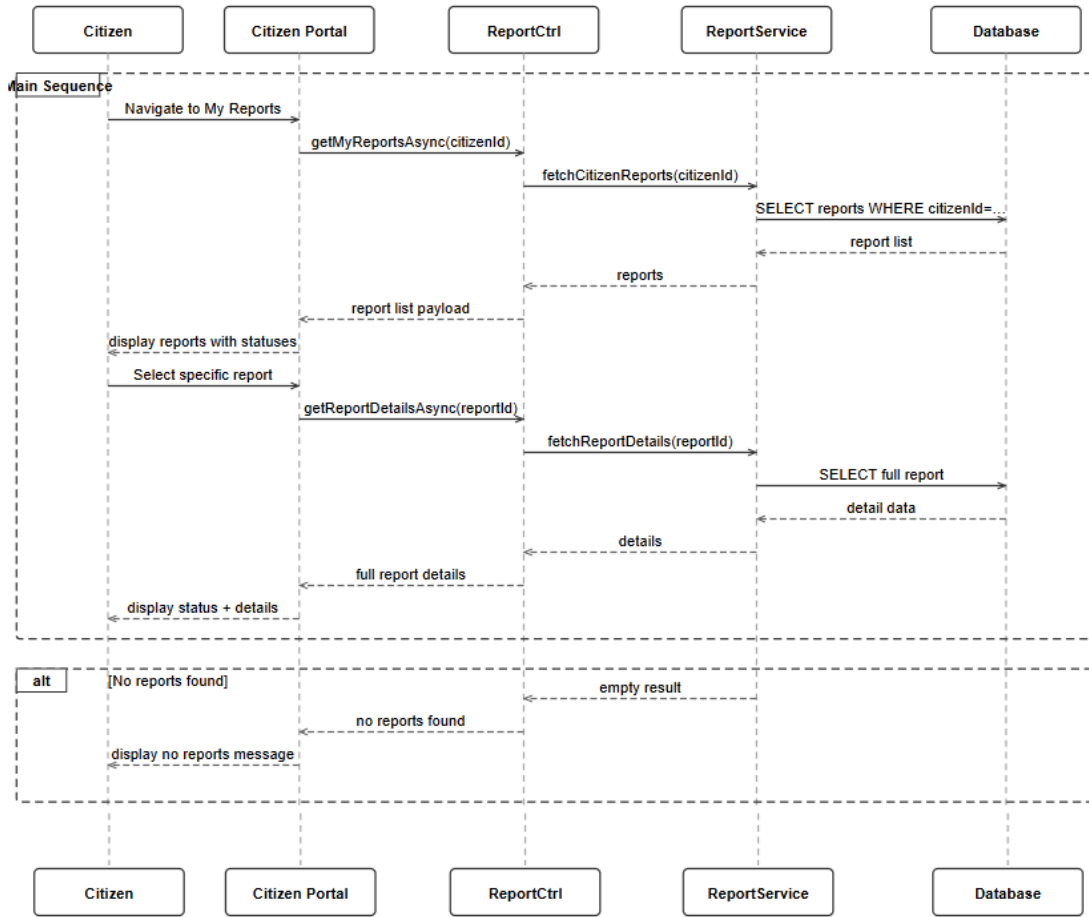
SeqD_31 – Submit a Complaint



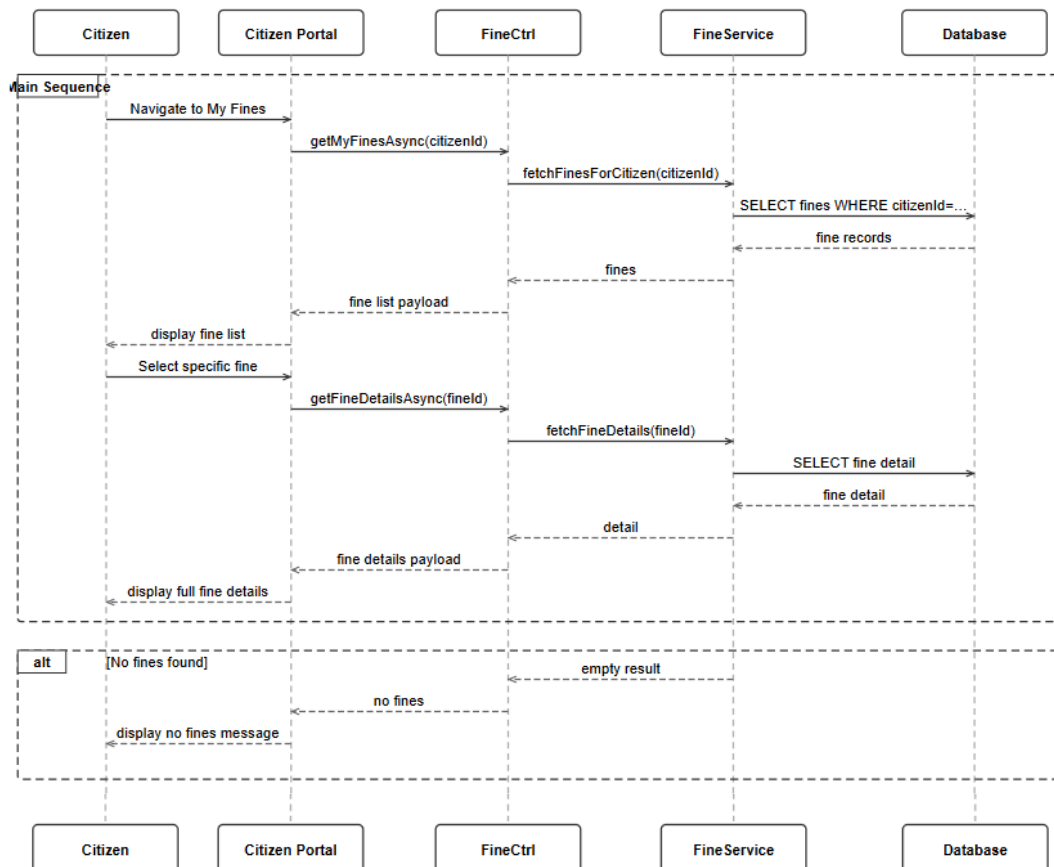
SeqD_32 – Report a Missing Person



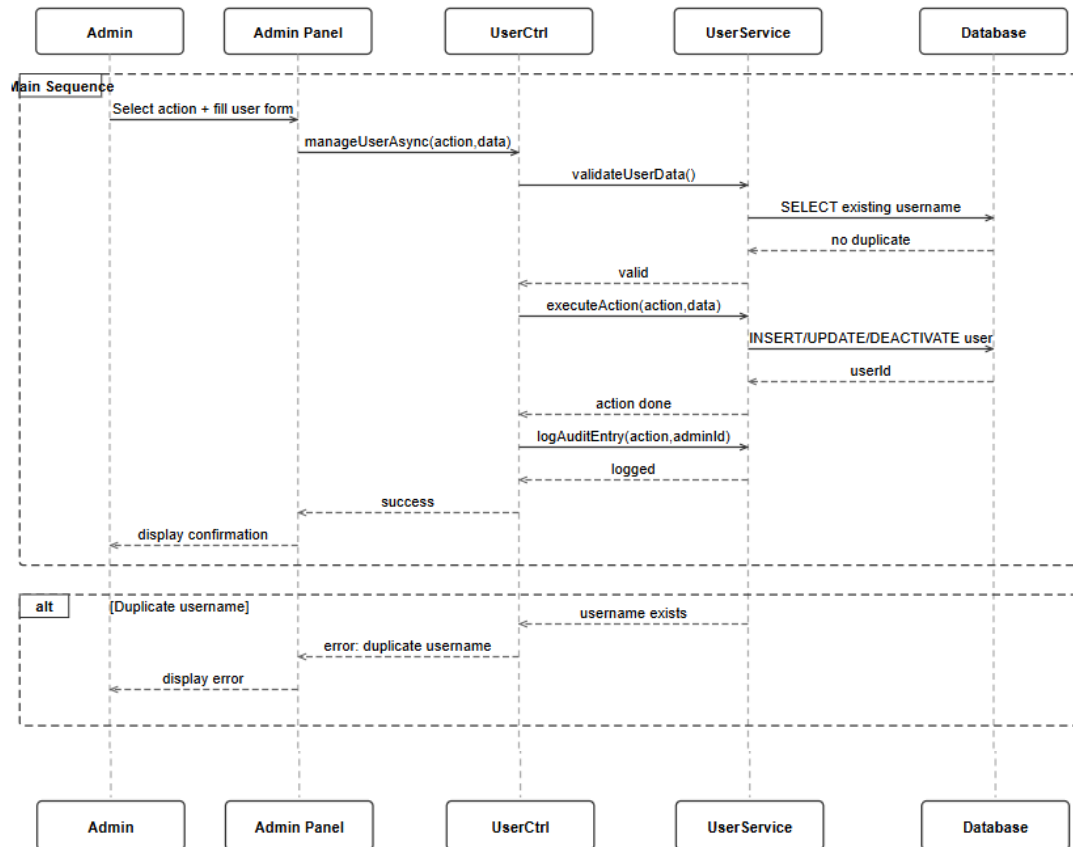
SeqD_33 – Check Case Status



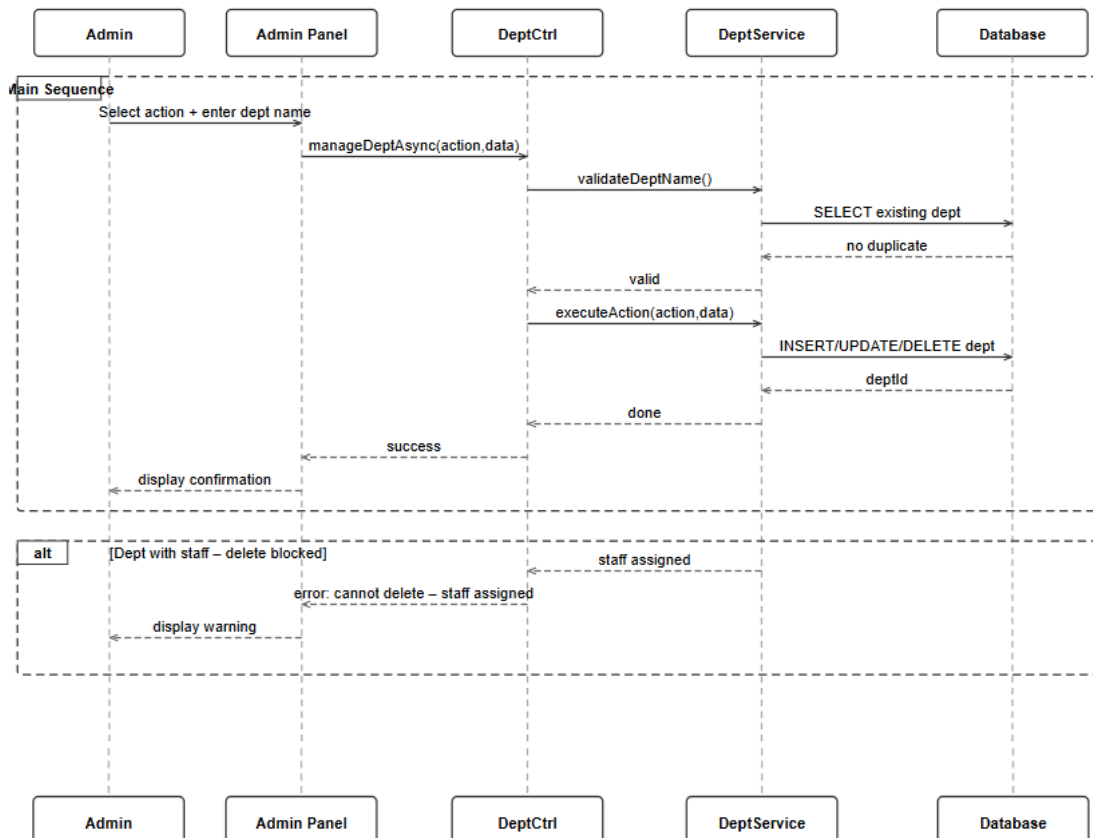
SeqD_34 – View Traffic Fines



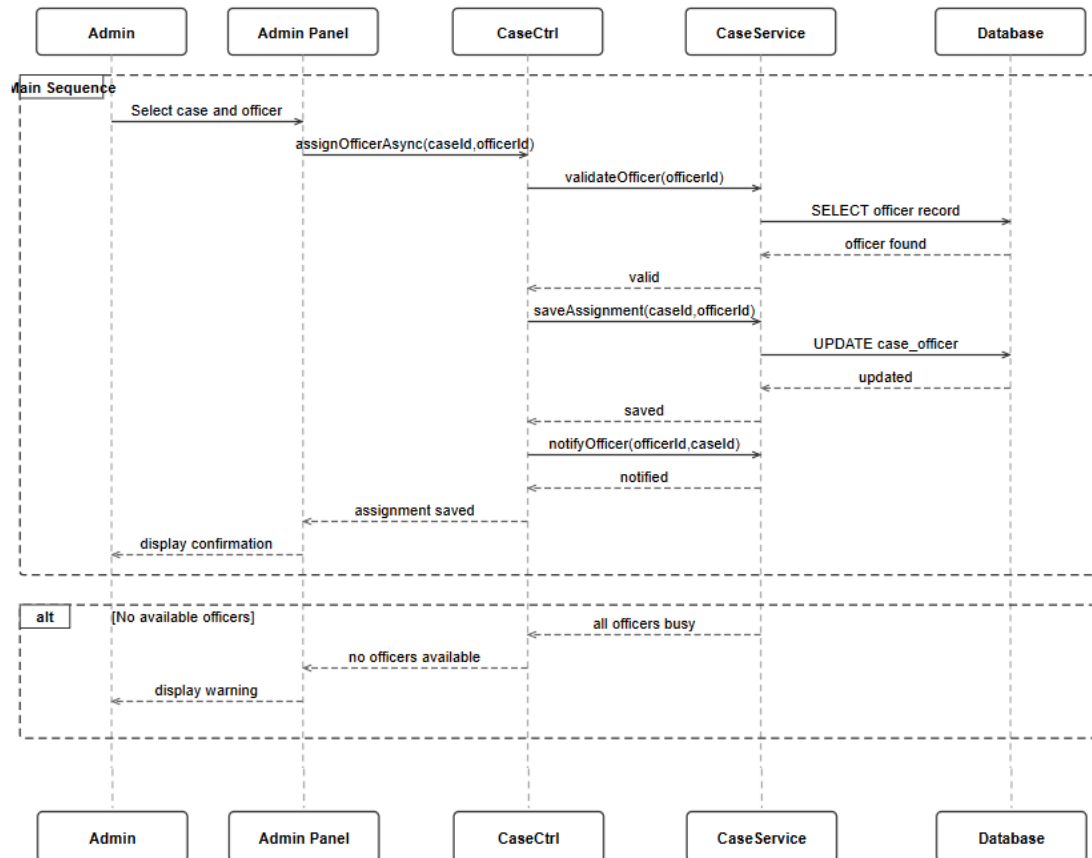
SeqD_35 – Manage Users and Roles



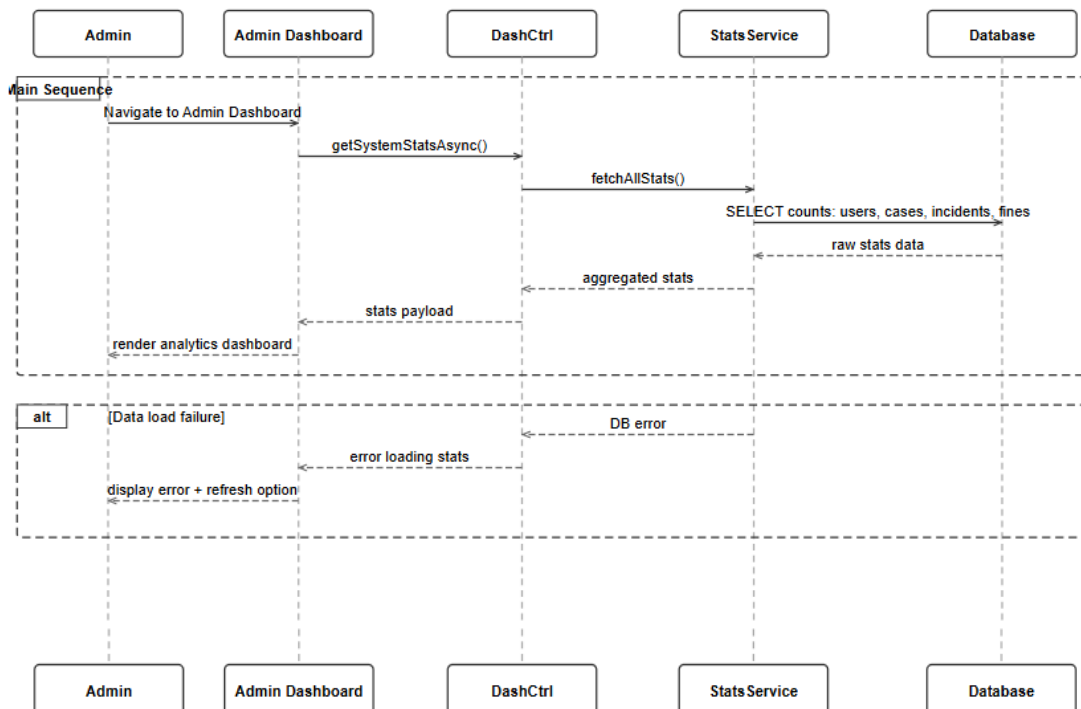
SeqD_36 – Manage Departments



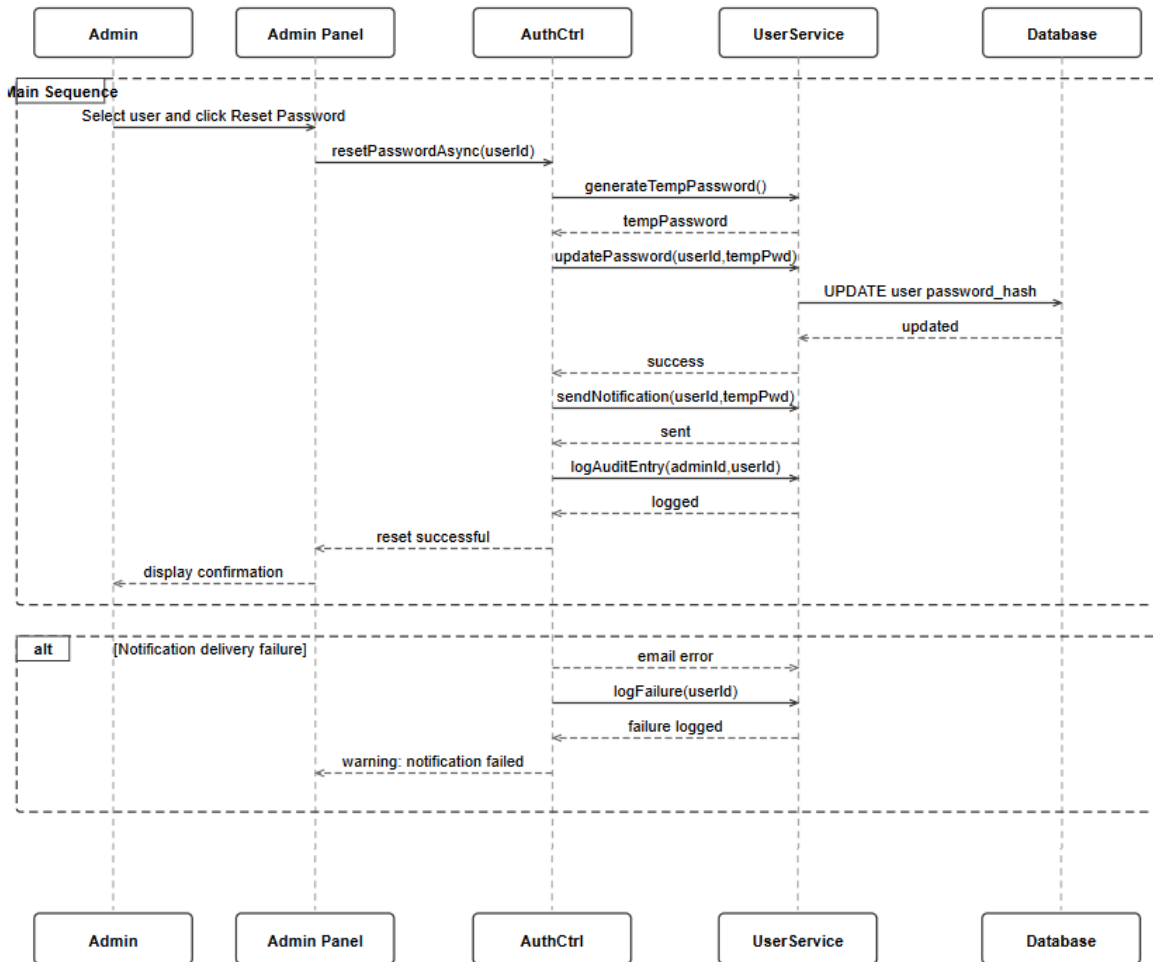
SeqD_37 – Assign Officer to Case



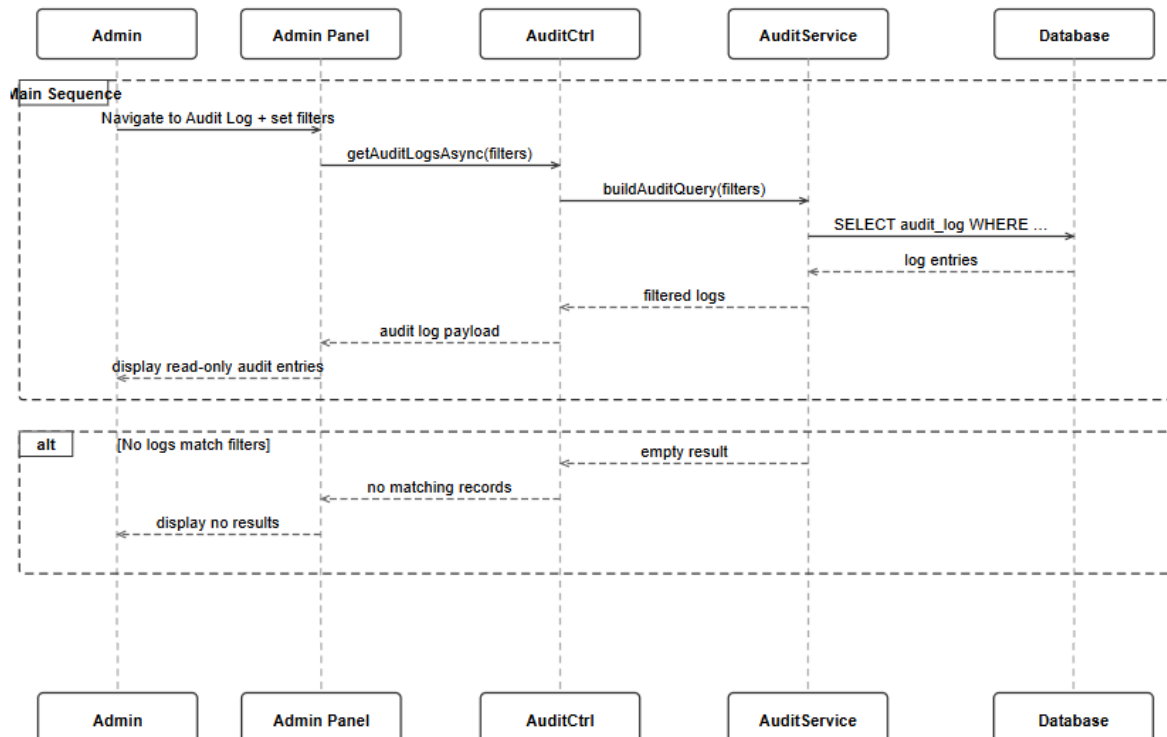
SeqD_38 – Monitor System Statistics



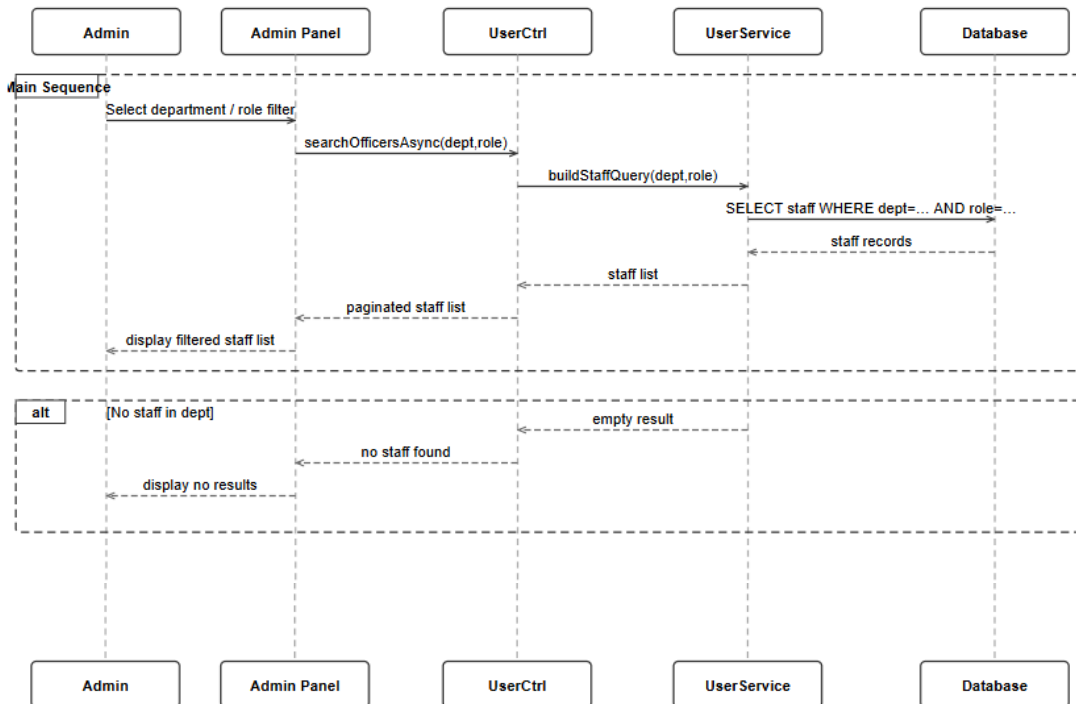
SeqD_39 – Reset User Password



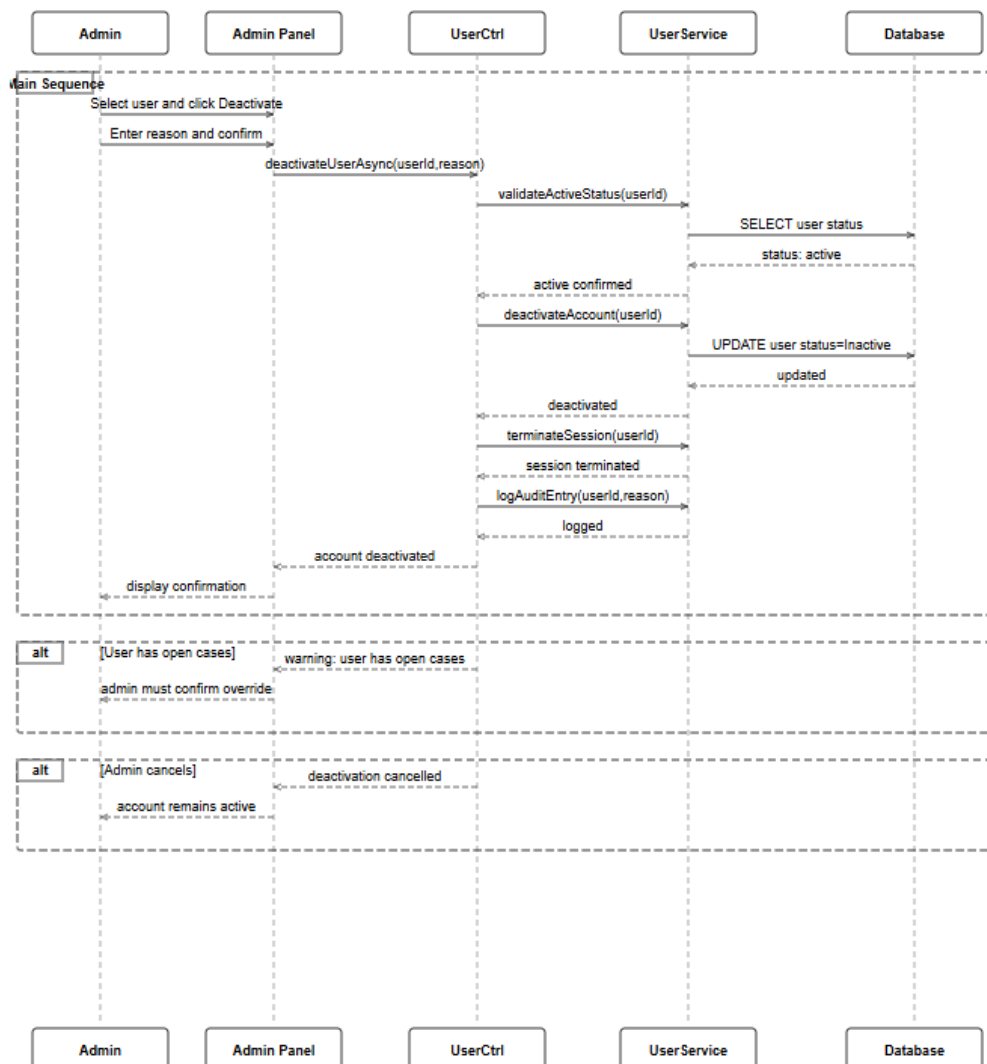
SeqD_40 – View System Audit Log



SeqD_41 – Search Officers by Department

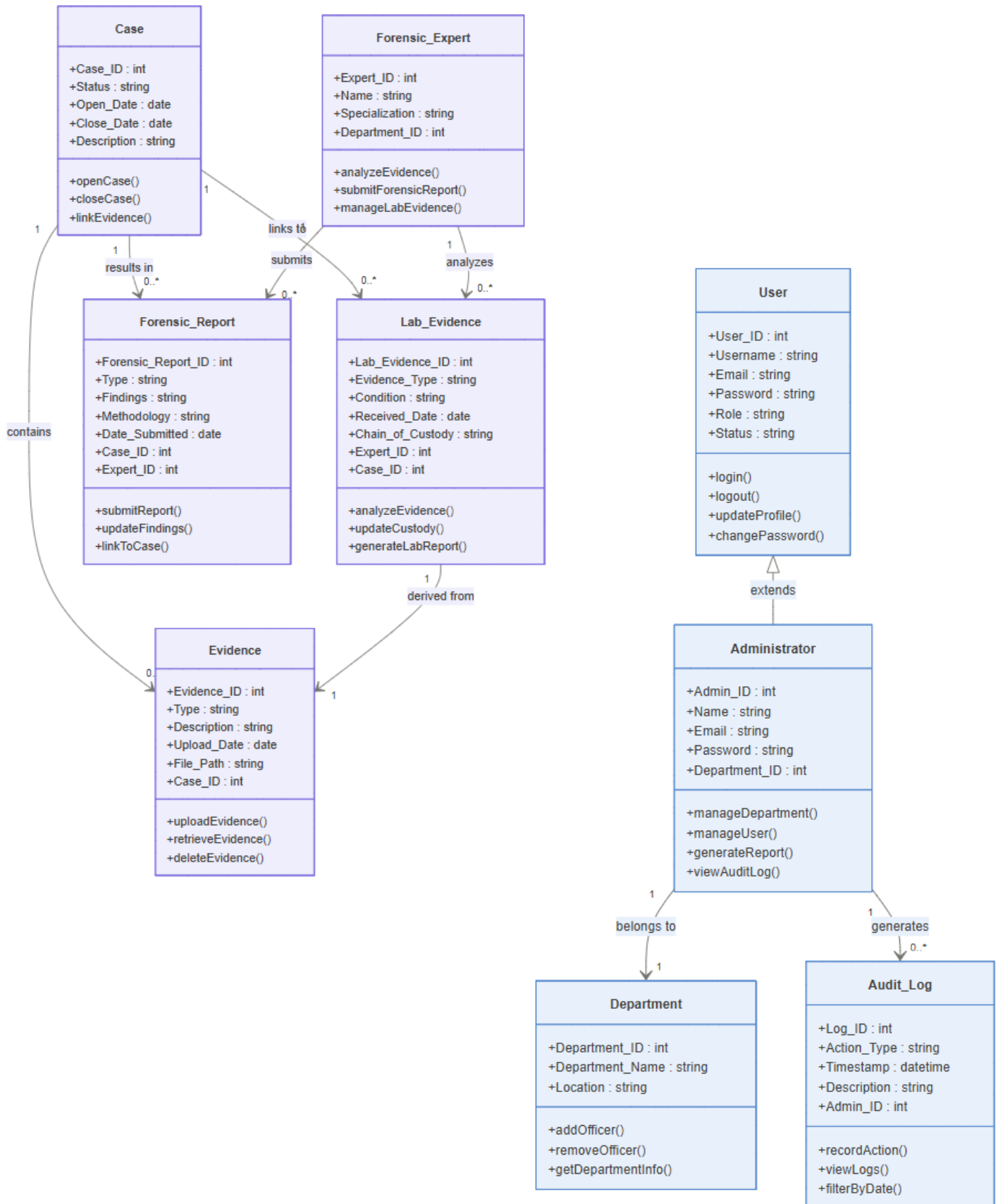


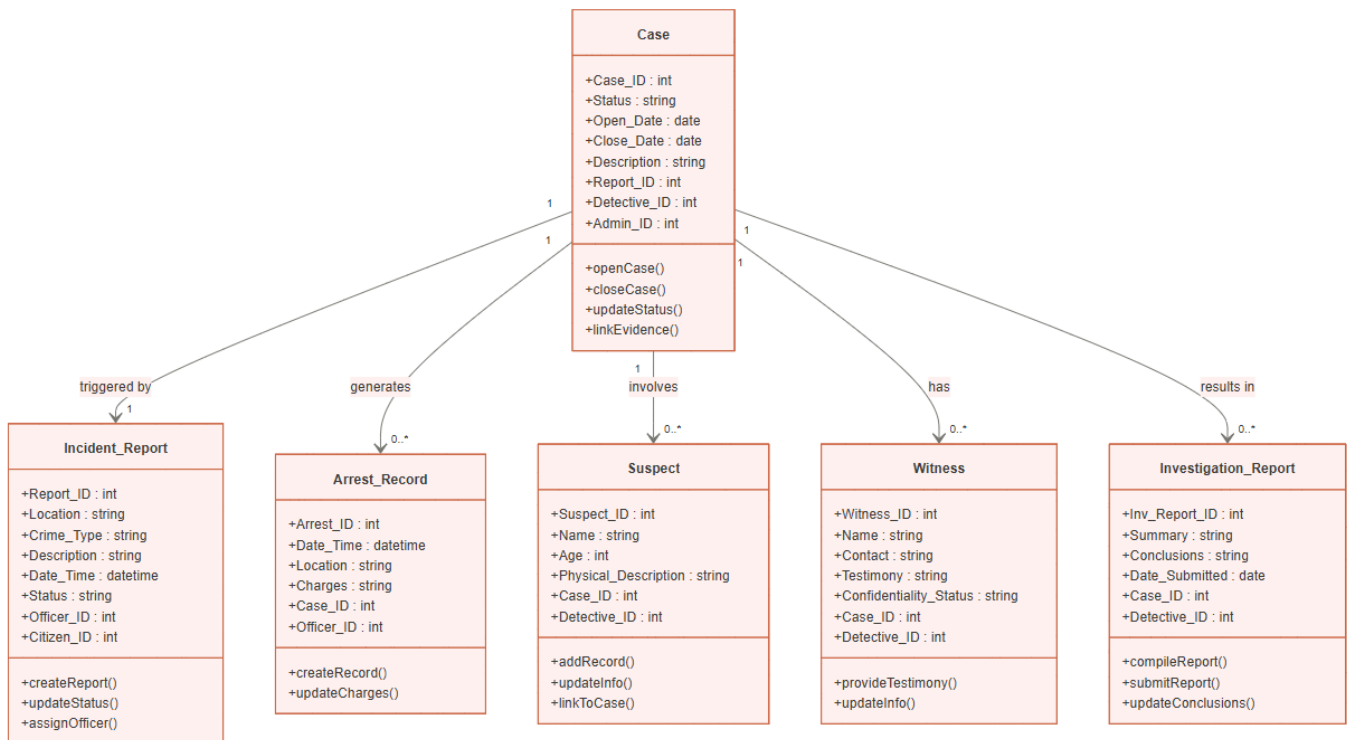
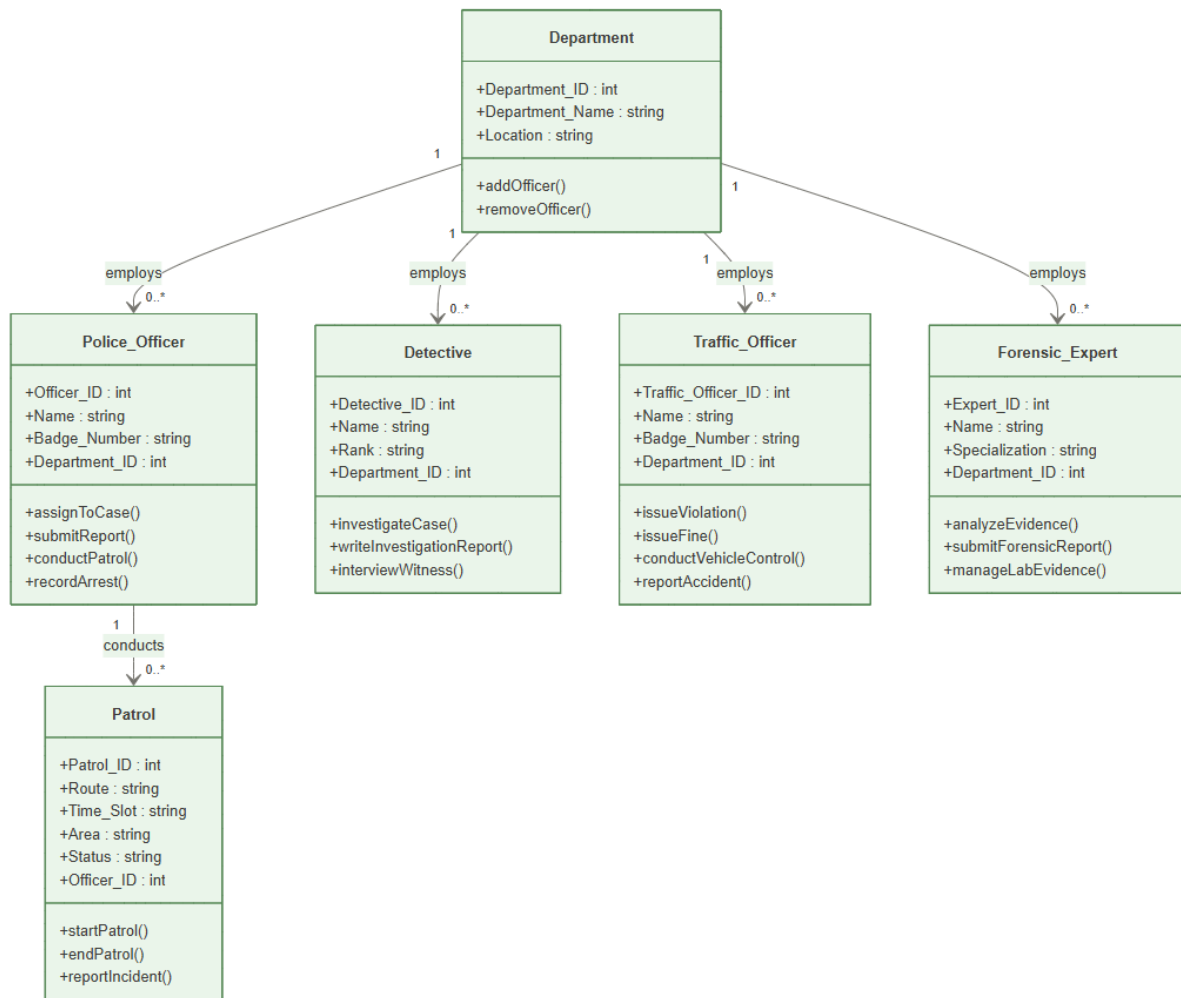
SeqD_42 – Deactivate User Account

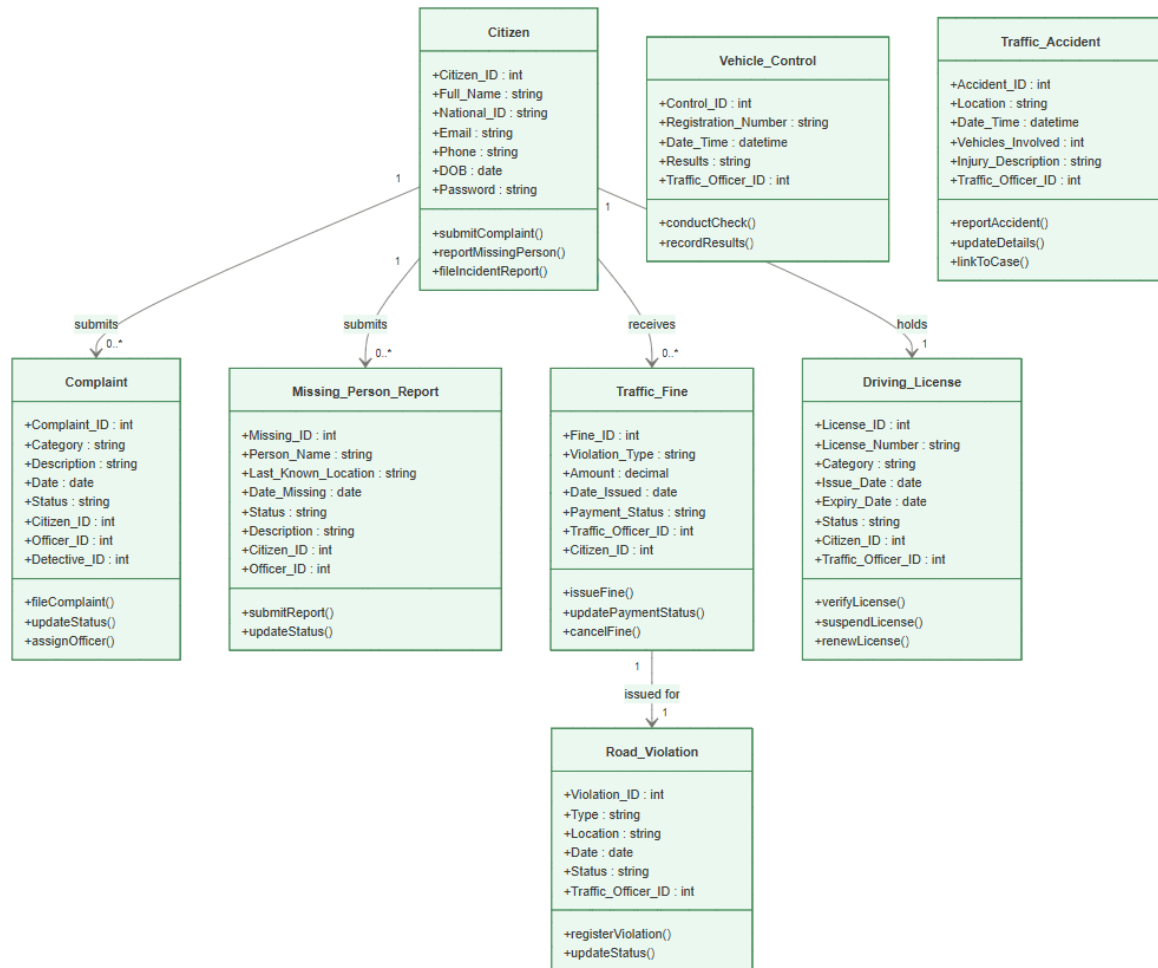


4.5 Structural Diagrams

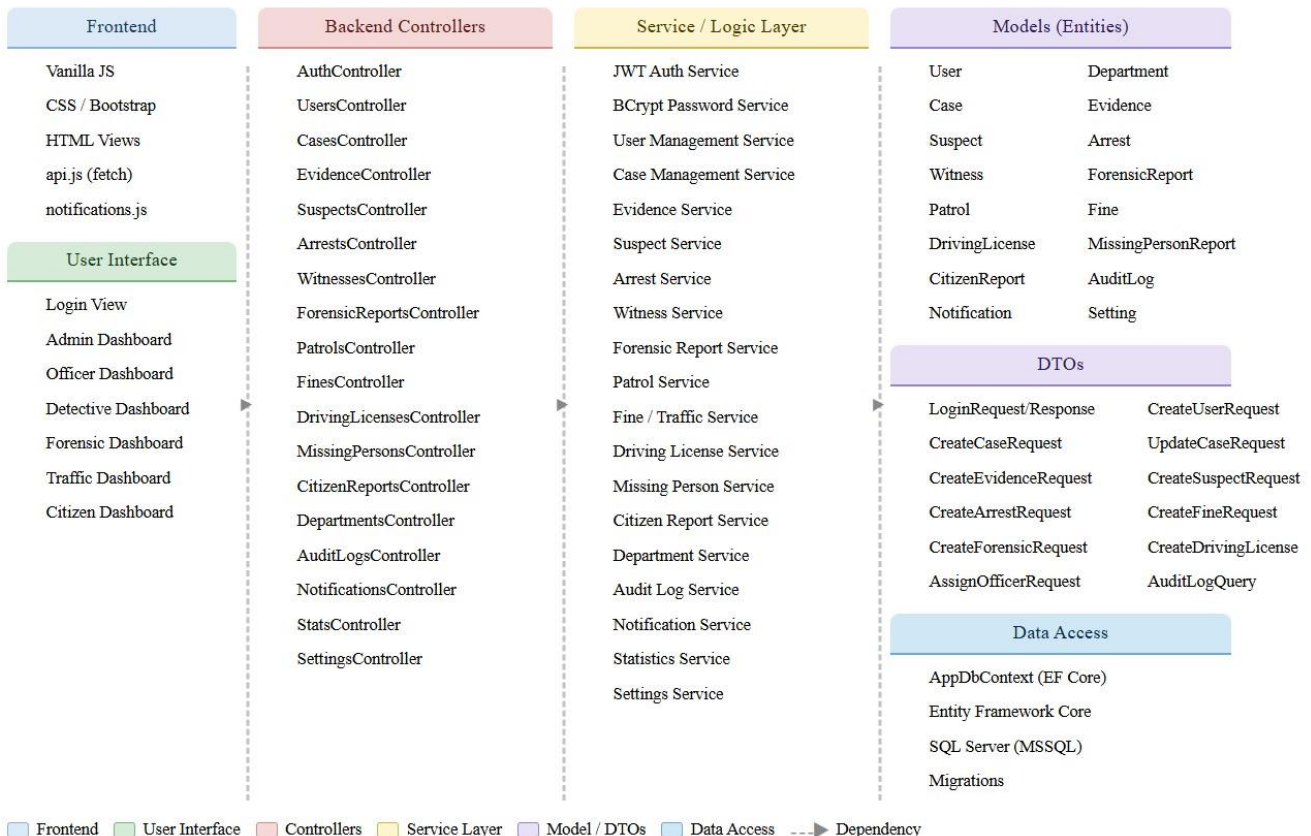
4.5.1 Class Diagram



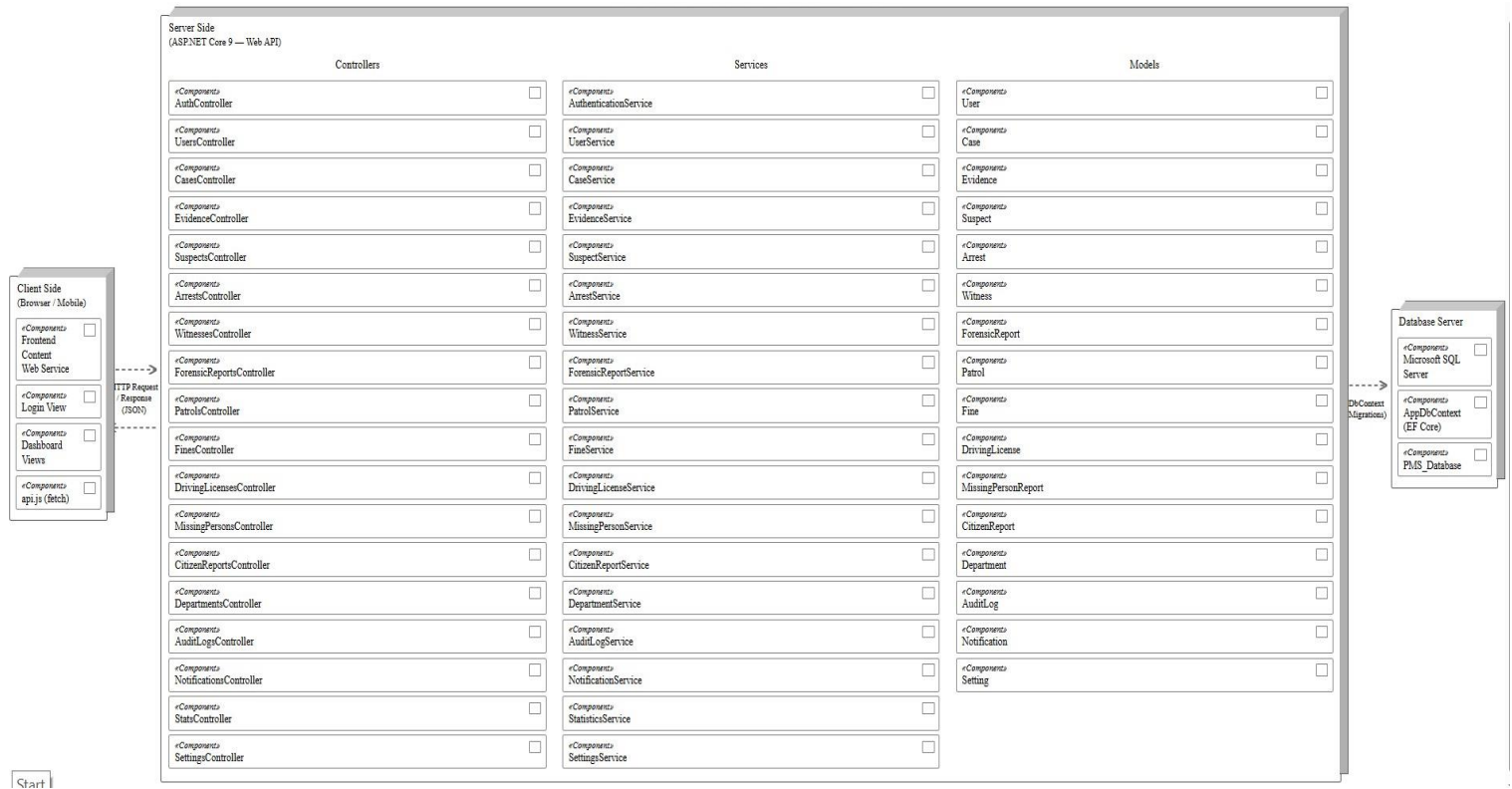




4.5.2 Component Diagrams



4.5.3 Deployment Diagram



5. Implementation

This chapter presents the implementation of the Police Management System (PMS), a full-stack web application developed using ASP.NET Core 9 Web API for the backend and plain HTML5/CSS3/JavaScript for the frontend. The system manages police operations including case management, evidence tracking, suspect profiling, forensic reporting, traffic enforcement, citizen services, patrol management, and comprehensive audit logging. The implementation directly reflects the functional and non-functional requirements defined in the earlier sections of this document.

The PMS was developed and tested on a Windows 10/11 environment using the following tools and technologies:

Category	Technology / Tool	Version / Detail
Backend Framework	ASP.NET Core Web API	.NET 9.0
Programming Language	C#	C# 13
ORM / Data Access	Entity Framework Core	EF Core 9 — Code First
Database	Microsoft SQL Server	SQL Server Express (local)
Authentication	JWT Bearer Tokens	Microsoft.AspNetCore.Authentication.JwtBearer
Password Hashing	BCrypt.Net-Next	Work factor 11 (default)
API Documentation	Swagger / OpenAPI	Swashbuckle.AspNetCore

Category	Technology / Tool	Version / Detail
Frontend	HTML5 / CSS3 / Vanilla JS	No build step required
Frontend HTTP Layer	Fetch API (api.js)	Custom wrapper with JWT injection
IDE	JetBrains Rider / Visual Studio	2024
Version Control	Git	Local repository
Package Manager	NuGet (.NET) / npm (tools)	—

The backend runs on localhost:5000 and exposes a RESTful API consumed by the frontend served as static files. CORS is configured in Program.cs to allow all origins during development, enabling the HTML/JS frontend to communicate with the API without proxy configuration

5.1 Project Structure

The solution is organized into a backend ASP.NET Core project and a frontend folder:

5.1.1 Backend Structure (PMS.sln)

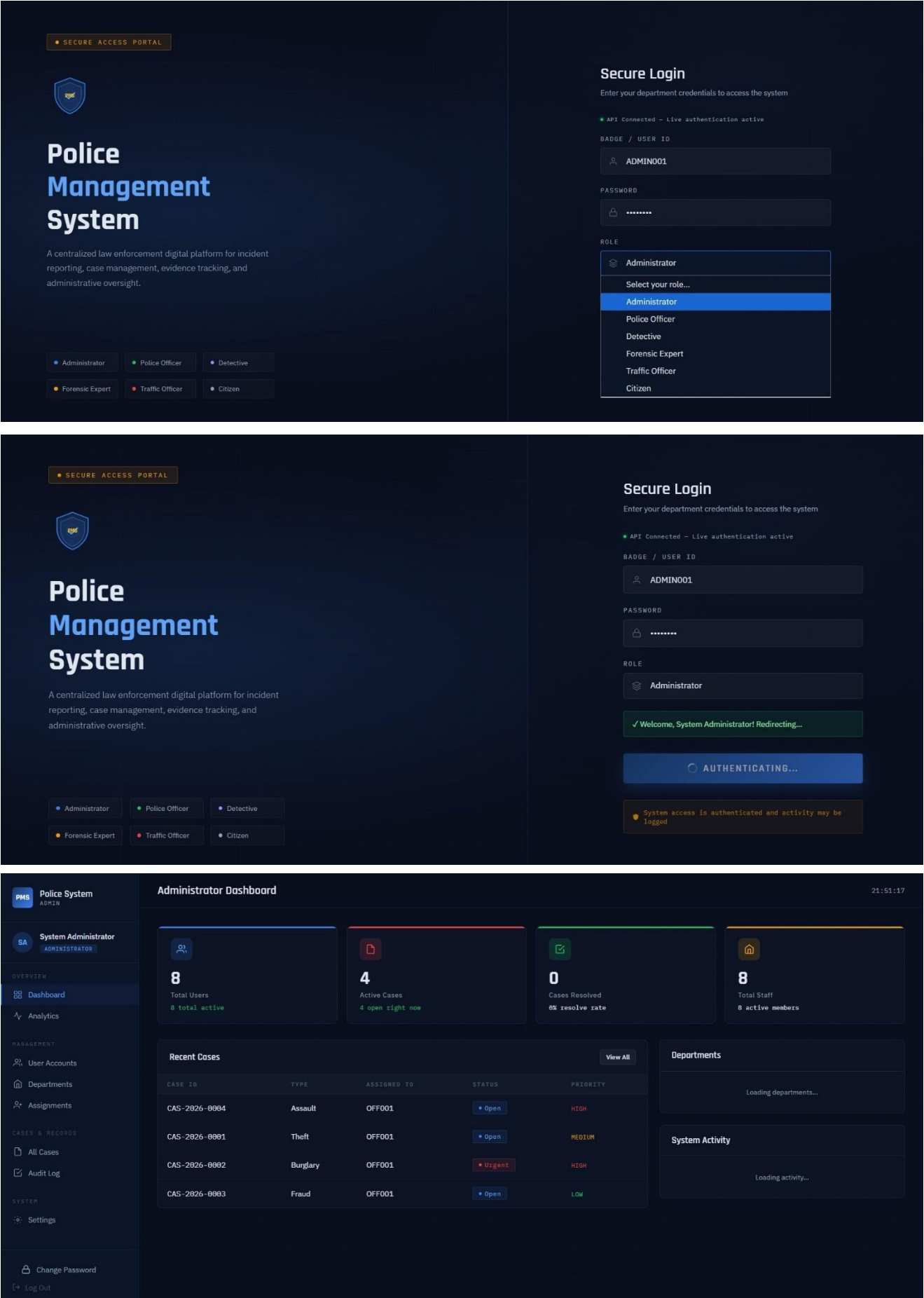
PMS/	
├─ Controllers/	← 18 API controllers (one per resource)
├─ Models/	← 16 EF Core entity classes
├─ DTOs/	← Request/Response data transfer objects (Dtos.cs)
├─ Data/	← AppDbContext (EF Core DbContext)
├─ Migrations/	← EF Core auto-generated migrations
├─ Program.cs	← App bootstrap, DI configuration, middleware pipeline
└─ appsettings.json	← Connection string, JWT config

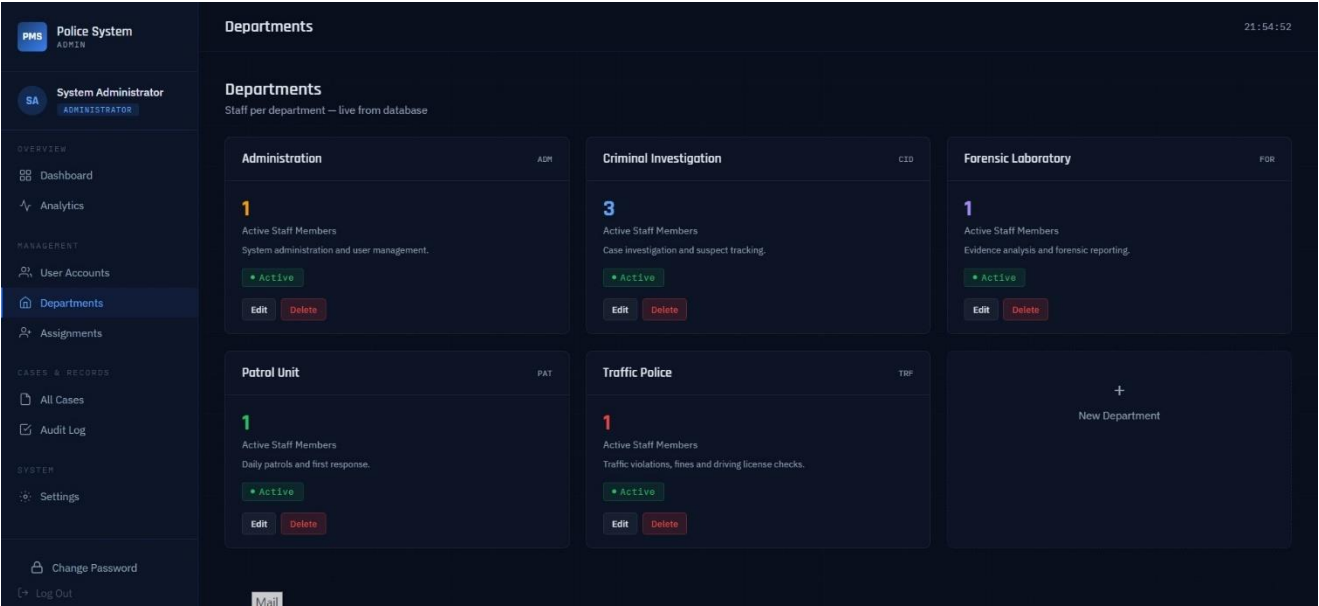
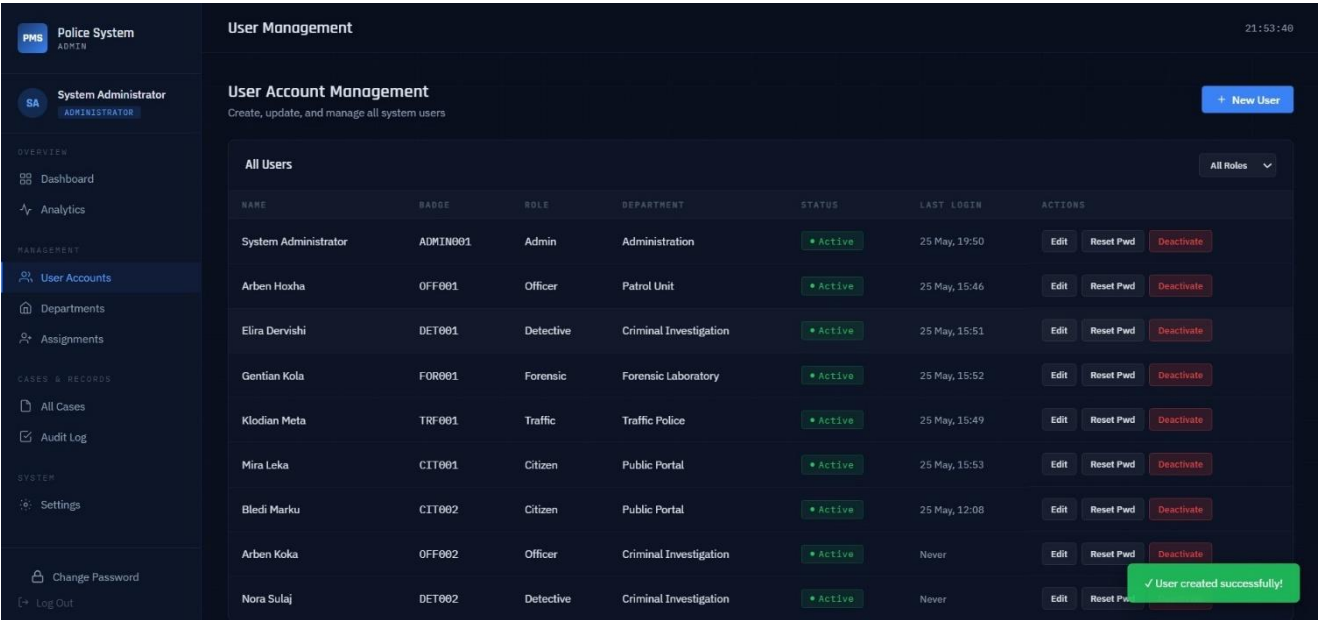
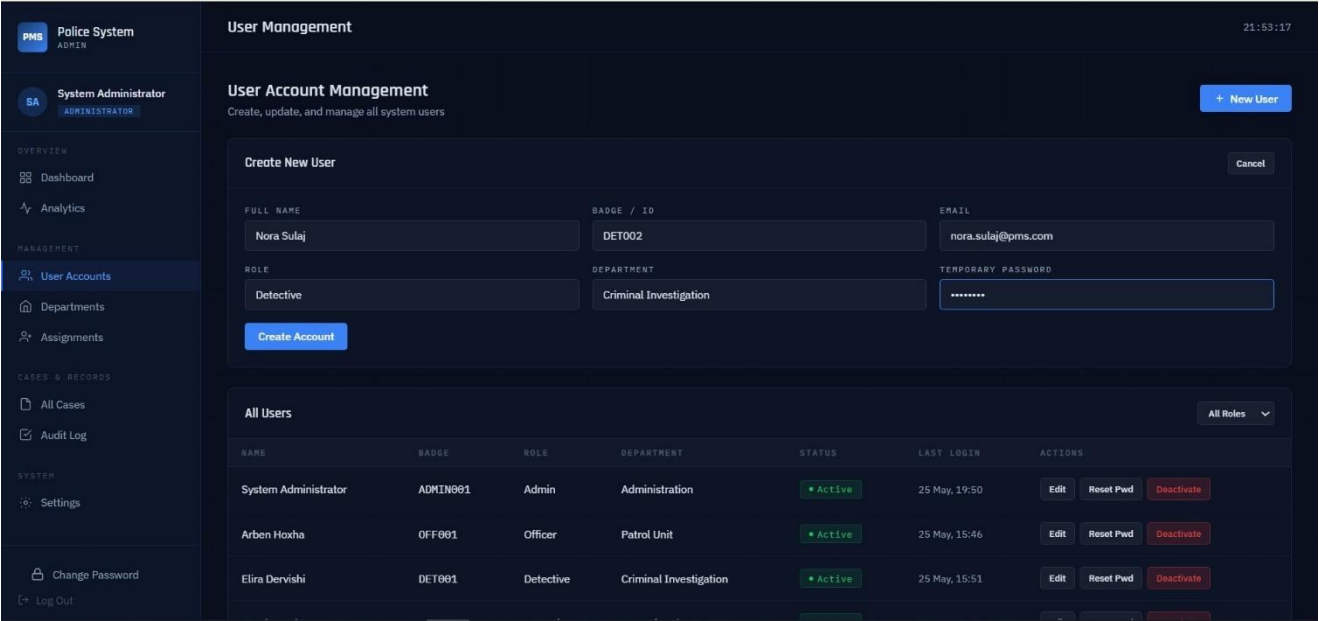
5.1.2 Frontend Structure

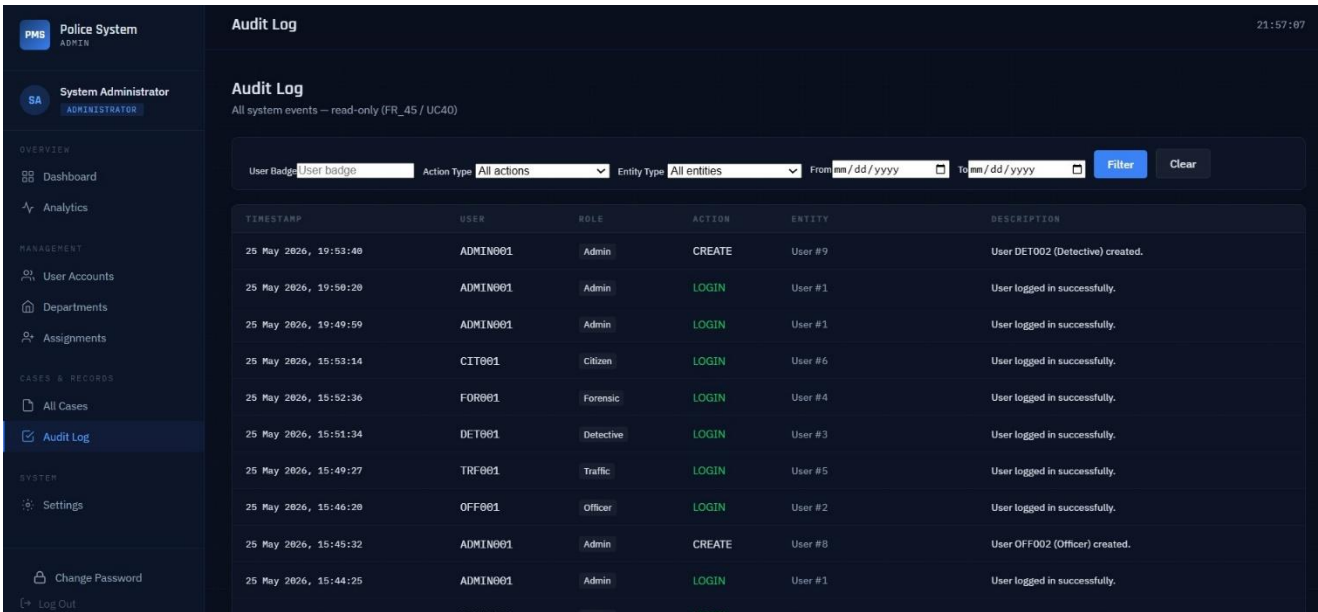
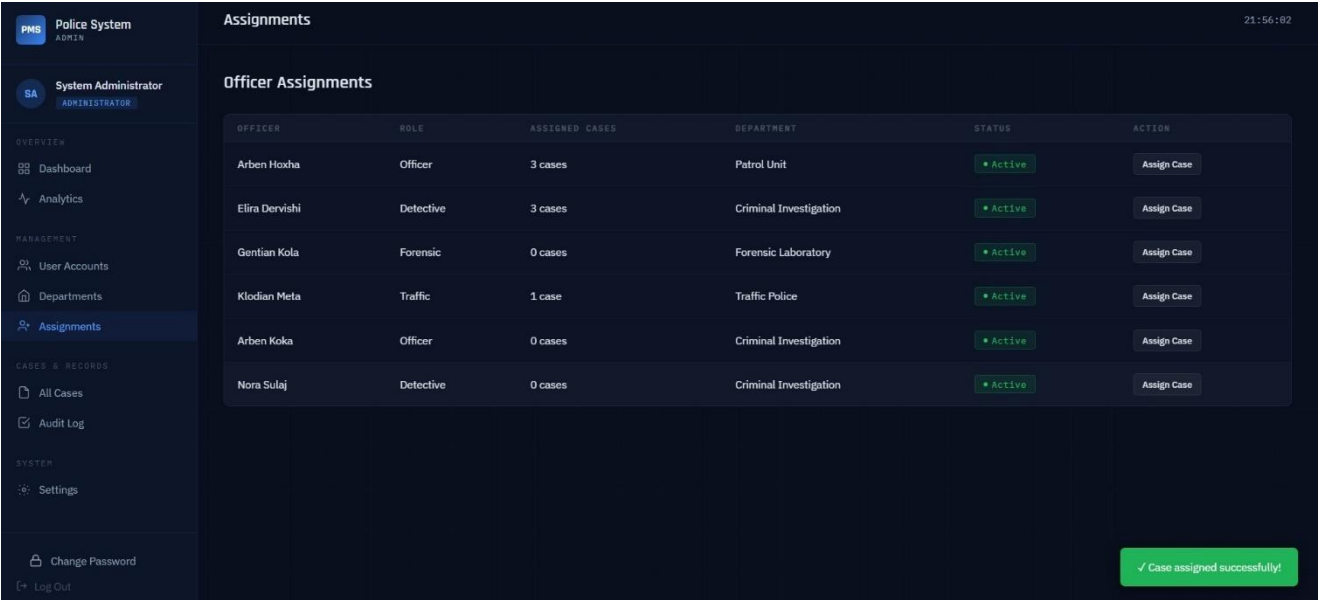
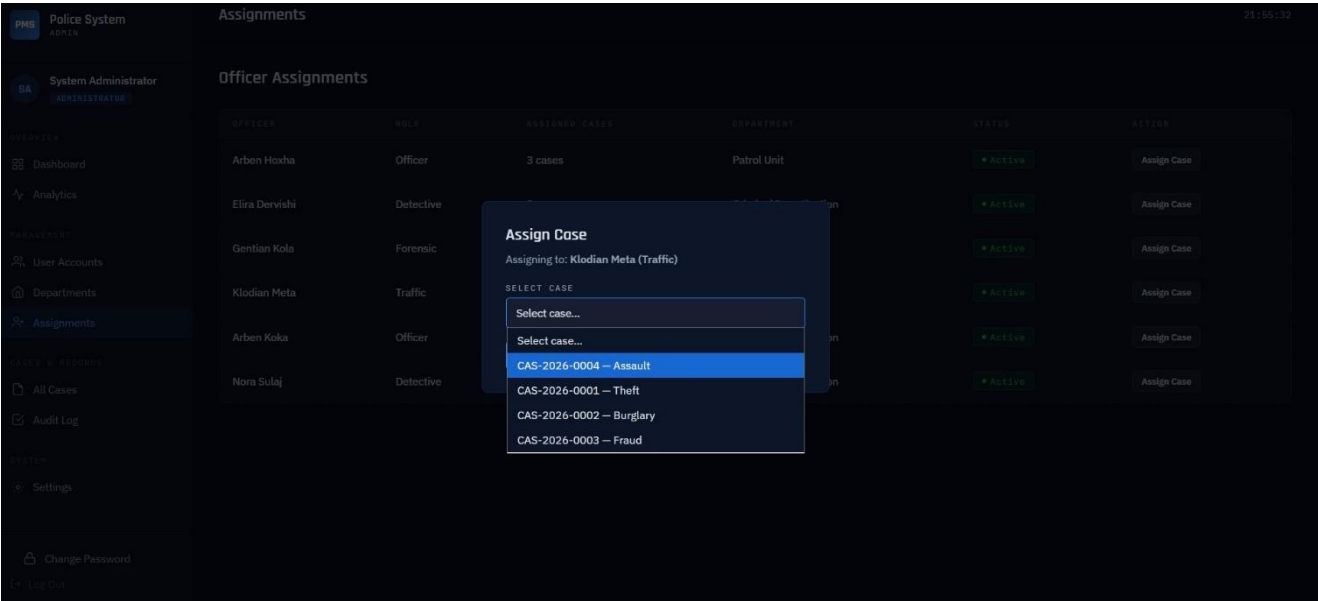
frontend/	
├─ index.html	← Login page (role-based redirect)
├─ dashboard-admin.html	← Administrator dashboard
├─ dashboard-officer.html	← Police Officer dashboard
├─ dashboard-detective.html	← Detective dashboard
├─ dashboard-forensic.html	← Forensic Expert dashboard
├─ dashboard-traffic.html	← Traffic Officer dashboard
├─ dashboard-citizen.html	← Citizen portal
├─ shared.css	← Global styles
└─ js/	
├─ api.js	← Centralized API layer (fetch wrapper + all API calls)
├─ admin.js	← Admin dashboard logic
├─ officer.js	← Officer dashboard logic
├─ detective.js	← Detective dashboard logic
├─ forensic.js	← Forensic dashboard logic
├─ traffic.js	← Traffic dashboard logic
├─ citizen.js	← Citizen portal logic
└─ notifications.js	← Real-time notification polling

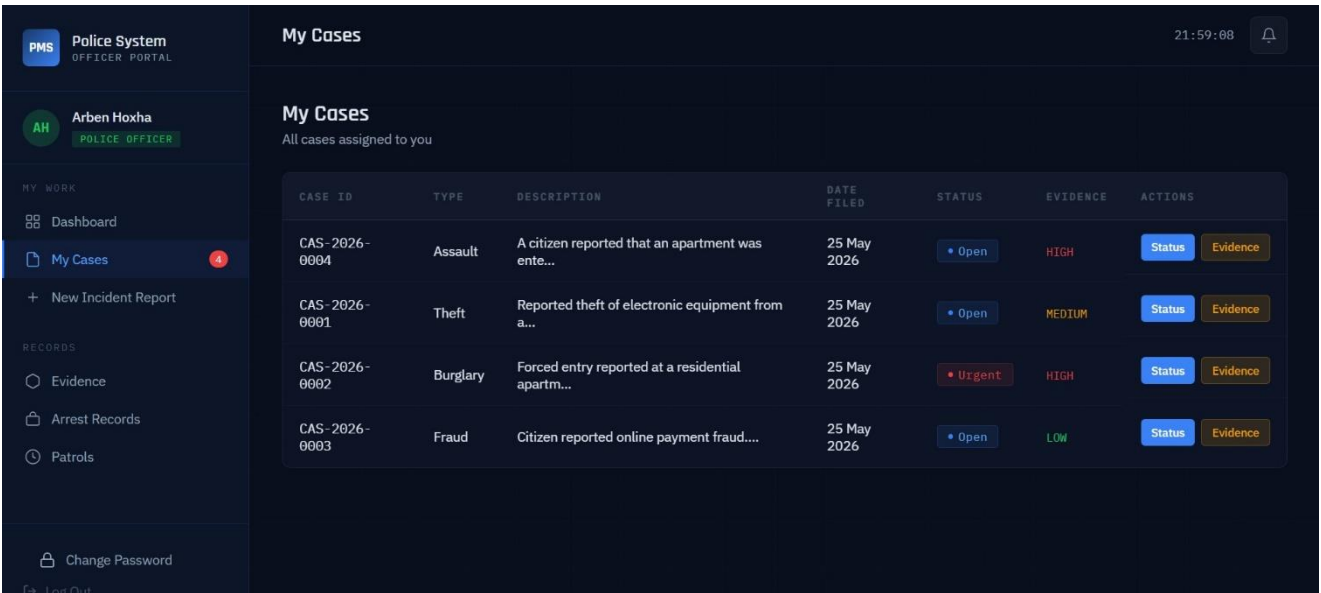
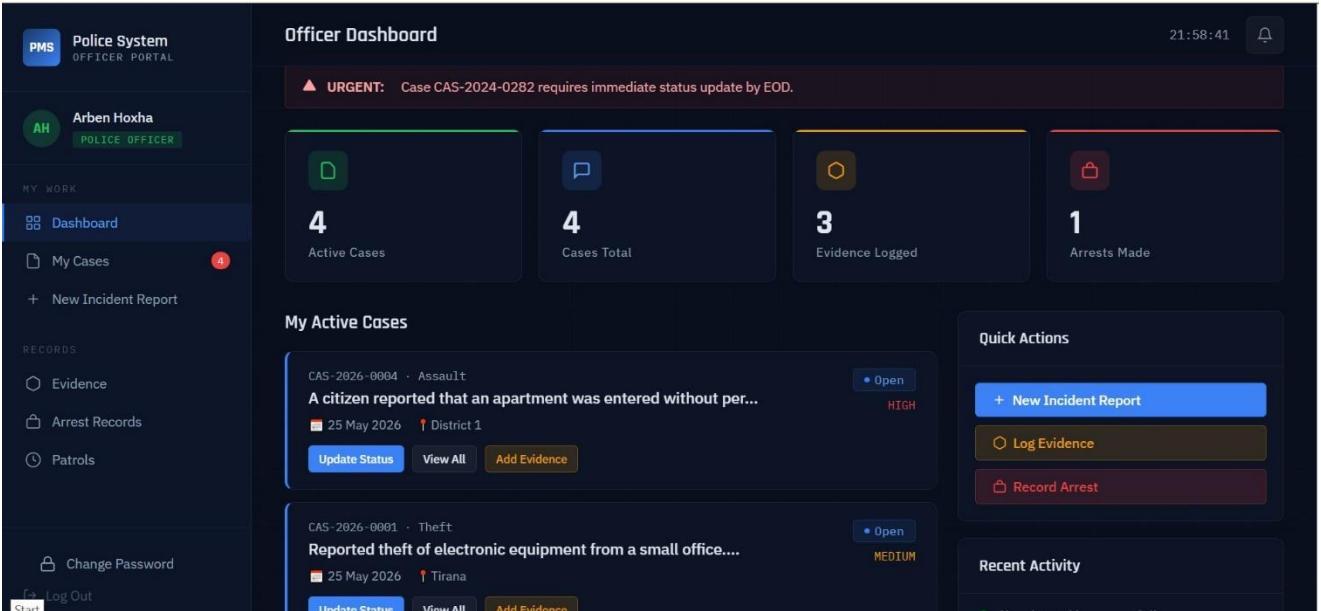
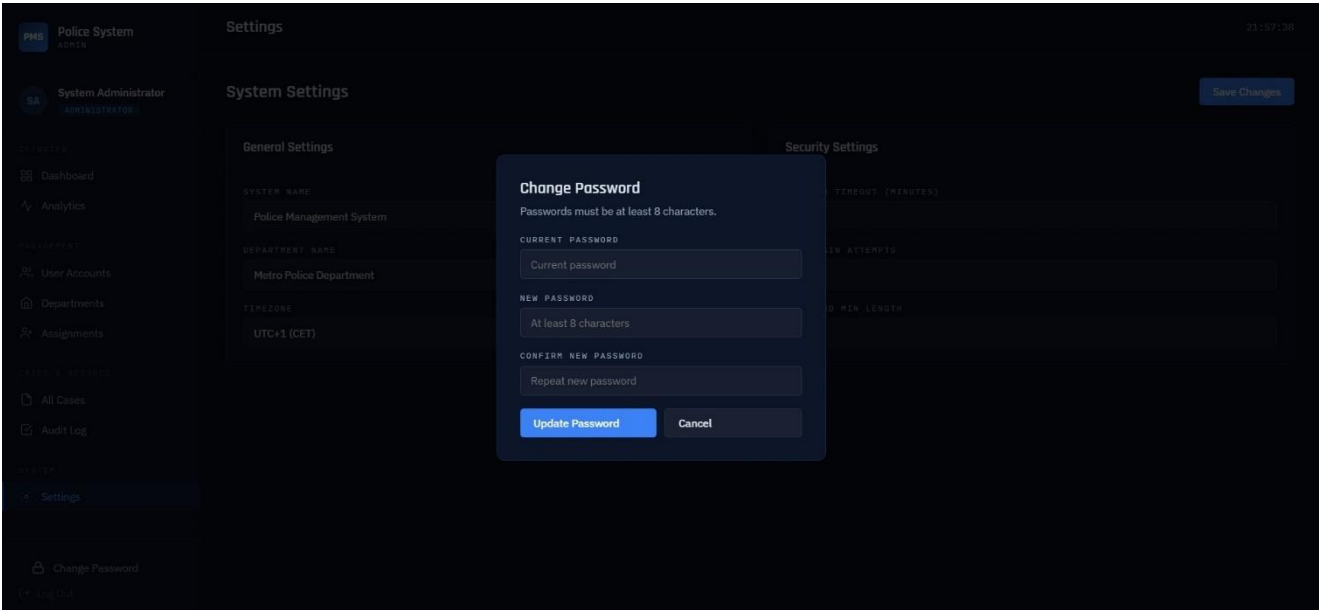
Github link : https://github.com/exheleshi23-sys/Software_Project

Appendix - Detailed Designs









PMS Police System
OFFICER PORTAL

21:59:44

Arben Hoxha
POLICE OFFICER

My Cases
All cases assigned to you

Log New Evidence

CASE
Select case...

EVIDENCE TYPE
Physical

DESCRIPTION
Describe the evidence in detail...

STORAGE LOCATION
e.g. Evidence Room B, Shelf 3

Log Evidence Cancel

STATUS	EVIDENCE	ACTIONS
+ Open	HIGH	Status Evidence
+ Open	MEDIUM	Status Evidence
+ Urgent	HIGH	Status Evidence
+ Open	LOW	Status Evidence

PMS Police System
OFFICER PORTAL

22:00:26

Arben Hoxha
POLICE OFFICER

New Incident Report

Incident Details

CRIME TYPE
Burglary

DATE OF INCIDENT
mm/dd/yyyy

TIME OF INCIDENT
--:--:--

LOCATION / ADDRESS
Street address or landmark

DISTRICT / ZONE
District 1

INCIDENT DESCRIPTION
Provide a detailed description of the incident, circumstances, and any immediate actions taken...

REPORTING OFFICER
Arben Hoxha (OFF001)

WITNESSES
0

SUSPECTS IDENTIFIED
No

PRIORITY
Low

Submit Report Save Draft

PMS Police System
OFFICER PORTAL

22:02:21

Arben Hoxha
POLICE OFFICER

Arrest Records

Record Arrest

RELATED CASE
CAS-2026-0004 — Assault

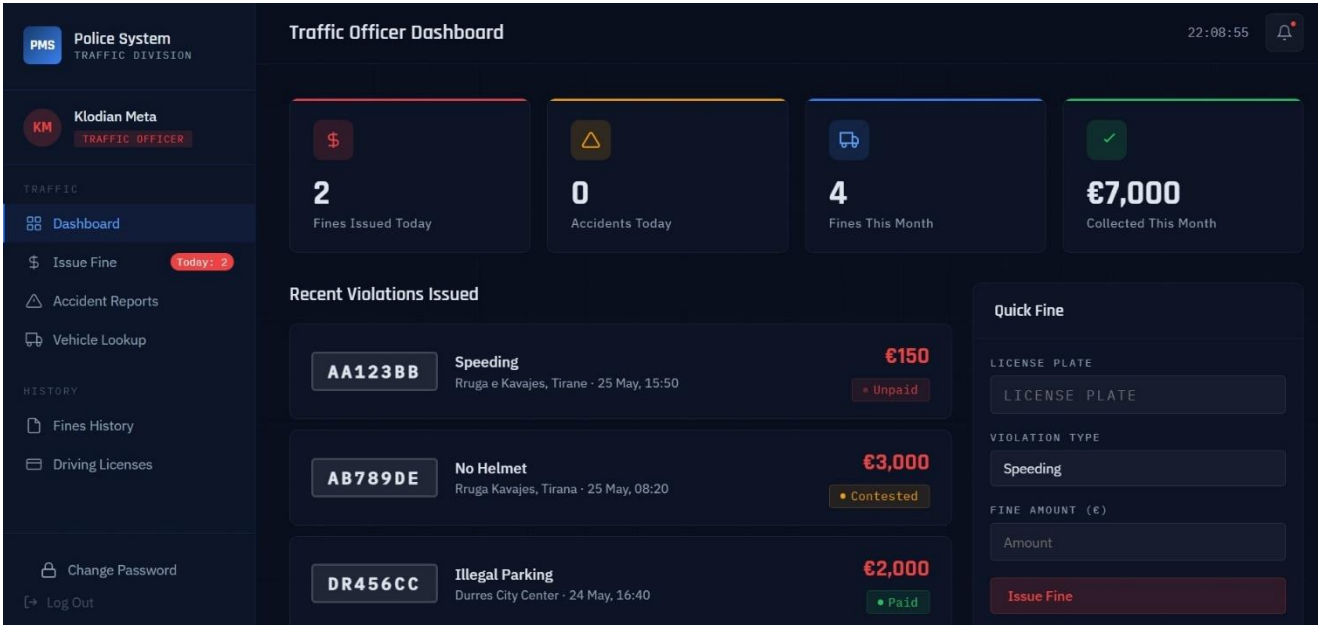
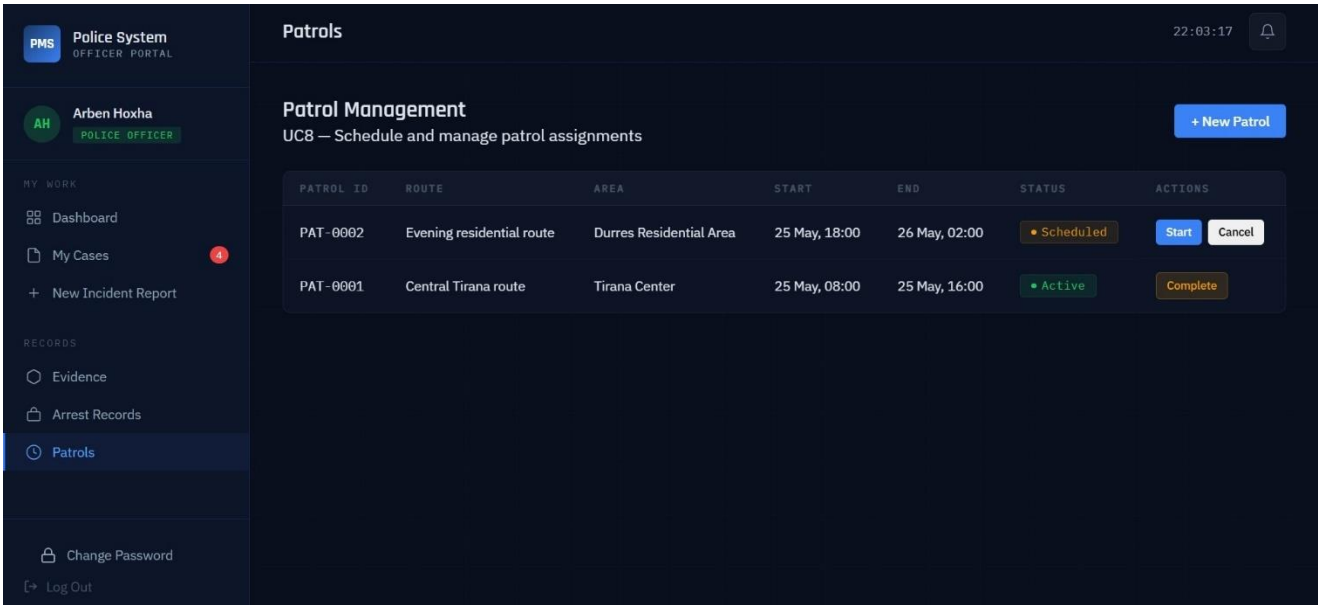
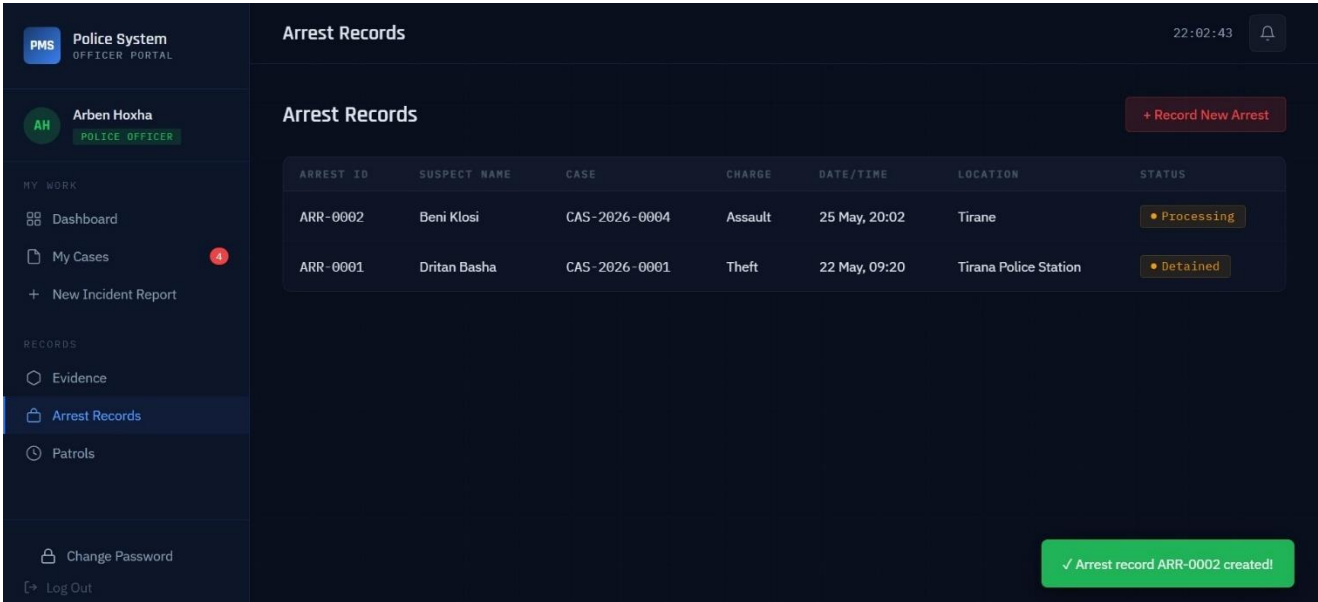
SUSPECT NAME
Beni Klosi

CHARGE
Assault

ARREST LOCATION
Tirane

Record Arrest Cancel

ARREST ID	LOCATION	STATUS
ARR-0001	Tirana Police Station	+ Detained



PMS Police System
TRAFFIC DIVISION

KM Klodian Meta
TRAFFIC OFFICER

TRAFFIC

- Dashboard
- Issue Fine Today: 2
- Accident Reports
- Vehicle Lookup

HISTORY

- Fines History
- Driving Licenses

Change PasswordLog Out

Vehicle Lookup

22:09:55

Vehicle Lookup

Check vehicle registration and fine history

LICENSE PLATE

AA123BB

Search

Results for: AA123BB

OWNER

Mira Leka

LICENSE NUMBER

DL-AL-0001

NATIONAL ID

J60412001A

STATUS

Valid

FINE HISTORY

FINE ID	VIOLATION	AMOUNT	DATE	STATUS
FIN-0004	Speeding	€150	25 May 2026	Unpaid
FIN-0001	Speeding	€5,000	23 May 2026	Paid

PMS Police Portal
CITIZEN ACCESS

ML Mira Leka
CITIZEN

MY PORTAL

- Home
- Report Incident
- Submit Complaint
- Report Missing Person

TRACKING

- My Reports
- My Fines
- Notifications

Change PasswordLog Out

Citizen Portal

22:06:58

Welcome, Mira 🙌

Use the Citizen Portal to report incidents, track your cases, view fines, and stay updated with official notifications.

+ Report an IncidentTrack My Reports

Report Incident

Report a crime or suspicious activity to police authorities

Track My Case

View the status and updates on your submitted reports

View Fines

Check and manage your traffic fines and penalties

File Complaint

Submit a formal complaint regarding police services

Notifications

Stay updated with important alerts and announcements

Emergency: 112

For emergencies, call 112 immediately. Do not use this portal.

PMS Police Portal
CITIZEN ACCESS

ML Mira Leka
CITIZEN

MY PORTAL

- Home
- Report Incident
- Submit Complaint
- Report Missing Person

TRACKING

- My Reports
- My Fines
- Notifications

Change PasswordLog Out

Report Incident

22:07:37

Report an Incident

All reports are reviewed by our officers within 24 hours

Incident Details

For emergencies in progress, please call 112 immediately instead of using this form.

INCIDENT TYPE

Theft/Robbery

DATE OF INCIDENT

mm/dd/yyyy

APPROXIMATE TIME

LOCATION

Street, building, or landmark

DESCRIPTION

Please describe the incident in as much detail as possible. Include any suspect descriptions, vehicle details, or other relevant information...

YOUR NAME (OPTIONAL - ANONYMOUS ALLOWED)

Leave blank to report anonymously

CONTACT PHONE (FOR FOLLOW-UP)

+355 ...

Submit ReportSubmit Anonymously

How It Works

1 Submit your report

Fill in the details and submit. You'll receive a report number for tracking.

2 Officer Review

A duty officer will review your report within 24 hours and classify it.

3 Investigation

If applicable, the case is assigned to a detective for further investigation.

4 Updates

You'll receive notifications when the status of your report changes.

Emergency Contacts

Emergency Number

112

March 26, 2026

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